

Continuum Structural Complexity: Target-Aligned Source Algebra, Physical Expenditure, and Singular Reachability

Guillem Duran Ballester
guillem@fragile.tech

Abstract

A public continuum presentation consists only of a displayed weak PDE or flow, its datum and boundary conditions, its standard local solution class, the canonical weak test core, native transformations, and the textbook formal energy, action, or entropy identity. Supplying a specialization is therefore only a transcription of public, target-blind data; no problem-specific estimate, compactness lemma, equality theorem, or auxiliary proof object is an input. From these data a recursive compiler assigns every native term and generated record to the inherited families `Geom`, `Caus`, `Abs`, `Lift`, `Bdry`. Weak limits, concentrations, oscillations, defect measures, tails, traces, and blow-up germs inherit the ancestry of their generating records; normalization and chart changes append only `Abs` or `Lift`. The post-saturation residual is defined as target-null and cannot carry singular progress.

The continuation graph generates the singular target, while its bounded weak evaluation coordinates generate a compact metrizable record space. Canonical target shells give a total measure identity on every pre-contact interval. Ordinary currents and every chain-rule or interface defect enter that same identity with inherited source tags. Contact therefore constructs infinitely many disjoint unit-progress shell records and forces a persistent target-aligned source without a compactness or margin premise. The native signed carrier identity cancels internal transport before taking positive parts and generates the sole physical expenditure measure and its orbitwise budget.

Sharp physical prices either contradict this finite measure or produce zero-price shell tails. The latter are not left as a scalar alternative. A canonical activity lift records their atomic, diffuse, concentration, tail, gauge, boundary, and graph-defect coordinates in one compact probability space. A witness-indexed successor relation preserves original event indices and prevents double counting. Iterating its zero-price viability operator produces finite-depth sets Z_n . If some Z_N is empty, a canonical finite rational-cylinder pruning transcript proves target avoidance from the generated graph relations. If every Z_n is nonempty, compactness constructs an infinite path and a stationary zero-price law whose support satisfies the full weak equation and every generated nonnegative saturation identity. The output is thus either global regularity or an explicit recurrent singularity mechanism with source ancestry, target pairing, and one cascade-origin witness. When contact generates the mechanism, that witness is a diagonal family of actual disjoint physical events.

Hamiltonian resolvents, signed carrier cohomology, covariant entropy cocycles, commutator squares, and compactness–rigidity identities are internally generated refinements of this successor algebra. A PDE specialization supplies no problem-specific estimate or Liouville theorem: all target-active values, including failed analytic relations, are iterated by the same structural mechanism.

Contents

Introduction	3
Visual architecture and proof-flow guide	4
Diagram map	5
Proof-dependency diagram	6
Node-by-node source audit	6
Monotone retained-fact table	7
Branch-closure audit	8
 I Closed Continuum Structural Algebra	 9
I.1 Physical continuum presentations	9
I.2 The five-source orbit algebra	13
I.3 Physical-budget-saturated sourcewise completion	17
I.4 Target quotient and the null residual	20
I.5 Prospective contact accountability	23
 II Physical Structural-Invariant Calculus	 29
II.1 Structural invariant reduction of analytic arguments	29
II.2 Coherent carrier passivity and structural successor closure	37
II.3 Covariant action and the physical budget	48
II.4 Hamiltonian meanings and chart covariance	50
II.5 Killed resolvents, resistance, and target realization	52
II.6 Full-generator response and physical aiming	56
II.7 Degenerate geometry and commutator-generated carriers	58
II.8 Compactness and rigidity as a price compiler	60
II.9 Constraint tails as finite structural obstructions	63
II.10 Covariant entropy cocycles and physical production	64
 III Structural Reachability and Regularity	 67
III.1 Singular targets and cascade compilers	67

III.2	The continuous structural regularity theorem	72
III.3	Structural evaluator coverage and failure semantics	79
III.4	A reusable PDE instantiation protocol	81
III.5	Conclusion	82
A	Total branch table for the main conclusions	83
B	Notation index	83

Introduction

Let u be an ordinary maximal branch of a fixed continuum evolution and let Σ be a target in its state or germ space. The central question is whether the equation can reach Σ while spending only a finite amount of the physical quantity controlled by its evolution law. Each orbit carries one nonnegative countably additive expenditure measure μ_u on a generated event space $\mathfrak{Q}_u$. The finite-budget stratum is

$$\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}} < \infty.$$

Action, dissipation, entropy production, forcing work, and boundary flux define this one measure; energy and conservation identities bound its total mass before the ceiling B_{phys} is set.

The source alphabet, ancestry discipline, saturation rule, and target-relative residual extend the structural algebra of Part I [2]. The continuum construction below derives its physical budget, source-preserving limit closure, prospective contact identity, signed carrier calculus, successor recurrence, Hamiltonian formulation, and saturation equations within this paper. The remaining citations identify standard analytic results used at explicitly named steps. These references and Part I constitute the complete logical dependency set.

The structural source alphabet records how target-directed behavior is generated. The native dissipative or reversible mechanism is **Geom**; the fixed directed or nonreversible transport is **Caus**; quotient and symmetry reduction are **Abs**; charts, renormalizations, and augmented state variables are **Lift**; and boundary, terminal, killing, and exterior-flux records are **Bdry**. A composite continuum object may carry several labels, but every elementary source belongs to this alphabet.

The completion is sourcewise. A limit object is placed in the closure of the cell that generated it. Thus a concentration measure retains the ancestry of the concentrating quantity, a blow-up profile retains the ancestry of the physical orbit together with its **Abs/Lift** chart, and a terminal tail trace acquires **Bdry** ancestry. Every source cell has an exact structural row value; its complementary value remains in the same cell and does not create an additional continuum source.

The target-relative residual is correspondingly strict. Let $\mathcal{N}_\Sigma$ be the common kernel of the target shell derivatives, target fluxes, and other declared prospective contact functionals. The residual coordinate is contained in $\mathcal{N}_\Sigma$. Hence

target-visible $\implies$ ancestry in $\{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}$; $J_{\text{res}} \implies$ target-null.

Every target-visible limit therefore retains ancestry in one of the five source families.

The central target-relative question is therefore not whether the entire solution is large, but which source current contributes positively to the formation of the singularity. For a normalized

shell crossing I_r , set

$$Q_{c,r} := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle dt.$$

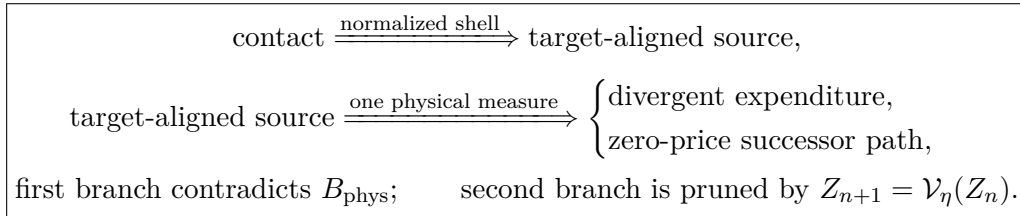
The compiled current identity gives the exact pre-contact identity on the ordinary shell branch

$$1 = \sum_{c \in \mathfrak{C}} Q_{c,r}, \quad \sum_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq 1, \quad \max_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq \frac{1}{5}.$$

The residual source term vanishes. Thus contact itself supplies the unit progress and proves the existence of target-aligned structure. If all source currents vanish in the target-active quotient, their magnitude is irrelevant: every shell observable is constant along the orbit, so contact is excluded without compactness, a physical-price gap, or a global estimate.

The aligned branch is simplified before any size estimate is attempted. A target-null or gauge direction is passed to the generated quotient and never reconsidered. For every surviving total-graph value the algebra forms its positive target-current measure, compact activity lift, and witness-indexed successors. The finite-depth sets $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ ask the structural question that replaces a global estimate: can this source occurrence sustain one more target-retaining zero-price step while satisfying the generated equation? Empty finite depth is a finite closure proof; survival at every depth constructs the recurrent mechanism itself.

The main expenditure argument then uses the same measure μ_u . A singular contact compiler produces infinitely many pairwise disjoint pre-contact events and selects a persistent aligned source cell. A divergent-price value contradicts countable additivity and $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}$. A finite-price value produces vanishing-price windows and activates the ordered successor calculus. It is therefore a reduction step, not a terminal singularity class.



Hamiltonian, resolvent, commutator, constraint-tail, scale, and entropy identities are functorial refinements generated when their typed expressions occur in the public equation. The symmetric form yields a spectral Hamiltonian and resistance; the full fixed-Caus generator retains its nonnormal current; signed carrier gluing cancels internal transport and exposes scale-feed cohomology; and a covariant entropy cocycle telescopes production across changing charts. A failed identity is lifted as a successor with the same ancestry. These modules operate on the five source families; they neither enlarge the public presentation nor alter the source algebra.

Visual architecture and proof-flow guide

The twenty diagrams are maps of proved implications and invariant interfaces. Blue nodes contain evolution data, violet nodes contain ancestry or coordinate transformations, teal nodes contain geometric or compactness data, amber nodes contain target uses or decisions, coral nodes contain reachability outputs, and gray nodes contain target-null classes or invariant branches. Solid arrows denote implications or constructions. Dashed arrows denote comparisons or derived evaluator routes.

The unique root of the global graph is the public continuum presentation. Rectangles are generated constructions, diamonds are exact structural tests, and ellipses are completed outputs. Every nonzero target-active value has an outgoing edge to the activity lift or successor relation. All locality, source labels, and physical balances established upstream remain active downstream. Node [6] forms the total shell identity, and node [17] performs finite-depth viability pruning.

Diagram map

No.	DAG node	Diagram	Principal internal sources
1	[1]–[18]	Constructive global-regularity proof flow	Proposition I.1.2, Definition II.2.13, and Theorem II.2.21
2	[1]–[2]	Public PDE to physical orbit records	Definition I.1.1 and Proposition I.1.2
3	[2]	Equation-generated physical expenditure	Definition I.1.7 and Proposition I.1.10
4	[2]	Five continuum source roles	Definition I.2.5 and Theorems I.2.6 and I.2.8
5	[3]	Sourcewise orbit completion	Definition I.3.2 and Theorem I.3.7
6	[4]	Limit ancestry and propagated reduction	Theorems I.3.4 and II.1.3
7	[5]	Target-visible re-entry and null residual	Definition I.4.2 and Theorem I.4.5
8	[6]–[7]	Normalized shell-crossing accountability	Theorems I.5.2 and I.5.14
9	[6]–[7]	Resolvent source-response factorization	Theorem I.5.5
10	[6]–[7]	Stochastic no-spontaneous-contact factorization	Theorems I.5.7 and I.5.14
11	[1], [9]–[10]	Carrier-to-physical-budget compiler	Theorems II.3.5, II.7.4 and II.7.6
12	[4]	Three Hamiltonian meanings	Proposition II.4.2
13	[4]	Same-current chart covariance	Theorem II.4.3 and Proposition II.4.4
14	[6], [9]	Killed arrival, resolvent, and resistance	Proposition II.5.3 and Theorems II.5.4 to II.5.6
15	[4], [6]	Full nonnormal target response	Definition II.6.1 and Proposition II.6.2
16	[6]–[9]	Physical aiming geometry	Theorem II.6.4 and Proposition II.6.5
17	[4], [14]–[17]	Branch-to-successor compiler	Definitions II.1.8 and II.2.13 and Theorem II.1.16
18	[1], [9]–[10]	Covariant entropy-cocycle production	Definition II.10.1 and Theorem II.10.2
19	[6]–[18]	Target-aligned scale-cascade and viability ledger	Proposition III.1.4 and Theorems II.2.19, II.2.20 and III.1.7

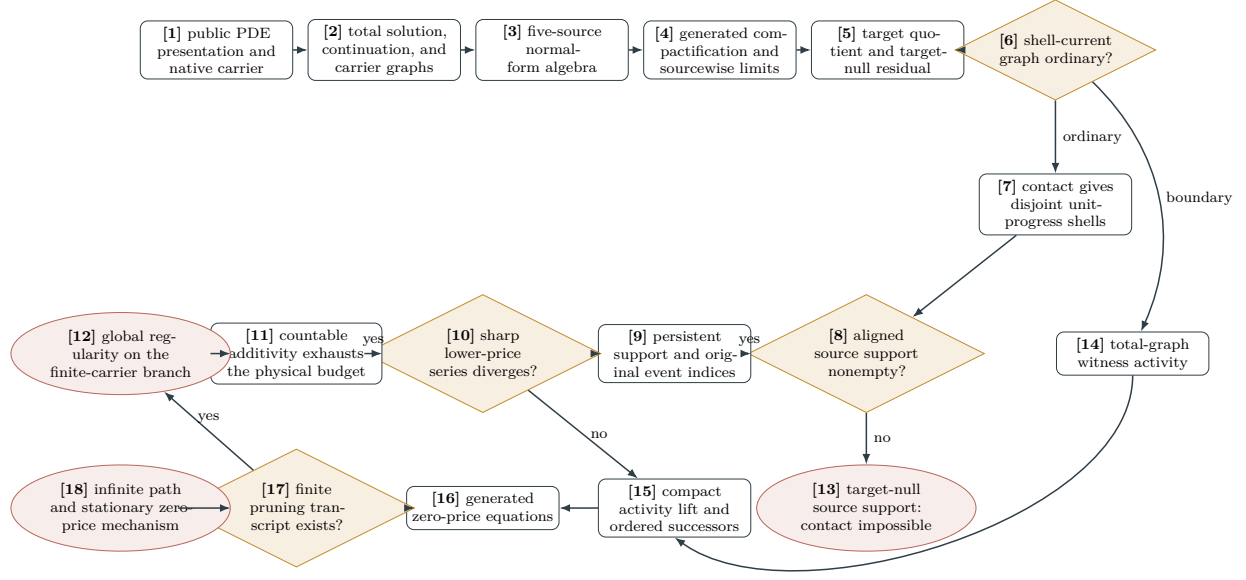


Figure 1: Global proof DAG. Public textbook data enter only at [1]. Nodes [2]–[6] generate the total graphs, source algebra, compactification, target quotient, and total shell graph. Its ordinary value gives unit progress at [7], while every boundary value enters [14]. Thus [8] forces aligned support. Nodes [9]–[11] close the divergent-price branch. Every ordinary or graph-boundary value also enters [14]; finite price joins it at [15]. The generated zero-price equations at [16] feed finite-depth viability pruning at [17]. A generated finite rational-cylinder transcript gives [12]. Failure of every finite pruning depth constructs the path and stationary mechanism at [18]. No live node is an unexpanded analytic label.

No.	DAG node	Diagram	Principal internal sources
20	[1]–[18]	Constructive structural regularity compiler	Theorems II.2.2, II.2.15, II.2.17 and II.2.21

Proof-dependency diagram

Node-by-node source audit

Node	State or test	Exact source	Successor
[1]	Public PDE, datum class, test core, operations, and native formal carrier	Definition I.1.1 and Proposition I.1.2	[2]
[2]	Total solution, continuation, and signed carrier graphs	Definitions I.1.1 and I.1.7	[3]
[3]	Five-source normal-form algebra	Definition I.2.1 and Theorem I.4.4	[4]

Node	State or test	Exact source	Successor
[4]	Generated evaluation compactification and sourcewise completion	Definition I.3.1 and Theorem I.3.4	[5]
[5]	Target-active quotient and null residual	Definition I.4.2 and Theorem I.4.5	[6]
[6]	Ordinary/boundary split of the canonical total shell-current relation	Definition I.5.1 and Theorem III.2.3	[7] or [14]
[7]	Contact-generated disjoint unit-progress shell family	Theorems I.5.2 and III.1.5	[8]
[8]	Singularity-formation support test	Theorem I.5.14 and Proposition III.1.4	[9] or [13]
[9]	Persistent source support with original disjoint event indices	Proposition III.1.10 and Lemma II.2.11	[10]
[10]	Canonical sharp lower-price series	Definition III.1.11 and Theorem III.1.7	[11] or [15]
[11]	Divergent series exhausts the finite physical measure	Theorems III.1.12 and III.1.14	[12]
[12]	Global regularity on the ordinary finite-carrier branch	Theorems II.2.21 and III.2.11	completed output
[13]	Every structural source current is target-null	Corollary I.5.3	completed output
[14]	Positive target activity of every total-graph witness	Definitions III.2.2 and II.2.8	[15]
[15]	Compact activity lift and witness-indexed successor relation	Definitions II.2.8 and II.2.10 and Theorem II.2.9	[16]
[16]	Full weak equation and generated nonnegative zero-price descendants	Definition II.2.16 and Theorem II.2.17	[17]
[17]	Finite-depth viability sets and canonical rational-cylinder pruning transcript	Definition II.2.13 and Theorems II.2.15 and III.2.8	[12] or [18]
[18]	Infinite zero-price path and stationary generated mechanism	Theorems II.2.12 and II.2.21	completed output

Monotone retained-fact table

Retained fact	Persistence rule
Physical orbit	Every normalization, window, limit, and prospective functional is attached to the same ordinary maximal branch or to a witnessed boundary sequence from that branch.

Physical budget	Restrictions to subintervals and pairwise disjoint event families use one nonnegative expenditure measure and cannot increase its total mass.
Source ancestry	Algebraic operations take unions of source words; sourcewise limits remain in the same closed cell.
Structural reduction	A closed target-null or otherwise excluded direction is propagated through every zero descent-defect quotient and is never re-estimated downstream.
Fixed Caus	Charts transport the declared nonreversible current and do not replace it by a target probe or an optimizer.
Target residual	The residual remains in the common target kernel under closure; a nonzero target pairing triggers re-entry into a source cell.
Target alignment	Every normalized contact shell has unit total progress, so at least one source pairing is at least $1/5$; target-null motion may be arbitrarily large without entering later estimates.
Source evaluation	Each persistent source cell is mapped to target-nullity, divergent physical expenditure, or the compact successor relation; the last is iterated by $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ and returns either its finite pruning transcript or its same-origin recurrent law.
Locality	Compactness, chain rules, lower semicontinuity, and price relations are evaluated on the generated windows and evaluation topology; failures are typed graph defects.
Invariant status	Every row returns its closing or complementary value; its source label remains fixed.

Branch-closure audit

Route	Branch	Constructive source	Output
[6]–[14]–[15]	an interface has a target-visible graph-boundary value	Theorem III.2.3 and Definition II.2.8	compact activity profile with witnessed successor
[8]–[13]	every compiled source pairing is target-null	Corollary I.5.3	target unreachable directly
[10]–[15]–[16]–[17]	the persistent aligned source has finite sharp lower-price series	Definitions III.1.11 and II.2.13 and Theorem II.2.20	finite-depth pruning or an infinite zero-price path
[10]–[11]–[12]	the persistent aligned source has divergent sharp lower-price series	Theorems III.1.14 and III.2.11	target unreachable
[17]–[12]	a finite pruning transcript covers the target-active lift	Theorems II.2.21 and III.2.8	global regularity on the ordinary finite-carrier branch

Part I

Closed Continuum Structural Algebra

I.1 Physical continuum presentations

Definition I.1.1 (Canonical physical continuum presentation). A canonical physical continuum presentation is the public tuple

$$\mathfrak{P}_{\text{phys}} = (\text{Eq}, u_0, \mathcal{D}, \text{Cont}, \text{Op}, \mathcal{K})$$

with the following data, all transcribed directly from a public textbook or standard reference statement of the initial–boundary value problem. A nonliteral entry below is admissible only when it is target-blind and already part of the standard local theory of the displayed equation; establishing a new estimate or theorem is not part of the specialization datum.

- (P1) **Eq** is the displayed differential equation with its domain, coefficients, boundary conditions, and declared regularity class; u_0 is its datum; and $\mathcal{D} = \{\varphi_n\}_{n \geq 1}$ is the canonical rational smooth test core. The datum class is accompanied by the standard local solution theorem used to define maximal branches. This local theorem must be target-blind in the sense of Definition I.1.3; no class selected by a desired global conclusion is part of the presentation.
- (P2) The *total local-solution graph* contains every maximal path $u : [0, T_*(u)) \rightarrow X_{\text{reg}}$ satisfying **Eq** against every φ_n , with the prescribed datum and physical boundary data. If the fiber over u_0 is empty, its value is the explicit record $\perp_{\text{loc}}$. Nonuniqueness retains all maximal paths. Consequently an empty solution fiber is a form defect and can never be read as target avoidance. On the public datum class, the standard local theorem makes the ordinary fiber nonempty; $\perp_{\text{loc}}$ audits presentation mismatch and data outside that class.
- (P3) **Cont** is the syntactic extension relation in the declared regularity class: two local paths are related when their restrictions agree. No continuation theorem is supplied. Its ordinary values are actual regular extensions and its graph-boundary values are terminal germs. The singular target is generated from this graph by Definition III.1.1; it is not an independently chosen favorable subset.
- (P4) **Op** = $\{O_n\}_{n \geq 1}$ is a countable typed list of the restriction, rational truncation, quotient, chart, augmentation, trace, and boundary operations syntactically available for **Eq**. An operation acts as a symmetry or gauge only on the ordinary value of its total covariance graph, where it preserves the weak equation, the carrier, and the continuation target. Thus the generated gauge family is the equalizer of these three actions; it is not declared independently.

(P5)

$$\mathfrak{C} = \{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}$$

is the inherited source alphabet of Definition I.2.1. Native terms are routed by the sector/gradient compiler and subsequent orbit-record operations acquire their source words by the constructor grammar; this is proved in Theorem I.4.4.

- (P6) $\mathcal{K}$ is the native textbook energy, action, or entropy density displayed with the equation before the continuation target is generated. It is not selected after a singular scenario has been inspected. Only its formal multiplier identity—obtained by testing Eq, integration by parts, and cancellation of equal internal faces—belongs to the public presentation. Testing Eq, summing with its native signs, and cancelling common internal faces generates a total signed carrier relation. Positivity, a lower carrier bound, finite exterior input, and finite total expenditure are values of that relation, not presentation data.
- (P7) Identify two states or germs when every bounded rational truncation of the weak pairings, carrier coordinates, continuation coordinates, and operations generated in (P1)–(P6) has the same value. Enumerate these coordinates as $(e_n)_{n \geq 1}$. They separate the resulting evaluation quotient by construction. The evaluation embedding and metric are

$$\iota(x) = (e_n(x))_{n \geq 1} \in [-1, 1]^{\mathbb{N}}, \quad d_{\text{ev}}(x, y) = \sum_{n \geq 1} 2^{-n} |e_n(x) - e_n(y)|. \quad (1)$$

The closure $X_{\text{ev}} := \overline{\iota(X_{\text{reg}} \sqcup X_{\text{germ}})}$ is compact metrizable. Every subsequence, probability law, and graph closure below is taken in this generated space.

The presentation is therefore fixed by public equation data and a formal multiplier calculation. The specialization step consists only of recording these statements and their public references; it contains no new mathematical argument. It may not contain a compactness theorem, coercive estimate, price lower bound, Liouville statement, target-exclusion lemma, special blow-up theorem, or exact invariant value. Such an insertion changes the equation signature and is rejected by the compiler.

Proposition I.1.2 (Public-presentation input contract). *After Eq, its public datum class, and its native formal carrier identity are fixed, every object used by the global-regularity track is generated functorially: the test core by rational enumeration, the solution and continuation graphs by satisfaction of the weak relation, the operation list by the typed syntax, the topology by bounded evaluation coordinates, the target by failed continuation, the source words by normal-form compilation, and the physical measure by the positive Jordan part of the signed carrier identity. A PDE specialization therefore supplies no additional problem-specific regularity input, proof object, favorable constant, or target-relative lemma.*

Proof. Each construction is the one stated in items (P1)–(P7). All subsequent objects are defined by graph closure, target quotient, Jordan decomposition, normalized shell formation, compact activity lifting, or the successor operator. These operations consume only the preceding generated objects. Inspection of the dependency chain shows that none has an input slot for any of the forbidden conclusions listed above. $\square$

Definition I.1.3 (Target-blind public input). Let $\text{Pre}(\text{Eq})$ be the signature formed before the continuation target, target shells, normalized bad windows, source quotients, or physical-price strata are constructed. It contains the displayed initial-boundary value problem, its declared local regularity class, weak test core, standard local solution theorem, native algebraic symmetries and scalings, and formal multiplier identities. It may use a fixed universal library of target-independent results such as distribution theory, integration by parts, Sobolev-space definitions, Jordan decomposition, compactness of countable products of compact metric spaces, weak compactness of probability measures on such spaces, and the spectral theorem.

A fact is an admissible public input precisely when it is expressible in $\text{Pre}(\text{Eq})$ and remains unchanged if the continuation target and all target observables are deleted. In particular, the

following are not public inputs:

- compactness of a target-approaching sequence, an epsilon-regularity or continuation conclusion,
 - a target price gap, a blow-up Liouville theorem, absence of a scale-soliton,
 - vanishing of a target quotient, or any decomposition selected after singular shells are formed.
- (2)

All objects in (2) must be generated and processed by the successor algebra.

Theorem I.1.4 (Syntactic no-smuggling criterion). *The dependency graph of the global-regularity compiler has exactly one edge from user data: the inclusion of $\text{Pre}(\text{Eq})$ into the canonical presentation. Every target, source, price, compactness, equality, and recurrence object lies strictly downstream of that edge. Hence no target-relative conclusion can enter as a public input. Adding any fact in (2) changes the signature and does not count as an evaluation of the original presentation.*

Proof. The target is generated from the continuation graph, the shells from the target distance in X_{ev} , the source cells from normal-form syntax, the price from the native carrier, and the recurrent equations from the successor fixed point. These constructions form a directed acyclic dependency chain beginning at $\text{Pre}(\text{Eq})$. None has a backward input slot. Every forbidden fact mentions a downstream object and therefore cannot be a term of the pre-target signature. $\square$

Corollary I.1.5 (Textbook-only specialization interface). *The external specialization datum ends with $\text{Pre}(\text{Eq})$. Its nonliteral mathematical content consists only of the cited standard local solution statement and the native formal carrier identity already attached to the equation. In particular, no later row has an argument slot for a PDE-specific compactness proof, shell estimate, price bound, equality analysis, continuation criterion, or prescribed value of Z_n . The compiler itself generates the total graphs, all complementary values, the rational-cylinder pruning transcript, and, when pruning never closes, the same-origin infinite successor mechanism.*

Proof. The first sentence is Definition I.1.3. The dependency statement follows from Theorem I.1.4. Totalization and the successor relation generate both values of every downstream relation; the finite pruning transcript is constructed in Theorem II.2.15. None of these constructions adds an input edge to the dependency graph. $\square$

Proposition I.1.6 (Non-vacuity of the total solution graph). *For every datum, exactly one of the following occurs: the total local-solution fiber contains at least one maximal orbit, or it contains the nonzero form defect $\perp_{\text{loc}}$. In particular, no reachability or regularity functional defined from the presentation assigns the value zero because the orbit fiber is empty.*

Proof. The total graph is the disjoint union of its ordinary solution values and its explicit failure value. The empty ordinary fiber therefore selects $\perp_{\text{loc}}$. Subsequent quotients retain that record unless its target projection is proved to vanish. $\square$

Definition I.1.7 (Generated physical expenditure and budget). Let u be an ordinary value of the total solution graph. Evaluate the total signed carrier relation generated by $\mathcal{K}$. On its ordinary branch, signed gluing gives

$$K_u(t) + \mu_u((s, t]) + W_u^-((s, t]) = K_u(s) + W_u^+((s, t]), \quad 0 \leq s < t < T_*(u), \quad (3)$$

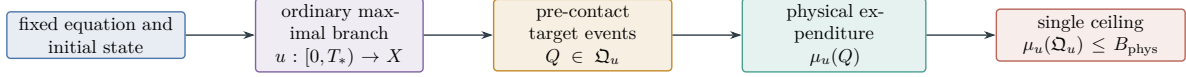


Figure 2: One physical-budget architecture. The equation generates the orbit, target approach selects pre-contact events, and all such events spend the same countably additive measure whose total mass is bounded by B_{phys} .

where $W_u^\pm$ are the Jordan parts of true exterior work and μ_u is the sum, with the native coefficients of the identity, of all generated nonnegative production and balance-defect measures. Internal transport, chart, gauge, and interface currents have already cancelled and do not enter μ_u .

If $K_u \geq K_- > -\infty$ and $W_u^+([0, T_*(u))) < \infty$, define the single orbit budget

$$B_{\text{phys}}(u_0, W^+) := K_u(0) - K_- + W_u^+([0, T_*(u))) < \infty. \quad (4)$$

Then (3) derives

$$\mu_u(\Omega_u) \leq B_{\text{phys}}(u_0, W^+). \quad (5)$$

Failure of the carrier identity, nonnegativity, the lower bound, or finite exterior input is the corresponding total-graph value. Its target-visible part enters the activity lift and finite-depth successor recursion; that recursion returns a finite violated carrier identity or a recurrent zero-price carrier-defect mechanism. The value is neither discarded nor misread as finite-budget regularity.

The measure is localizable. For pairwise disjoint $Q_j \in \mathcal{A}_u$,

$$\sum_j \mu_u(Q_j) = \mu_u\left(\bigsqcup_j Q_j\right) \leq B_{\text{phys}}. \quad (6)$$

No target-dependent expenditure may be appended. A Hamiltonian, entropy, commutator, or localized identity refines μ_u only after it has been proved to be an algebraic decomposition of (3). If the basic carrier assigns finite price to a singular cascade, that cascade proceeds to the zero-cost successor calculus; the budget is not strengthened after the fact.

Example I.1.8 (Energy–work budget). On the zero-defect branch of the lower-energy, dissipation-balance, and finite-work rows, an energy $\mathcal{E}$, lower value E_- , and nonnegative dissipation measure μ_{diss} satisfy

$$\mathcal{E}(u(t)) + \mu_{\text{diss}}([0, t]) \leq \mathcal{E}(u_0) + W^+([0, t]),$$

and $W^+([0, \infty)) < \infty$. Then

$$\mu_{\text{diss}}([0, \infty)) \leq \mathcal{E}(u_0) - E_- + W^+([0, \infty)).$$

Taking $\mu_u = \mu_{\text{diss}}$ gives the single budget

$$B_{\text{phys}} = \mathcal{E}(u_0) - E_- + W^+([0, \infty)).$$

If this physical measure assigns finite total price to a singular cascade, the cascade enters the zero-price successor calculus. No stronger measure or second budget is introduced after the target has been inspected.

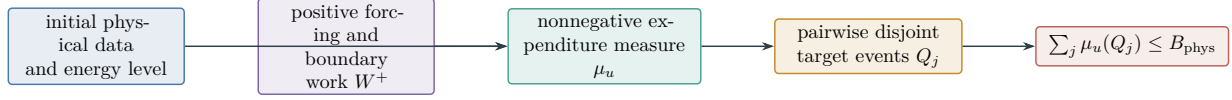


Figure 3: Physical expenditure accounting. The initial state and legitimate positive input bound one countably additive measure. Every disjoint target event spends that same measure, whose total mass is the sole budget.

Definition I.1.9 (Fixed-Caus gradient-flow presentation). Let $\mathcal{V} \hookrightarrow \mathcal{H} \hookrightarrow \mathcal{V}^*$ be a Gelfand triple. A fixed-Caus gradient-flow presentation consists of a differentiable energy $\mathcal{E}$, a symmetric nonnegative mobility $\mathbb{K}(u) : \mathcal{V}^* \rightarrow \mathcal{V}$, a fixed constitutive transport $A_{\text{Caus}}(u) \in \mathcal{V}^*$, and an exterior input $f_{\text{Bdry}} \in \mathcal{V}^*$, with equation

$$\dot{u} + A_{\text{Caus}}(u) = -\mathbb{K}(u)D\mathcal{E}(u) + f_{\text{Bdry}}. \quad (7)$$

The Caus term is energy-skew when

$$\langle D\mathcal{E}(u), A_{\text{Caus}}(u) \rangle = 0. \quad (8)$$

Write

$$\|\mathbb{K}(u)^{1/2}\xi\|^2 := \langle \xi, \mathbb{K}(u)\xi \rangle$$

for $\xi \in \mathcal{V}^*$; this notation does not assume a bounded inverse or a globally defined square root on $\mathcal{V}^*$. Here **Geom** is the mobility current $-\mathbb{K}D\mathcal{E}$, **Caus** is the declared transport, and **Bdry** contains the exterior input and boundary flux. Quotients and symmetry reductions applied to this orbit have **Abs** ancestry; charts, blow-up rescalings, and augmented variables have **Lift** ancestry.

Proposition I.1.10 (Fixed-Caus expenditure bound). *Let u satisfy Eqs. (7) and (8) on $[0, T]$, with the chain rule for $\mathcal{E}(u)$. Then*

$$\frac{d}{dt}\mathcal{E}(u(t)) = -\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\|^2 + \langle D\mathcal{E}(u), f_{\text{Bdry}} \rangle. \quad (9)$$

If $\mathcal{E} \geq E_-$, then

$$\int_0^T \|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\|^2 dt \leq \mathcal{E}(u_0) - E_- + \int_0^T \langle D\mathcal{E}(u), f_{\text{Bdry}} \rangle_+ dt. \quad (10)$$

For a weak solution with an energy inequality, the same statement holds with an additional nonnegative energy-defect measure on the left.

Proof. Pair (7) with $D\mathcal{E}(u)$. The chain rule gives the left side of (9), the Caus pairing vanishes by (8), and positivity of $\mathbb{K}$ gives the squared norm. Integrate, discard the terminal energy above E_- , and replace exterior work by its positive part. The weak form is the increment formula for the energy inequality; its defect is nonnegative. $\square$

I.2 The five-source orbit algebra

Definition I.2.1 (Continuum source alphabet). The source alphabet is

$$\mathfrak{C} = \{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}.$$

A raw continuum record is an orbit occurrence together with a finite ancestry word $\text{Anc}(r) \in \mathfrak{C}^*$. Its source support $\text{Src}(r) \subseteq \mathfrak{C}$ is the set of letters appearing in that word. The primitive and closure rules are:

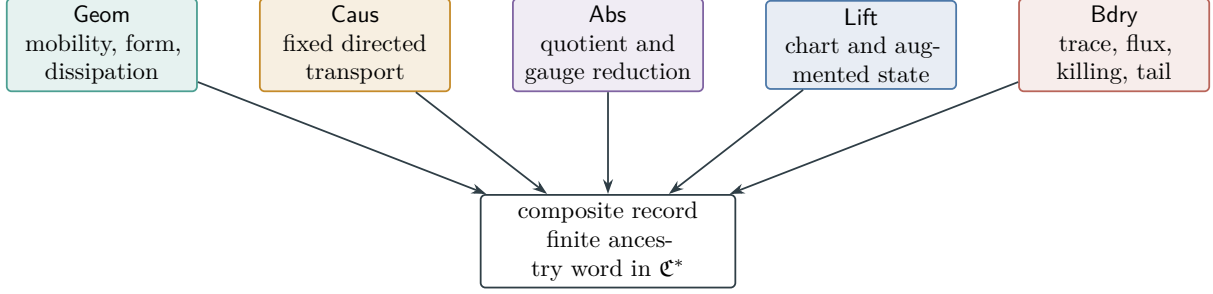


Figure 4: The five continuum source roles. **Geom** and **Caus** are native current mechanisms; **Abs** and **Lift** record transformations of those mechanisms; **Bdry** records exterior and terminal effects. Composite records retain the union of their ancestries.

- (S1) symmetric mobility, Dirichlet-form, reversible, and dissipative current records start in **Geom**;
- (S2) the fixed directed transport and its current start in **Caus**;
- (S3) quotient, symmetry reduction, gauge removal, and coarse variables append **Abs**;
- (S4) charts, renormalizations, blow-up variables, auxiliary-state extensions, and Young- or defect-variable lifts append **Lift**;
- (S5) boundary traces, exterior fluxes, killing, terminal values, and tails read at the boundary of a compactification append **Bdry**;
- (S6) finite algebraic combination, restriction, and concatenation take the union of the input source supports.

The symbol J_{res} is absent from this source alphabet. It is introduced only as the target-null coordinate of Definition I.4.2.

Remark I.2.2 (Channel and source terminology). The channels **Geom** and **Abs** carry intrinsic reusable structure, while **Caus**, **Lift**, and **Bdry** orient, reorganize, and read out that structure. A *source cell* is a provenance cell in the complete five-channel normal form; this terminology does not promote the three derived channels to independent creators of intrinsic structure.

Remark I.2.3 (Forcing and constraints). The source compiler routes a volume force according to its constitutive role. A fixed directed body force may be **Caus**, while exterior work or a boundary supply is **Bdry**. A Lagrange multiplier implementing a quotient constraint has **Abs** ancestry. The five labels are assigned from the declared weak equation, not from the sign of a later estimate.

Definition I.2.4 (Operational weak-PDE signature). An operational weak-PDE signature is formed downstream from the public presentation. It consists of a Gelfand triple $V \hookrightarrow H \hookrightarrow V^*$, a separating test core $\mathcal{D} \subset V$, an initial datum, the continuation target Σ generated by Definition III.1.1, and a weak evolution relation

$$\langle \dot{u}, \varphi \rangle_{V^*, V} = \mathcal{A}(u; \varphi) + \mathcal{F}_{\text{ext}}(u; \varphi), \quad \varphi \in \mathcal{D}. \quad (11)$$

The native spatial relation is entered before any realization property is asserted. Its totalized realization relation has the following ordinary values.

(W1) A fixed-Caus gradient realization:

$$\mathcal{A}(u; \varphi) = -\langle \mathbb{K}(u) D\mathcal{E}(u), \varphi \rangle - \langle A_{\text{Caus}}(u), \varphi \rangle,$$

with $\mathbb{K}(u)$ symmetric and nonnegative.

(W2) A sector-form realization on a real L^2 -space:

$$\mathcal{E} = \mathcal{E}^s + \mathcal{E}^a, \quad \mathcal{E}^s(v, w) = \frac{1}{2}\{\mathcal{E}(v, w) + \mathcal{E}(w, v)\}, \quad \mathcal{E}^a(v, w) = \frac{1}{2}\{\mathcal{E}(v, w) - \mathcal{E}(w, v)\},$$

where $\mathcal{E}^s$ is a Dirichlet form and $\mathcal{E}^a$ satisfies the sector bound.

If neither ordinary value exists, the realization relation returns $\perp_{\text{nat}}$. The raw fixed bulk current remains a **Caus** occurrence and the failed attempted **Geom**–**Caus** split is a form-defect record with support $\{\text{Geom}, \text{Caus}\}$. Its target-active quotient is evaluated by the form row. Thus absence of a gradient, Dirichlet, positivity, or sector realization is admitted as structure rather than excluded from the signature. The signature also declares the symmetry, quotient, chart, auxiliary-state, and trace operations that may be applied to (11). It contains no assertion of compactness, regularity, target avoidance, positive physical price, or global continuation.

Definition I.2.5 (Continuum source compiler). Let $\text{Syn}(\mathfrak{P})$ be the smallest typed syntax containing the native terms of (11) and closed under finite sum, restriction, concatenation, quotient, chart, state augmentation, weak limit, and trace. The source compiler

$$C_{\mathfrak{P}} : \text{Syn}(\mathfrak{P}) \longrightarrow 2^{\mathfrak{C}} \setminus \{\emptyset\}$$

is defined recursively by

$$\begin{aligned} C_{\mathfrak{P}}(-\mathbb{K}D\mathcal{E}) &= \{\text{Geom}\}, & C_{\mathfrak{P}}(\mathcal{E}_{\text{local/jump}}^s) &= \{\text{Geom}\}, \\ C_{\mathfrak{P}}(-A_{\text{Caus}}) &= \{\text{Caus}\}, & C_{\mathfrak{P}}(\mathcal{E}^a) &= \{\text{Caus}\}, \\ C_{\mathfrak{P}}(\mathcal{F}_{\text{ext}}) &= \{\text{Bdry}\}, & C_{\mathfrak{P}}(\perp_{\text{nat}}) &= \{\text{Geom}, \text{Caus}\}. \end{aligned} \tag{12}$$

Finite sums take unions. Quotients append **Abs**; charts and auxiliary variables append **Lift**; traces, killing, terminal values, and compactification boundaries append **Bdry**. Restriction, concatenation, and sourcewise weak closure retain support. A constraint multiplier is compiled through the operation that generates it: quotient constraints append **Abs**, while an exterior constraint supply appends **Bdry**.

Theorem I.2.6 (Operational native-source exhaustion). *For every operational weak-PDE signature, $C_{\mathfrak{P}}$ is defined on every finite dynamically generated record and has image in the five inherited source families. Its evaluation commutes with the weak equation: for every $\varphi \in \mathcal{D}$,*

$$\langle \dot{u}, \varphi \rangle = \sum_{c \in \mathfrak{C}} \langle J^c(u), \varphi \rangle. \tag{13}$$

The source word of a transformed record is the union of the native word and the letters appended by the transformation. There is no untyped native or finite-constructor remainder.

Proof. On ordinary branch (W1), the native relation has exactly the mobility, directed-transport, and exterior terms displayed in the definition. Nonnegativity of $\mathbb{K}$ identifies the first as **Geom**; the fixed transport and exterior relation give **Caus** and **Bdry**. On ordinary branch (W2), polarization uniquely separates the symmetric and antisymmetric forms. The Beurling–Deny representation uniquely separates the symmetric form into strongly local, jump, and killing terms; the first two are **Geom** and killing is **Bdry**. The antisymmetric form is **Caus**.

On the failure branch, retain the weak bulk current itself as a fixed **Caus** occurrence, so (13) still holds, and record the failed attempted split with composite $\{\text{Geom}, \text{Caus}\}$ -ancestry. The form row evaluates its target-active quotient without asserting the missing realization.

Structural induction over $\text{Syn}(\mathfrak{P})$ proves the remaining claim. Each constructor is covered by exactly one recursive clause in Definition I.2.5; finite combination takes unions and cannot erase an ancestor. Applying the evaluation map at each induction step gives (13). Weak closure is treated sourcewise, so it changes the analytic representative but not the cell label. $\square$

Corollary I.2.7 (Failure is compiled structure). *If an operation in $\text{Syn}(\mathfrak{P})$ lacks a classical value on a limit record, its totalized graph defect has the source support assigned to the approximating graph, together with Lift or Bdry when the failure occurs at a chart or terminal interface. Consequently lack of closed range, compactness, a classical trace, or a strong nonlinear limit cannot create a sixth source.*

Proof. Apply the compiler before graph closure, then use the sourcewise closure rule. The graph defect is a limit of records in that tagged cell. The chart and terminal cases use the corresponding recursive clauses. $\square$

Theorem I.2.8 (Sector-form source compiler). *Let X be locally compact and separable, let m be a Radon measure of full support, and let $(\mathcal{E}^s, D(\mathcal{E}))$ be a regular symmetric Dirichlet form on the real space $L^2(X, m)$. Write its Beurling–Deny decomposition, with normalization absorbed into J , as*

$$\mathcal{E}^s(u, v) = \mathcal{E}^c(u, v) + \int_{X \times X \setminus \text{diag}} (u(x) - u(y))(v(x) - v(y)) J(\mathrm{d}x, \mathrm{d}y) + \int_X uv \kappa(\mathrm{d}x). \quad (14)$$

Let $\mathcal{E}^a$ be antisymmetric on the same domain and satisfy the sector bound

$$|\mathcal{E}^a(u, v)| \leq C_{\text{sec}} \mathcal{E}_1^s(u, u)^{1/2} \mathcal{E}_1^s(v, v)^{1/2}. \quad (15)$$

Then the form $\mathcal{E} = \mathcal{E}^s + \mathcal{E}^a$ is closed and sectorial on $D(\mathcal{E})$, and its source assignment is exhaustive: $\mathcal{E}^c$ and the symmetric jump form are Geom, $\mathcal{E}^a$ is the fixed Caus form, and the killing measure κ is Bdry. The uniqueness of (14) leaves no additional symmetric Markovian primitive. Quotients and state-space enlargements append Abs and Lift as usual.

Proof. The Beurling–Deny theorem gives (14) and uniqueness of its strongly local, jump, and killing measures [3]. Antisymmetry gives $\mathcal{E}^a(u, u) = 0$, while (15) makes it continuous in the $\mathcal{E}_1^s$ -norm. Hence $\mathcal{E}$ has the same form norm completeness as $\mathcal{E}^s$ and satisfies the sector condition. The stated routing is exactly the source assignment of Definition I.2.1. A quasi-regular form has the same conclusion after a declared quasi-homeomorphic regular realization. $\square$

Proposition I.2.9 (Exact-drift symmetrization). *Let M be a Riemannian domain, let $\Psi \in C^2(M)$. On the locally-finite-weight branch let $\pi(\mathrm{d}x) = e^{-\Psi(x)} \mathrm{d}x$ is a locally finite measure. For*

$$L_\Psi g = \Delta g - \nabla \Psi \cdot \nabla g,$$

totalize the integration-by-parts trace. On its zero boundary-defect branch, the resulting form core satisfies

$$-\langle f, L_\Psi g \rangle_{L^2(\pi)} = \int_M \nabla f \cdot \nabla g \, \mathrm{d}\pi. \quad (16)$$

Thus an exact drift is absorbed into a symmetric Geom form by a weighted Abs-gauge. If an exact/nonexact decomposition of a general drift is proved on the declared domain, only its nonexact remainder retains independent Caus ancestry. No global boundedness of $e^{\pm\Psi}$ is inferred; comparison with another physical measure is evaluated by its own totalized physical-transfer row.

Proof. The weighted divergence identity

$$L_\Psi g = e^\Psi \operatorname{div}(e^{-\Psi} \nabla g)$$

and the stated trace condition give (16). Symmetry and nonnegativity of the right side prove the source statement; changing the reference measure is the declared gauge operation. $\square$

Proposition I.2.10 (Wiener lift creates no additional source). *Let H and U be separable Hilbert spaces and let a legal Itô evolution have drift $b : H \rightarrow H$ and Hilbert–Schmidt noise coefficient $\sigma : H \rightarrow \mathcal{L}_2(U, H)$. Totalize the generator relation on bounded C^2 cylinder functions. Its ordinary branch is*

$$LF(u) = \langle DF(u), b(u) \rangle + \frac{1}{2} \operatorname{Tr}[\sigma(u)\sigma(u)^* D^2 F(u)]. \quad (17)$$

Then the carré du champ is

$$\Gamma(F) := L(F^2) - 2FLF = \|\sigma^* DF\|_U^2 \geq 0. \quad (18)$$

*Thus the diffusion covariance and predictable quadratic variation are **Geom**; the drift retains its declared **Geom/Caus/Bdry** decomposition; and adjoining the Wiener path is **Lift**. Noise therefore creates no sixth continuum source.*

Proof. Apply the second-order chain rule to F^2 in (17); the drift terms cancel and the remaining trace is $\|\sigma^* DF\|_U^2$. Itô’s formula gives the same integrand for the predictable quadratic variation of the martingale part. The source labels follow from positivity of Γ , the fixed drift decomposition, and the state-space augmentation rule in Definition I.2.1. $\square$

I.3 Physical-budget-saturated sourcewise completion

The closed continuum algebra contains every mathematical limit generated by an ordinary finite-budget orbit. Its topology is generated canonically by the equation, target, physical balance, and constructor grammar. Strong convergence of a particular nonlinear term is subsequently read as a graph invariant in this completion.

Definition I.3.1 (Structural test system). For each nonempty $A \subseteq \mathfrak{C}$ and derived physical ceiling B_{phys} , let $\mathcal{R}_A^0(B_{\text{phys}})$ be the raw orbit records with source support equal to A . The structural test system $\mathfrak{F}_A = (\phi_n^A)_{n \geq 1}$ is generated recursively from the countable coordinates (1). At every finite grammar depth it contains:

- (T1) the weak PDE pairings and source-current pairings;
- (T2) the target shells and target fluxes;
- (T3) every truncation of the physical-expenditure and entropy-production coordinates generated by the fixed balance;
- (T4) every chart, quotient, boundary, and tail coordinate generated by the source grammar;
- (T5) the total graphs of every finite algebraic operation generated by that grammar, encoded by rational distance-to-graph truncations.

An unbounded real-valued test is represented by its rational bounded truncations. Enumerating grammar depth and rational parameters gives one countable family with values in $[-1, 1]$. No map or topology may be appended after a row value is inspected. Failure of an operation to extend as an ordinary continuous map is recorded by the boundary of its total graph.

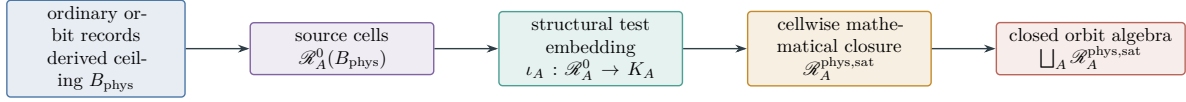


Figure 5: Sourcewise physical saturation. Closure is taken separately in each ancestry cell before the cells are assembled, and every cell retains the same physical ceiling B_{phys} .

Definition I.3.2 (Physical-budget-saturated closed orbit algebra). Let

$$\iota_A(r) = (\phi_n^A(r))_{n \geq 1} \in K_A := [-1, 1]^{\mathbb{N}}.$$

The space K_A is compact metrizable in the product metric. The sourcewise saturated cell and the full physical orbit algebra are

$$\mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}}) := \overline{\iota_A(\mathcal{R}_A^0(B_{\text{phys}}))}^{K_A}, \quad \mathcal{R}^{\text{phys,sat}}(B_{\text{phys}}) := \bigsqcup_{\emptyset \neq A \subseteq \mathfrak{C}} \mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}}). \quad (19)$$

A point of the closure is a mathematical limit record. Its ancestry support is the cell label A . A finer ancestry word may also be retained when it is retained by the generated test system. The closure contains all sequential limits of ordinary orbit records seen by the generated tests. Here Src denotes provenance support, not the smallest support of an analytically simplified formula.

Remark I.3.3 (Why the disjoint union is retained). The same analytic denotation can arise through different source histories. The closed algebra retains those histories as distinct records. Forgetting ancestry is an evaluation map from the disjoint union to the ambient analytic space; it is not part of the source decomposition.

Theorem I.3.4 (Source-preserving continuum completion). *Every sequence (r_i) of ordinary orbit records with $\text{Src}(r_i) = A$ that converges in the structural test topology has its limit record in $\mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}})$. Consequently*

$$\text{Src}(\lim_i r_i) = A = \bigcup_i \text{Src}(r_i). \quad (20)$$

If the sequence first passes through quotient or chart maps, Abs or Lift, respectively, is appended before taking the limit. If the limit is read as a boundary or tail trace, Bdry is appended. Thus defect measures, oscillation variables, concentration profiles, closure directions, blow-up germs, and tails create no source outside $\mathfrak{C}$.

Proof. By definition, $\mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}})$ is the sequential closure of the test image of the raw cell with provenance support A ; compact metrizability makes this equal to its topological closure. Hence every limit of such a sequence belongs to that cell. The algebraic rules of Definition I.2.1 take unions of source supports and append the labels of quotient, chart, and boundary operations before the closure is formed. The new coordinates of a limit record are limits of tests on the same tagged sequence; they do not supply a new primitive letter. This proves (20) and each listed case. $\square$

Corollary I.3.5 (Mixed sequences). *Every sequence of raw records has source support in the finite set $2^{\mathfrak{C}} \setminus \{\emptyset\}$. It has a subsequence with constant source support. Each convergent such subsequence has its limit therefore contained in an existing saturated cell.*

Proof. There are only $2^5 - 1$ nonempty source supports. One support occurs infinitely often. Apply Theorem I.3.4 to that subsequence. $\square$

Definition I.3.6 (Source realization map). Let $\mathcal{P}_A$ and $\mathcal{J}_A$ be Hilbert spaces of potentials and currents for a fixed source support $A \subseteq \mathfrak{C}$. A *source realization map* is a densely defined closed operator

$$G_A : D(G_A) \subseteq \mathcal{P}_A \longrightarrow \mathcal{J}_A$$

whose range consists of the currents carrying an actual potential witness in that source cell. Its saturated current space is $\overline{\text{Ran } G_A}^{\mathcal{J}_A}$. The closure has the same source support A , independently of whether every closure direction has a potential witness.

Theorem I.3.7 (Closed-range realization criterion). *Let $G : D(G) \subseteq \mathcal{P} \rightarrow \mathcal{J}$ be a source realization map. The following conditions are equivalent.*

(R1) $\text{Ran } G$ is closed in $\mathcal{J}$.

(R2) There is $\gamma > 0$ such that

$$\|G\phi\|_{\mathcal{J}}^2 \geq \gamma \text{dist}_{\mathcal{P}}(\phi, \ker G)^2, \quad \phi \in D(G). \quad (21)$$

(R3) The restriction of G^*G to $(\ker G)^\perp$ has spectrum in $[\gamma, \infty)$ for some $\gamma > 0$.

(R4) The Moore–Penrose inverse G^+ is bounded on $\text{Ran } G$.

When these conditions hold, every saturated current in $\overline{\text{Ran } G}$ has a potential witness, unique modulo $\ker G$. When they fail, elements of $\overline{\text{Ran } G} \setminus \text{Ran } G$ remain limit records in the same source cell by Theorem I.3.4; failure of potential realization does not define another source family.

Proof. The closed-range theorem for densely defined closed operators gives the equivalence of (R1), (R2), and boundedness of G^+ . On $(\ker G)^\perp$, $\|G\phi\|^2 = \langle G^*G\phi, \phi \rangle$, so the spectral theorem makes (R2) equivalent to (R3). Closed range gives $\overline{\text{Ran } G} = \text{Ran } G$, and the kernel is exactly the ambiguity in a potential witness. The final assertion is Theorem I.3.4 applied to the source cell generated by G . $\square$

Theorem I.3.8 (Target-relative gradient exhaustiveness). *Let $G : D(G) \subseteq \mathcal{P} \rightarrow \mathcal{J}$ be a densely defined closed realization of one inherited source cell, and let $P_{\overline{G}}$ be the orthogonal projection onto $\overline{\text{Ran } G}$. Then every $j \in \mathcal{J}$ has the unique split*

$$j = P_{\overline{G}}j + (I - P_{\overline{G}})j, \quad P_{\overline{G}}j \in \overline{\text{Ran } G}, \quad (I - P_{\overline{G}})j \in \ker G^*. \quad (22)$$

The first term belongs to the saturated cell generated by G , including when it lies in $\overline{\text{Ran } G} \setminus \text{Ran } G$. Relative to the target family Γ_Σ , the second term has the further canonical split into its class in $\ker G^*/(\ker G^* \cap \mathcal{N}_\Sigma)$ and its target-null part. Hence failure of closed range creates neither a new source nor a target-visible residual: it creates a closure direction in the inherited cell, while every orthogonal target-visible class remains a current of the native compiled equation and retains its native ancestry.

Proof. The range closure is a closed Hilbert subspace, so the projection theorem gives (22); the identity $(\text{Ran } G)^\perp = \ker G^*$ identifies its complement. The definition of the sourcewise completion and Theorem I.3.4 assign $P_{\overline{G}}j$ to the same tagged cell as the approximating gradients. Quotienting the complementary space by its intersection with $\mathcal{N}_\Sigma$ is exact. Its kernel is target-null by definition, and a nonzero quotient class has a nonzero target pairing. Such a class cannot be renamed residual; its ancestry is the ancestry of the native current from which j was compiled by Theorem I.4.4. These alternatives exhaust $\mathcal{J}$. $\square$

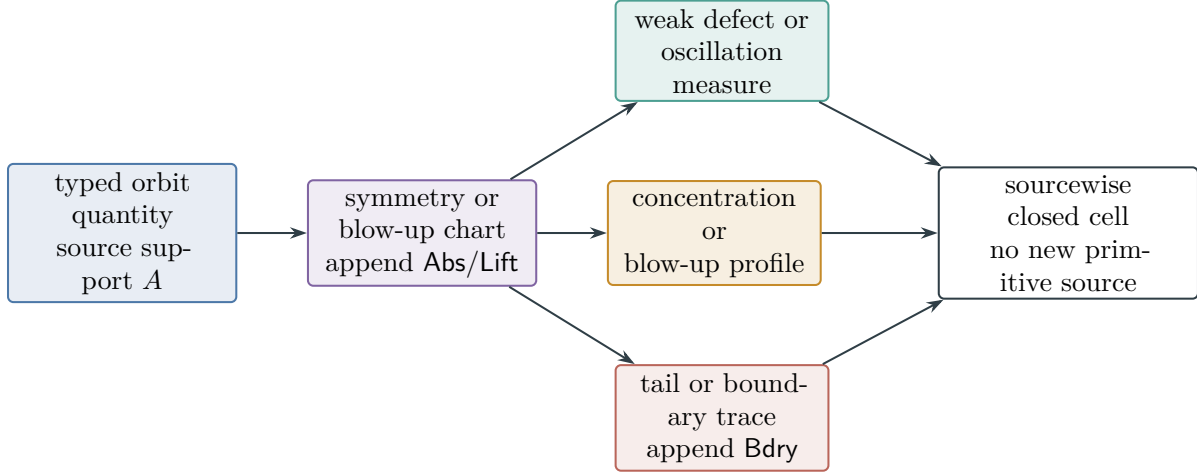


Figure 6: Continuum limit phenomena inherit ancestry. Normalization changes the source word in the declared Abs/Lift manner; a terminal trace appends Bdry. Weakness or concentration changes the analytic realization, not the source alphabet.

Corollary I.3.9 (Regularized realization). *For $j \in \mathcal{J}$ and $\lambda > 0$, the functional*

$$\phi \mapsto \|G\phi - j\|_{\mathcal{J}}^2 + \lambda \|\phi\|_{\mathcal{P}}^2$$

has a unique minimizer $\phi_\lambda \in D(G)$. The resulting current $G\phi_\lambda$ is a stable regularized realization, with the parameter λ recording its bias. Existence of this minimizer for every $\lambda > 0$ does not imply any of the equivalent conditions in Theorem I.3.7.

Proof. On $D(G)$, the form $\langle G\phi, G\psi \rangle + \lambda \langle \phi, \psi \rangle$ is closed and coercive in its form norm. The functional $\psi \mapsto \langle j, G\psi \rangle$ is continuous in that norm, so the Riesz representation theorem gives a unique minimizer. The final statement follows because adding λI supplies coercivity even when the unregularized range is not closed. $\square$

Remark I.3.10 (Structural and evaluator topologies). The evaluation compactification closes the structural algebra. A nonlinear PDE estimate may require a stronger topology to pass a particular term to the limit or to keep a bad set closed. Such a topology must be generated by PDE-derived seminorms, and its compactness statement evaluates the boundary row in Section II.8; failure remains in the same source cell.

I.4 Target quotient and the null residual

Definition I.4.1 (Generated prospective target family). Let $\mathcal{J}^{\text{phys}, \text{sat}}(B_{\text{phys}})$ be the current-occurrence part of the closed orbit algebra. The finite-current locus is replaced, when necessary, by its Hausdorff locally convex envelope under the continuously extended addition and scalar multiplication; the kernel of the envelope map is retained as a tagged closure defect. Records at an infinite compactification endpoint are treated as boundary or tail records and are tested by their bounded readouts, not inserted into this vector space. Let $\rho_\Sigma(x) = d_{\text{ev}}(x, \Sigma)$, where Σ is the complete continuation boundary of Definition III.1.1. The *target-shell algebra* is the least countable algebra containing every rational ramp of ρ_Σ , its pullback by every ordinary generated covariance map, and the rational cylindrical approximants supplied by the coordinates (1).

The prospective target family Γ_Σ is the closed normalized cone generated by every ordinary current differential of this shell algebra, together with the killed-flux and predictable-compensator differentials when their types occur in the fixed equation. In particular it contains

$$J \longmapsto \langle D\Phi_r(z), J \rangle$$

whenever that ordinary differential exists. A missing differential is the explicit value of the total shell-current graph and is tested by its endpoint shell increment. Thus neither the probes nor their normalization are application data: they are generated by the continuation target and the equation's verified covariance grammar.

Definition I.4.2 (Target-null residual). The target-null space and target-active quotient are

$$\mathcal{N}_\Sigma := \bigcap_{\Lambda \in \Gamma_\Sigma} \ker \Lambda, \quad \mathcal{J}_\Sigma := \mathcal{J}^{\text{phys,sat}}(B_{\text{phys}}) / \mathcal{N}_\Sigma. \quad (23)$$

The post-saturation residual is any closed current coordinate containing zero and satisfying

$$\mathcal{J}_{J_{\text{res}}}^{\text{phys,sat}} \subseteq \mathcal{N}_\Sigma. \quad (24)$$

It is an audit coordinate for target-invisible transport, not a sixth source family. A current is target-visible precisely when its quotient class in $\mathcal{J}_\Sigma$ is nonzero.

Definition I.4.3 (Compiled continuum current). A continuum current is *compiled* when it is generated from the native terms of the fixed equation by the finite constructor rules (S1)–(S6) of Definition I.2.1. Its evaluation has the normal form

$$J = J^{\text{Geom}} + J^{\text{Caus}} + J^{\text{Abs}} + J^{\text{Lift}} + J^{\text{Bdry}} + J^{J_{\text{res}}}, \quad J^{J_{\text{res}}} \in \mathcal{N}_\Sigma, \quad (25)$$

as an equality of weak current pairings. Empty components are allowed. The residual is an optional regrouping in the common target kernel after the five source components have been evaluated; the choice $J^{J_{\text{res}}} = 0$ is always admissible. It is not a constructor and carries no source letter.

Theorem I.4.4 (Continuum normal-form compilation). *For every operational weak-PDE signature, totalize the native realization relation and adjoin the quotient, chart, auxiliary-state, trace, forcing, and boundary operations generated by Definition I.2.1. Every finite orbit-current expression is compiled and has (25). A failed native realization additionally generates its target-active form-defect class with $\{\text{Geom}, \text{Caus}\}$ -ancestry. After sourcewise completion, every finite target pairing has the same normal form. Thus five-channel exhaustion does not presume a gradient or sector realization.*

Proof. On the fixed-Caus ordinary branch, positivity of $\mathbb{K}$, the directed transport, and the exterior term give **Geom**, **Caus**, and **Bdry**, respectively. On the form ordinary branch, the sector-form compiler makes the same assignment to the symmetric mobility, antisymmetric directed, and killing terms. Structural induction on the syntax of an orbit-current expression now proves the claim. A quotient appends **Abs**, a chart or auxiliary variable appends **Lift**, a trace appends **Bdry**, and finite combination takes the union of the already assigned supports. These exhaust the transformed ordinary branches.

On the native failure branch, retain the raw fixed bulk current in **Caus** and its totalized attempted split in the composite **Geom**–**Caus** form-defect cell, as in Theorem I.2.6. This additional case remains inside the five-source alphabet.

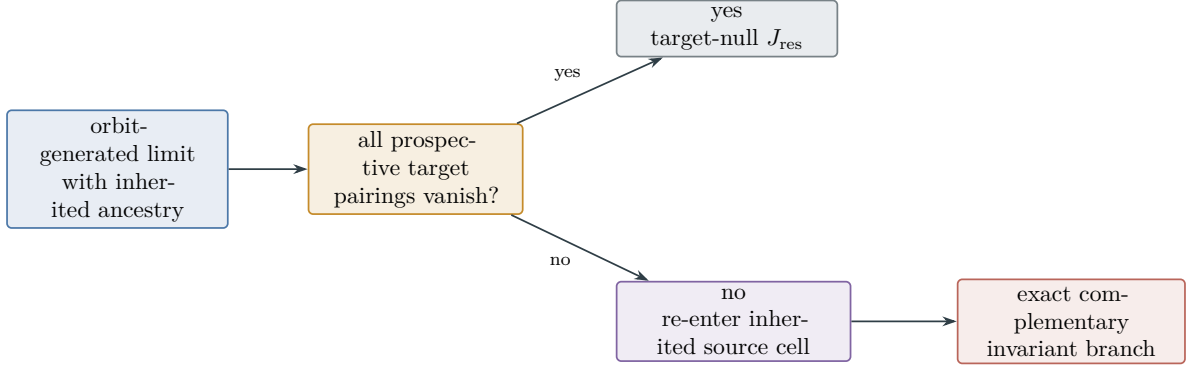


Figure 7: Target-visible limits retain their inherited source cell. The only residual output is target-null. A nonzero pairing fixes structural ancestry before any attempt is made to estimate the corresponding source profile.

For a sourcewise convergent net, each prospective target functional is a coordinate of the structural test system. Its value therefore passes to the limit. By Theorem I.3.4, the limit remains in the same tagged source cell. The image in the quotient by $\mathcal{N}_\Sigma$ is therefore generated by the five cell images. Taking $J^{J_{\text{res}}} = 0$, or regrouping any already identified target-null summand as $J^{J_{\text{res}}}$, gives (25) without requiring a complemented kernel and without adding a primitive source. $\square$

Theorem I.4.5 (No post-saturation target-visible residual). *Let $\mathfrak{P}_{\text{phys}}$ be generated by the compiled continuum algebra of Theorem I.4.4. Every target-visible current record in $\mathcal{J}^{\text{phys},\text{sat}}(B_{\text{phys}})$ has nonempty ancestry in $\mathfrak{C}$, and its target-active quotient is generated by the images of the five saturated source cells. The residual source profile vanishes:*

$$G_{B_{\text{phys}},\Sigma}^{J_{\text{res}},\text{src},\text{sat}} := \sup_{J \in \mathcal{J}_{J_{\text{res}}}^{\text{phys},\text{sat}}} \sup_{\Lambda \in \Gamma_\Sigma} (\Lambda J)_+ = 0. \quad (26)$$

If a provisional limit object pairs nontrivially with a target functional, then Theorem I.3.4 places it in the existing source cell inherited from its generating records. Thus every dynamically generated target-visible limit belongs to one of the five source families.

Proof. For raw occurrences, apply the quotient map $q_\Sigma : \mathcal{J}^{\text{phys},\text{sat}} \rightarrow \mathcal{J}_\Sigma$ to (25). The residual term vanishes because it lies in $\mathcal{N}_\Sigma$. Continuity of every $\Lambda \in \Gamma_\Sigma$ passes the same identity to each sourcewise limit. The source-preserving theorem assigns every nonresidual limit term to the closure of its existing cell. Thus a nonzero quotient class is generated by the five cell images. Finally, every functional in Γ_Σ vanishes on $\mathcal{J}_{J_{\text{res}}}^{\text{phys},\text{sat}}$, so both nested suprema in (26) are zero. $\square$

Corollary I.4.6 (Source type and invariant value are independent). *Each saturated source cell has a value in every applicable fast-track invariant: null or nonnull quotient, finite or infinite radius, positive or zero price, zero or nonzero equality remainder, zero or positive capacity, finite or infinite resistance, and divergent or finite cascade price. These values change neither the source word nor the target quotient. Failure of a compactness or resolvent invariant therefore returns the complementary invariant branch; it never creates an untyped or residual source.*

I.5 Prospective contact accountability

Target contact may produce a retrospective object, such as a singular germ or a completed hitting distribution. Source accountability is instead read from the evolution strictly before contact. The next three theorems give deterministic, semigroup, and stochastic versions of the same factorization.

Definition I.5.1 (Canonical target shells and total shell current). In the compact metric space X_{ev} , put $\rho_{\Sigma}(x) = d_{\text{ev}}(x, \Sigma)$, $U_r(\Sigma) = \{\rho_{\Sigma} < r\}$, and, for rational $r > 0$,

$$\Phi_r(x) = \begin{cases} 1, & \rho_{\Sigma}(x) \leq r, \\ 2 - \rho_{\Sigma}(x)/r, & r < \rho_{\Sigma}(x) < 2r, \\ 0, & \rho_{\Sigma}(x) \geq 2r. \end{cases} \quad (27)$$

Then Φ_r is continuous and r^{-1} -Lipschitz, $\bigcap_{r \in \mathbb{Q}_+} U_{2r}(\Sigma) = \Sigma$, and

$$\Phi_r = 0 \quad \text{on } X_{\text{ev}} \setminus U_{2r}(\Sigma), \quad \Phi_r = 1 \quad \text{on } U_r(\Sigma). \quad (28)$$

For a regular interval $I = [a, b]$, totalize the relation

$$\Phi_r(u_b) - \Phi_r(u_a) = \sum_{c \in \mathfrak{C}} Q_{c,r}(I) + \eta_{r,u}(I), \quad (29)$$

where the ordinary value is

$$Q_{c,r}(I) = \int_I \langle D\Phi_r(u_t), J_t^c \rangle dt, \quad \eta_{r,u}(I) = 0, \quad (30)$$

whenever the cylindrical approximation, integrability, and chain relations converge. On every other branch, $\eta_{r,u}(I)$ is by definition the endpoint increment minus the ordinary source values that exist; a missing source value is included in the same total-graph record. Sourcewise graph closure gives $\eta_{r,u}(I)$ the union of the ancestries of its endpoint and current witnesses. Hence a target-visible shell defect is an existing five-channel source occurrence, not an additional source and not a premise.

The contact time is $\tau_{\Sigma}(u) = \inf\{t : u(t) \in \Sigma\}$, with the germ interpretation when $T_*(u)$ is a noncontinuation time. If $u(0) \notin \Sigma$ and u contacts Σ , then for every sufficiently small r there are strictly pre-contact times

$$0 \leq t_r^- < t_r^+ < \tau_{\Sigma}(u), \quad u(t_r^-) \notin U_{2r}(\Sigma), \quad u(t_r^+) \in U_r(\Sigma). \quad (31)$$

The metric shells separate every point outside the closed target, while contact supplies the second time. A target at infinity is already a boundary point of X_{ev} ; an additional ordinary normalization merely appends its Abs/Lift ancestry.

Theorem I.5.2 (Total normalized shell-crossing accountability). *Let u be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$, on the contact branch. For every sufficiently small r , choose a crossing interval $I_r = [t_r^-, t_r^+]$ as in (31). Write the actual physical current as*

$$J = \sum_{c \in \mathfrak{C}} J^c + J^{\text{Jres}},$$

where $J^{\text{Jres}} \in \mathcal{N}_{\Sigma}$. Evaluate the total shell-current relation on I_r . Its ordinary signed source progress is

$$Q_{c,r}(u) := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle dt. \quad (32)$$

On every branch the shell crossing has the exact accountability identity

$$1 = \Phi_r(u(t_r^+)) - \Phi_r(u(t_r^-)) = \sum_{c \in \mathfrak{C}} Q_{c,r}(u) + \eta_{r,u}(I_r). \quad (33)$$

On the ordinary branch $\eta_{r,u}(I_r) = 0$ and

$$\int_{I_r} \langle D\Phi_r(u_t), J_t^{J_{\text{res}}} \rangle dt = 0. \quad (34)$$

Consequently the ordinary branch satisfies

$$\sum_{c \in \mathfrak{C}} (Q_{c,r}(u))_+ \geq 1, \quad \max_{c \in \mathfrak{C}} (Q_{c,r}(u))_+ \geq \frac{1}{5}. \quad (35)$$

On a nonordinary branch, either some ordinary source contribution is positive or $\eta_{r,u}(I_r) > 0$. In the second case its nonempty inherited source support is a target-aligned shell-defect occurrence. Thus, without requiring a classical chain rule,

$$\boxed{\begin{array}{l} u \text{ contacts } \Sigma \implies \\ \forall r \text{ sufficiently small } \exists \text{ a five-channel source occurrence } R_r : \text{Prog}_\Sigma(R_r) > 0. \end{array}} \quad (36)$$

Thus contact itself exposes a target-aligned pre-contact occurrence in an existing source cell at every sufficiently small shell.

Proof. The endpoint normalization gives the unit increment. Equation (29) gives the second equality on every total-graph branch. On the ordinary branch, insert the current normal form; the generated shell differential belongs to Γ_Σ , so residual target-nullity proves (34). The elementary positive-part argument gives (35). On a nonordinary branch, a sum of the existing ordinary values and the defect equals one; at least one summand is positive. Sourcewise closure assigns every positive defect summand a nonempty inherited support. $\square$

Corollary I.5.3 (No spontaneous deterministic contact). *Let u be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$. Suppose that every ordinary source current and every nonordinary shell-current defect is target-null on every pre-contact shell. On the ordinary branch this reads*

$$\langle D\Phi_r(u_t), J_t^c \rangle = 0 \quad \text{for a.e. } t < \tau_\Sigma(u), \quad c \in \mathfrak{C}. \quad (37)$$

Then u cannot contact Σ . Target avoidance also follows when every ordinary shell integral and every defect increment is nonpositive. For a singularity at infinity, these statements apply to the canonical boundary shells in the generated evaluation compactification.

Proof. If contact occurred, Theorem I.5.2 would produce, on every sufficiently small shell, either a positive ordinary source contribution or a positive target-visible defect with inherited source support. Both alternatives contradict the stated target-nullity. Under the stronger ordinary zero-pairing condition, the same conclusion follows directly from (29) because its defect is target-null. $\square$

Corollary I.5.4 (Normalized contact at infinity). *On the branch where singular contact is defined after generated normalizations $v_r = \mathcal{S}_r u$ into a state or germ compactification carrying normalized*

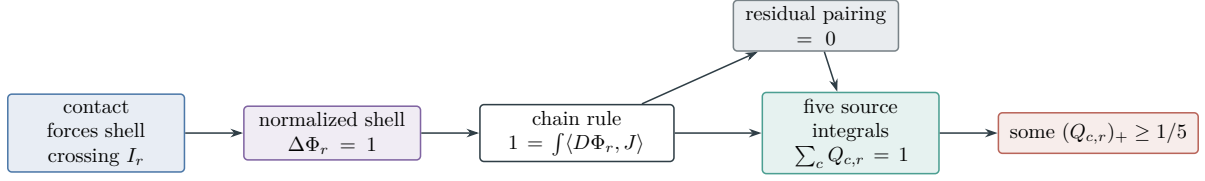


Figure 8: Normalized shell-crossing accountability. Contact forces a unit increase of the shell observable along the actual pre-contact current. The target-null residual contributes zero, so one of the five source integrals has positive part at least $1/5$.

shells $\tilde{\Phi}_r$. If the normalized current $\tilde{J}_r$ has the compiled normal form, including every chart derivative with its Abs/Lift ancestry, then every normalized crossing interval satisfies

$$1 = \sum_{c \in \mathfrak{C}} \int_{I_r} \langle D\tilde{\Phi}_r(v_r(t)), \tilde{J}_{r,t}^c \rangle dt, \quad \max_{c \in \mathfrak{C}} \left(\int_{I_r} \langle D\tilde{\Phi}_r(v_r(t)), \tilde{J}_{r,t}^c \rangle dt \right)_+ \geq \frac{1}{5}.$$

Thus blow-up at infinity cannot occur without target-aligned structure in an existing source cell.

Proof. Apply Theorem I.5.2 in the normalized state space. Source preservation under the legal chart is Theorem I.3.4. $\square$

Theorem I.5.5 (Resolvent source-response comparison). *Let L and L_0 be closed killed generators on the same Banach space, and let $\lambda \in \rho(L) \cap \rho(L_0)$. Write*

$$R_L(\lambda) = (\lambda - L)^{-1}, \quad R_0(\lambda) = (\lambda - L_0)^{-1}.$$

Totalize domain membership of $R_0(\lambda)k_\Sigma$ and the generator difference. On their ordinary branch,

$$L - L_0 = \sum_{c \in \mathfrak{C}} K^c + K^{J_{\text{res}}} \quad (38)$$

on that vector. Then, for every continuous initial functional ℓ_0 ,

$$\begin{aligned} \langle \ell_0, (R_L(\lambda) - R_0(\lambda))k_\Sigma \rangle &= \sum_{c \in \mathfrak{C}} \langle R_L(\lambda)^* \ell_0, K^c R_0(\lambda)k_\Sigma \rangle \\ &\quad + \langle R_L(\lambda)^* \ell_0, K^{J_{\text{res}}} R_0(\lambda)k_\Sigma \rangle. \end{aligned} \quad (39)$$

If the last pairing is target-null, it vanishes. If the real part p of the left side is positive, some existing source cell has real pairing at least $p/5$. Any domain or generator-split failure is returned by the form/realization row and appears only as the ordinary value of that specialized row.

Proof. The second resolvent identity gives

$$R_L(\lambda) - R_0(\lambda) = R_L(\lambda)(L - L_0)R_0(\lambda).$$

Pair with ℓ_0 and k_Σ , then insert (38). Target-nullity removes the residual term. A positive sum of five real numbers has a summand at least one fifth of that sum. $\square$

Remark I.5.6 (Arrival semantics). Equation (39) is an operator identity. It becomes a difference of discounted contact responses when the killed semigroups admit the target-flux realization of Theorem II.5.4. No probabilistic meaning is inferred from the algebraic resolvent alone.

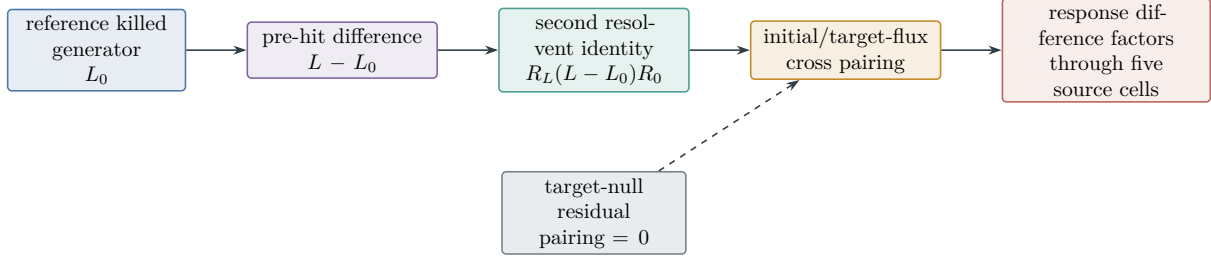


Figure 9: Resolvent source-response comparison. A discounted target-response difference is sandwiched around the pre-hit generator difference and therefore inherits its five-source decomposition. The no-spontaneous-contact theorem itself comes from normalized shell crossing.

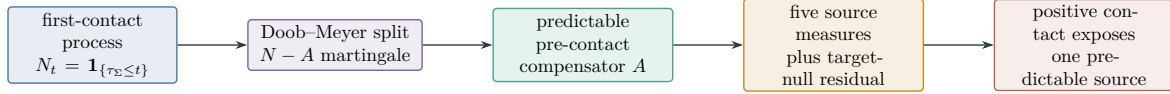


Figure 10: Stochastic no-spontaneous-contact accountability. First contact is charged by its predictable pre-contact compensator. A positive contact probability forces a positive contribution from one of the five source measures before the stopping time.

Theorem I.5.7 (Stochastic no-spontaneous-contact accountability). *Let τ_Σ be a stopping time and $N_t = \mathbf{1}_{\{\tau_\Sigma \leq t\}}$. Totalize the Doob–Meyer and source-split relations. On the branch where $N - A$ is a martingale, the predictable compensator on $[0, T]$ has the finite-variation decomposition*

$$A = \sum_{c \in \mathfrak{C}} A^c + A^{J_{\text{res}}}, \quad (40)$$

where all measures are supported on $\{t < \tau_\Sigma\}$, the signed source terms are integrable, every term starts from zero, $\mathbb{P}\{\tau_\Sigma = 0\} = 0$, and $\mathbb{E}A_T^{J_{\text{res}}} = 0$. Then

$$\mathbb{P}\{\tau_\Sigma \leq T\} = \sum_{c \in \mathfrak{C}} \mathbb{E}A_T^c. \quad (41)$$

If $p_T := \mathbb{P}\{\tau_\Sigma \leq T\} > 0$, one pre-contact source compensator satisfies $\mathbb{E}A_T^c \geq p_T/5$. In particular, all five source expectations cannot vanish when contact has positive probability. Failure of the martingale, finite-variation, support, or integrability relation is its stochastic total-graph defect and remains in the fluctuation row.

Proof. Taking expectations of the integrable martingale $N - A$, using the stated zero initial values, gives $\mathbb{E}N_T = \mathbb{E}A_T$. Insert (40) and use the zero expectation of the residual term. The final assertion follows from the five-term pigeonhole argument. $\square$

Corollary I.5.8 (Continuum prospective source exhaustion). *Normalized deterministic contact and positive-probability stochastic contact both force a positive pre-contact occurrence with ancestry in $\mathfrak{C}$. A positive killed-resolvent response difference has the same source-extraction property. The post-saturation residual contribution is zero in all three interfaces.*

Definition I.5.9 (Saturated prospective source profile). Fix one of the three prospective interfaces, restricted to records satisfying its stated target-null residual condition, and fix its normalization.

Write $\text{Prog}_\Sigma^c(R)$ for the signed c -source contribution: $Q_{c,r}$ for a deterministic shell, $\mathbb{E}A_T^c$ for a stochastic contact record, and the real c -pairing in (39) for a resolvent comparison. Set

$$\text{Prog}_\Sigma(R) := \sum_{c \in \mathfrak{C}} \text{Prog}_\Sigma^c(R). \quad (42)$$

Define

$$\mathsf{T}_{\Sigma,c}^{\text{phys,sat}}(B_{\text{phys}}) := \sup_{R \in \mathcal{R}^{\text{phys,sat}}(B_{\text{phys}})} (\text{Prog}_\Sigma^c(R))_+, \quad (43)$$

$$\mathsf{T}_\Sigma^{\text{phys,sat}}(B_{\text{phys}}) := \sum_{c \in \mathfrak{C}} \mathsf{T}_{\Sigma,c}^{\text{phys,sat}}(B_{\text{phys}}). \quad (44)$$

The supremum is taken over the ordinary graph values of the selected interface and may equal $+\infty$. Inputs outside that graph enter the corresponding totalized defect cell, so every input has a structural value.

Definition I.5.10 (Source-support and realization-mode profiles). For each nonempty $A \subseteq \mathfrak{C}$, set

$$\text{Prog}_\Sigma^A(R) := \sum_{c \in A} \text{Prog}_\Sigma^c(R) \mathbf{1}_{\{\text{Src}(R)=A\}}.$$

Every compiled record also carries the subset of realization-mode flags

$$\mathfrak{M} = \{\text{exact}, \text{closure}, \text{harmonic}, \text{concentration}, \text{boundary}\}$$

determined by its construction:

- (M1) **exact**: it has a witness in the range of its native realization map;
- (M2) **closure**: it lies in the sourcewise range closure, possibly outside the range;
- (M3) **harmonic**: its class in $\ker G_A^*/(\ker G_A^* \cap \mathcal{N}_\Sigma)$ is nonzero;
- (M4) **concentration**: it is read through a Young measure, defect measure, profile, or blow-up-state lift;
- (M5) **boundary**: it is a trace, killing, tail, terminal germ, or graph-boundary record.

Flags may coexist; they refine ancestry and do not form new sources. For bookkeeping fix the order

$$\text{exact} < \text{closure} < \text{harmonic} < \text{concentration} < \text{boundary}$$

and let $m_*(R)$ be the maximal flag carried by R . Thus the complete flag set is retained, while every record belongs to exactly one *primary-mode* cell. Generated operations only append flags, so m_* is monotone along a successor path and hence eventually constant. For $m \in \mathfrak{M}$, define

$$\mathsf{T}_{\Sigma,A,m}^{\text{phys,sat}}(B_{\text{phys}}) := \sup_{\substack{R \in \mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}}) \\ m_*(R)=m}} (\text{Prog}_\Sigma^A(R))_+, \quad (45)$$

with supremum 0 for an empty class.

Theorem I.5.11 (Realization-mode exhaustion). *Every target-active continuum record belongs to a nonempty source cell $A \subseteq \mathfrak{C}$, has at least one flag in $\mathfrak{M}$, and has exactly one primary mode. Moreover,*

$$\mathsf{T}_\Sigma^{\text{phys,sat}}(B_{\text{phys}}) \leq \sum_{\emptyset \neq A \subseteq \mathfrak{C}} \sum_{m \in \mathfrak{M}} \mathsf{T}_{\Sigma,A,m}^{\text{phys,sat}}(B_{\text{phys}}). \quad (46)$$

Thus exact currents, nonclosed-range limits, harmonic classes, concentrations, and boundary defects exhaust the analytic realization modes of target progress without enlarging the five-source alphabet.

Proof. Native records are exact. A sourcewise limit is in the closure mode. The orthogonal decomposition $\mathcal{J}_A = \overline{\text{Ran } G_A} \oplus \ker G_A^*$, followed by the target quotient, assigns every target-active orthogonal record to the harmonic mode. The compiler marks every profile lift and every boundary operation with the last two flags. These cases cover the finite syntax and its sourcewise completion. Every positive source contribution is therefore bounded by at least one term on the right of (46); summing gives the result. Multiple flags only enlarge the right side. $\square$

Theorem I.5.12 (Source-only prospective envelope). *Every record in the domain of a prospective interface satisfies*

$$(\text{Prog}_\Sigma(R))_+ \leq \mathsf{T}_\Sigma^{\text{phys,sat}}(B_{\text{phys}}). \quad (47)$$

The envelope consists solely of the five source profiles. Any finite upper bounds on those profiles give a complete prospective progress bound. A normalized deterministic contact record satisfies $\text{Prog}_\Sigma(R) = 1$.

Proof. Each of Theorems I.5.2, I.5.5 and I.5.7 has, after removal of its target-null residual pairing, the form

$$\text{Prog}_\Sigma(R) = \sum_{c \in \mathfrak{C}} \text{Prog}_\Sigma^c(R).$$

Take the positive part, bound every signed summand by its positive part, and then apply (43). $\square$

Definition I.5.13 (Singularity-aligned structural support). For a pre-contact record R in any prospective interface, define

$$\text{Align}_\Sigma(R) := \{c \in \mathfrak{C} : \text{Prog}_\Sigma^c(R) > 0\} \cup \begin{cases} \text{Src}(R), & R \text{ is a total-graph defect with } \text{Prog}_\Sigma(R) > 0, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (48)$$

Equivalently, in the deterministic interface an occurrence is aligned when the singularity-descending covector $D\Phi_r(u_t)$ has positive integrated pairing with its source current on the crossing interval. The word “aligned” is target-relative: it does not rename a source as **Caus**, and it is evaluated only along the fixed equation.

Theorem I.5.14 (No-spontaneous-progress principle). *For every prospective record, including total-graph defect records,*

$$\text{Prog}_\Sigma(R) > 0 \implies \text{Align}_\Sigma(R) \neq \emptyset. \quad (49)$$

For an ordinary five-term additive record one also has

$$\max_{c \in \mathfrak{C}} (\text{Prog}_\Sigma^c(R))_+ \geq \frac{\text{Prog}_\Sigma(R)}{5}. \quad (50)$$

If every ordinary source occurrence and every total-graph defect represented by R lies in $\mathcal{N}_\Sigma$, then

$$\text{Prog}_\Sigma(R) = 0, \quad (51)$$

regardless of the size, oscillation, or compactness of those target-null currents. More generally, if every source contribution is nonpositive, then positive target progress is impossible. For a normalized deterministic shell crossing, contact yields either an ordinary source contribution at least $1/5$, or a positive shell defect with nonempty inherited support.

Proof. On an ordinary graph value, (42) is the compiled five-term identity. A positive sum has a positive summand, and its largest positive part is at least one fifth of the sum. On a defect graph value, positive progress inserts $\text{Src}(R)$ into $\text{Align}_\Sigma(R)$ by definition; source-preserving graph closure makes this support nonempty. If every ordinary and defect occurrence lies in the common target kernel, every term vanishes. The shell assertion is Theorem I.5.2. $\square$

Corollary I.5.15 (No-spontaneous-contact regularity). *Let the target have a normalized shell presentation, and let every legal orbit be generated by the compiled algebra on its regular pre-contact interval. If $\text{Align}_\Sigma(R) = \emptyset$ for every legal shell record, then the target is unreachable. In particular, if all five source currents have zero class in $\mathcal{J}_\Sigma$ along every pre-contact shell, their ancestry is retained while their target contribution lies in $\mathcal{N}_\Sigma$, and contact is impossible:*

$$\boxed{q_\Sigma(J^c) = 0 \text{ on every pre-contact target shell for all } c \in \mathfrak{C} \implies \tau_\Sigma = \infty.} \quad (52)$$

This conclusion requires neither compactness of the full orbit nor a positive physical-price estimate.

Proof. If an orbit contacted Σ , Theorem I.5.2 would provide a normalized shell record with unit progress. By Theorem I.5.14, its aligned support would be nonempty, contrary to the displayed null-source condition. Under that quotient value, every shell source integral vanishes, so the same contradiction follows directly from (33). $\square$

Remark I.5.16 (Why the target quotient is stronger than a full norm bound). A classical energy estimate commonly controls every component of a current, including directions irrelevant to the singular target. The quotient $\mathcal{J}_\Sigma$ discards a current only after all declared prospective target pairings vanish. The discarded current may be large or noncompact; no estimate of it is needed for (51). Only the target-aligned quotient directions proceed to resistance, compactness, rigidity, or physical-price invariants. The quotient strictly reduces the regularity problem because target-null directions are absent from all downstream estimates.

Part II

Physical Structural-Invariant Calculus

II.1 Structural invariant reduction of analytic arguments

Remark II.1.1 (Ordinary-branch convention for analytic modules). Every conditional identity in this part describes the ordinary value of a total graph generated from the public signature. Its operator, form, chart, or multiplier must already occur in the typed syntax or in the fixed target-independent textbook library. The condition is never appended as specialization data: when it does not hold, its complementary graph value is sent to the activity lift and successor relation. Hamiltonian, commutator, compactness, tail, rigidity, and entropy modules can shorten a pruning transcript, but the core contact-to-successor theorem does not require their ordinary values.

The closed source algebra separates the origin of a continuum quantity from the estimate used to control it. Once a source family has been placed in a closed target-null quotient or removed from the admissible orbit family by a proved physical exclusion, later invariants are evaluated on the remaining reduced space. The analytic calculation then has only three possible jobs: close a remaining class, compare two nonnegative forms, or resolve the equality set of a sharp comparison.

Definition II.1.2 (Quotient-compatible structural reduction). Let Y be a Hausdorff locally convex space of currents, profiles, or limit records, and let $K \subseteq Y$ be a closed linear subspace already annihilated by the downstream target and invariant maps. Write

$$\pi_K : Y \longrightarrow Y/K.$$

A downstream map, form, evaluation closure, or source transport is *compatible with the reduction* when it is constant on cosets of K and therefore factors continuously through π_K . A previously excluded closed family is removed from the admissible domain before the quotient is formed. Thus a reduction never discards a target-visible family without the proved invariant identity that excludes that family.

Theorem II.1.3 (Propagation of structural reductions). *Let $K_0 \subseteq K_1 \subseteq Y$ be closed subspaces. Totalize every later map relative to both reductions. On the zero descent-defect branch, the canonical map is a topological linear isomorphism*

$$(Y/K_0)/(K_1/K_0) \cong Y/K_1. \quad (53)$$

On a nonzero descent-defect branch that defect is retained in its inherited source cell. Consequently a source direction closed at one stage may be propagated into the next quotient and omitted from every later compatible calculation. A later invariant computed after reduction has the same value as the corresponding invariant on Y after quotienting by all previously closed directions.

Proof. The quotient map $Y/K_0 \rightarrow Y/K_1$ is continuous, open, and has kernel K_1/K_0 . The topological first isomorphism theorem gives (53). Every zero-defect downstream object factors through these quotient maps, so its reduced and unreduced evaluations agree under the displayed isomorphism. A failure to factor is the nonzero descent-defect branch stipulated in the theorem. $\square$

Theorem II.1.4 (Target-quotient coercivity). *Let Y be a Hilbert source space, let $K \subseteq Y$ be a closed target-null subspace, and let $T : Y \rightarrow Z_\Sigma$ be bounded with $K \subseteq \ker T$. Let $q : Y \rightarrow [0, \infty]$ be an extended nonnegative quadratic form with finite domain $D(q)$ satisfying $D(q) + K \subseteq D(q)$, and define its relaxed quotient form by*

$$q_{\text{red}}([y]) := \inf_{k \in K} q(y + k), \quad [y] \in Y/K. \quad (54)$$

If, on the required reduced domain,

$$\|[y]\|_{Y/K}^2 \leq C q_{\text{red}}([y]), \quad (55)$$

then

$$\|Ty\|_{Z_\Sigma}^2 \leq \|\tilde{T}\|^2 C q_{\text{red}}([y]) \leq \|\tilde{T}\|^2 C q(y), \quad (56)$$

where $\tilde{T} : Y/K \rightarrow Z_\Sigma$ is the descended target map. No bound for the full norm $\|y\|_Y$ follows or is needed.

Proof. Because T annihilates K , the quotient universal property gives the bounded map $\tilde{T}$ with $T = \tilde{T}\pi_K$. Hence

$$\|Ty\| \leq \|\tilde{T}\| \|[y]\|_{Y/K}.$$

Apply (55); the final inequality follows by using $k = 0$ in (54). $\square$

Example II.1.5 (Strict gain over full-state coercivity). Let $Y = K \oplus Z_\Sigma$, let $T(k, z) = z$, and let $q(k, z) = \|z\|^2$. Then (55) holds with $C = 1$, while no inequality $\|(k, z)\|_Y^2 \leq C'q(k, z)$ can hold when $K \neq \{0\}$. The sequence $(nk_0, 0)$ may diverge in the full state norm but is exactly target-null. Thus the quotient theorem proves every target estimate available from q without controlling that divergent motion.

Definition II.1.6 (Structural evaluation invariants). Work on a zero-descent-defect reduced space Y_{red} , with a continuous target projection $\Pi_\Sigma : Y_{\text{red}} \rightarrow Z_\Sigma$. The structural invariants used below have the following forms.

(I1) For a candidate family $\mathcal{F} \subseteq Y_{\text{red}}$, its *target closure invariant* is

$$\mathfrak{H}_{\mathcal{F}}^\Sigma := \overline{\text{span } \Pi_\Sigma(\mathcal{F})}^{Z_\Sigma}. \quad (57)$$

(I2) For nonnegative functionals D, N_+ on a reduced admissible family $\mathcal{P}$, the *sharp domination radius* is

$$\alpha_*(N_+ \mid D) := \sup_{\substack{V \in \mathcal{P} \\ D(V) > 0}} \frac{N_+(V)}{D(V)}. \quad (58)$$

The value is $+\infty$ when some V has $D(V) = 0 < N_+(V)$. For a normalized family $\mathcal{N} \subseteq \mathcal{P}$, the *physical-price gap* is

$$\delta_*(D \mid \mathcal{N}) := \inf_{V \in \mathcal{N}} D(V). \quad (59)$$

(I3) When $\alpha_* < \infty$, let $\widehat{\mathcal{E}}_{\alpha_*}$ be the closure in the canonical evaluation compactification of the normalized equality family

$$\mathcal{E}_{\alpha_*} := \{V \in \mathcal{P} : D(V) > 0, N_+(V) = \alpha_* D(V)\}.$$

Totalize the remainder relation generated by the equality identities. Its ordinary graph image has *equality-rigidity invariant*

$$\mathfrak{H}_{\text{eq}}^\Sigma := \overline{\text{span } \Pi_\Sigma \mathcal{R}_{\text{eq}}(\widehat{\mathcal{E}}_{\alpha_*})}^{Z_\Sigma}. \quad (60)$$

and its failure and graph-boundary values enter $\mathfrak{H}_{\text{sat}}^\Sigma$. For $\alpha_* = +\infty$, the equality row is canonically inapplicable and the positive null-cost or unbounded-feeding class is already returned by the radius row. The equality identities may be saturation equations, conservation laws, modulation equations, Pohozaev identities, or Liouville constraints.

Each invariant is computed after all zero-descent-defect source, gauge, trace, and target-null reductions already proved have been propagated.

Theorem II.1.7 (Structural normal form for invariant evaluation). *Within the interfaces of this paper, every analytic input to target exclusion has one of the following exact structural meanings.*

- (N1) A candidate family is target-null on the reduced interface exactly when $\mathfrak{H}_{\mathcal{F}}^\Sigma = 0$.
 - (N2) The domination inequality $N_+ \leq aD$ holds on $\mathcal{P}$ exactly when $\alpha_*(N_+ \mid D) \leq a$.
 - (N3) The normalized family $\mathcal{N}$ has a uniform positive physical price exactly when $\delta_*(D \mid \mathcal{N}) > 0$.
 - (N4) A sharp equality branch has no target-visible remainder exactly when $\mathfrak{H}_{\text{eq}}^\Sigma = 0$.
- Source closure and target-null residual calculations use (N1); resolvent, flux, and carrier identities generate (N2); physical prices generate (N3); and zero-cost equations generate (N4). None is an endpoint. Its nonclosing value supplies a normalized witness to the activity lift, after which the successor and viability operators continue the calculation.

Proof. Item (N1) is the definition of the closed target projection in (57). For (N2), divide by $D(V)$ when it is positive; the convention for a positive D -null mode gives the remaining case. Item (N3) is the definition of the infimum in (59). Item (N4) follows from (60): its vanishing is precisely the vanishing of every target projection of the equality remainder. These are the four structural coordinates occurring in Theorems I.4.5 and I.5.12, Definition II.8.2, and Lemma II.8.3. $\square$

Definition II.1.8 (Operational fast-track evaluator). For a source/mode cell (A, m) , propagate all previously vanished closed subspaces and let $Y_{A,m}^\Sigma$ be the resulting target-active quotient. Introduce the ordered row set

$$\begin{aligned} \mathcal{R}_{\text{FT}} := \{ & \text{form, grad, chain, bdry, tail,} \\ & \text{target, feed, price, route, capacity,} \\ & \text{scale, sat, fluct, cov, succ} \}. \end{aligned} \quad (61)$$

Its fast-track value is the row-indexed tuple

$$\mathcal{J}_{A,m}^\Sigma := (\mathcal{J}_{A,m}^{\Sigma,r})_{r \in \mathcal{R}_{\text{FT}}}, \quad (62)$$

where the first five and covariance entries are target-active total-graph defect quotients; $\mathfrak{H}_{A,m}^\Sigma$ is the target closure; α^* is the feeding radius; δ^* is the physical-price infimum; $\mathcal{R}$ is effective resistance; Cap is target capacity; $\mathcal{S}$ is the pulled-back nonnegative scale-price series; $\mathfrak{H}_{\text{sat}}^\Sigma$ is the projected saturation or equality remainder; and χ^Σ is the target-active covariance, bracket, or fluctuation radius when the evolution is stochastic. An inapplicable row has the singleton value $\{\perp_{\text{n/a}}\}$ precisely when its defining type is absent from the fixed PDE signature, as for the fluctuation row of a deterministic equation. Applicability is not a choice. Every other entry is an exact quotient, graph defect, infimum, supremum, kernel, capacity, resistance, or series value. No favorable analytic property occurs in the definition.

Definition II.1.9 (Structural property compiler). For a row $r \in \mathcal{R}_{\text{FT}}$, let F_r^{tot} be its totalized interface on the current reduced cell and let $\mathcal{K}_r^\Sigma$ be the closed ledger already propagated into that cell. The structural property compiler $\mathcal{P}_\Sigma$ assigns the following exact invariant statements:

$$\mathcal{P}_\Sigma[\text{native realization or local existence}] := \mathfrak{H}_{\text{form},A,m}^\Sigma = 0, \quad (63)$$

$$\mathcal{P}_\Sigma[\text{target-relative closed gradient range}] := \mathfrak{H}_{\text{grad},A,m}^\Sigma = 0, \quad (64)$$

$$\mathcal{P}_\Sigma[\text{target-relevant compactness of generated limits}] := \mathfrak{H}_{\text{bdry},A,m}^\Sigma = 0, \quad (65)$$

$$\mathcal{P}_\Sigma[\text{pre-contact chain rule}] := \mathfrak{H}_{\text{chain},A,m}^\Sigma = 0, \quad (66)$$

$$\mathcal{P}_\Sigma[\text{constraint-tail closure}] := \mathfrak{H}_{\text{tail},A,m}^\Sigma = 0, \quad (67)$$

$$\mathcal{P}_\Sigma[\text{generated operation descends}] := \mathfrak{H}_{\text{cov},A,m}^\Sigma = 0, \quad (68)$$

$$\mathcal{P}_\Sigma[N_+ \leq aD] := \alpha_{A,m}^*(N_+ \mid D) \leq a, \quad (69)$$

$$\mathcal{P}_\Sigma[\inf_{\mathcal{N}} D > 0] := \delta_{A,m}^*(D \mid \mathcal{N}) > 0, \quad (70)$$

$$\mathcal{P}_\Sigma[\text{routable target current}] := \mathcal{R}_{A,m}(\Sigma) < \infty, \quad (71)$$

$$\mathcal{P}_\Sigma[\text{target polar}] := \text{Cap}_{A,m}(\Sigma) = 0, \quad (72)$$

$$\mathcal{P}_\Sigma[\text{finite-budget scale exclusion}] := \mathcal{S}_{A,m}^\Sigma = +\infty, \quad (73)$$

$$\mathcal{P}_\Sigma[\text{sharp equality or saturation closes}] := \mathfrak{H}_{\text{sat},A,m}^\Sigma = 0, \quad (74)$$

$$\mathcal{P}_\Sigma[\text{fluctuation is target-null or physically routed}] := \chi_{A,m}^\Sigma = 0 \text{ or its resistance is finite.} \quad (75)$$

The successor row is the finite-depth sequence $Z_{n+1,A,m} = \mathcal{V}_\eta(Z_{n,A,m})$; its two values are finite pruning with its rational-cylinder transcript and the compact inverse-limit path of Theorem III.2.8. Here every object is formed after quotienting by $\mathcal{K}_r^\Sigma$. Any algebraic identity derivable from the public weak equation may refine a row into nonnegative descendants. The compiler has no input slot for an external compactness, coercivity, Liouville, or realization statement.

Definition II.1.10 (Propagated physical fast-track ledger). Order the rows as in (61). Begin with the closed target-null ledger $\mathcal{K}_0^\Sigma = \mathcal{A}_\Sigma$. If row i has closed branch C_i and transport T_i into the next row, set

$$\mathcal{K}_{i+1}^\Sigma := \overline{\mathcal{K}_i^\Sigma + T_i(C_i)}. \quad (76)$$

All spaces, totalized interfaces, target projections, physical forms, and subsequent invariants are then replaced by their descended objects on the quotient by $\mathcal{K}_{i+1}^\Sigma$. If a map fails to descend, its nonzero commutator with the quotient map is the next row's target-active graph defect. Thus quotient compatibility is tested by the construction rather than imposed on it.

Theorem II.1.11 (Analytic-property compilation). *Every analytic property used by the continuum argument to decide whether a source/mode cell can contribute to singular contact is equivalent to the corresponding structural invariant statement in Definition II.1.9. The opposite analytic property is equivalent to the complementary invariant value and produces a tagged activity profile and successor. In particular, the fast track never presupposes closed range, compactness, a chain rule, a coercive estimate, a positive price, rigidity, polarity, finite resistance, summable tails, fluctuation control, orbit realization, or continuation.*

Proof. Existence and regularity properties are membership statements in the graph of a partial map. Totalization makes their negations explicit graph values, and quotienting their closed graph boundary by $\mathcal{A}_\Sigma$ proves (63)–(67) and (68). In the compactness row only target-visible loss of compactness survives; target-null escape is already in J_{res} . Thus the invariant tests exactly the compactness needed downstream rather than ambient compactness of the orbit. The radius and gap equivalences follow directly from the supremum and infimum definitions, including positive D -null directions and zero-price sequences. The spectral theorem gives the resistance/range/harmonic trichotomy, with the harmonic part retained in the target quotient. Capacity zero is the definition of the polar branch; positive capacity is its complementary target-compatible class. A nonnegative scale-price series either diverges or is finite. Saturation equations define a remainder map, so closure of the equality branch is exactly vanishing of its target quotient. Finally, stochastic drift, bracket, covariance, and conditional-expectation contributions are currents in the lifted state space; quotienting their target-null part and applying the same radius/resistance construction gives (75). Every complementary value enters the successor row, whose finite-depth/inverse-limit alternatives are exhaustive by compactness. These cases exhaust (61). $\square$

Definition II.1.12 (Target-relevant analytic statement grammar). Let $\text{An}_\Sigma(\mathfrak{P})$ be the least grammar generated from the fixed PDE signature by:

- (A1) membership in, or failure of, a native, limit, chain, trace, tail, normalization, or continuation graph;
- (A2) vanishing or nonvanishing after the target quotient;
- (A3) comparison of two nonnegative physical forms, including an infimum or sharp radius;
- (A4) range, kernel, resistance, capacity, and harmonic statements for the generated operator or form;
- (A5) convergence or divergence of a nonnegative physical-price series;

- (A6) a saturation/equality remainder after sharp comparison;
 - (A7) covariance, bracket, compensator, or conditional-expectation statements in a generated stochastic lift;
 - (A8) membership in a finite-depth successor set or its compact inverse-limit path space.
- The grammar is closed under finite conjunction, sourcewise limit, pullback by a generated chart, and descent through a generated quotient. These closure operations are themselves totalized.

Theorem II.1.13 (Exhaustive fast-track compilation). *The property compiler extends uniquely to a total map*

$$P_\Sigma : \text{An}_\Sigma(\mathfrak{P}) \longrightarrow \prod_{\substack{\emptyset \neq A \subseteq \mathfrak{C} \\ m \in \mathfrak{M}}} \prod_{r \in \mathcal{R}_{\text{FT}}} \text{Val}(\mathfrak{J}_{A,m}^{\Sigma,r}). \quad (77)$$

Every statement that can affect the contact identity or the physical expenditure contradiction occurs in exactly one primitive row, possibly with several source/mode tags. Its negation is the complementary value of that same row. Hence no analytic property, compactness assertion, or equation-specific estimate is an unregistered premise of the fast track.

Proof. Clauses (A1)–(A8) map respectively to the total-graph rows; target row; feeding/price rows; gradient/routing/capacity rows; scale row; saturation row; fluctuation row; and finite-depth successor row. Source and realization tags are assigned before evaluation by Theorems I.2.6 and I.5.11. Finite conjunction gives a product value. Sourcewise limits remain in their tagged closures. Chart pullback appends Abs/Lift, and quotient descent either factors or produces its totalized descent defect. Structural induction therefore defines P_Σ on the whole grammar and leaves no unmatched constructor.

By normalized shell accountability, contact affects the argument only through a target-active current or a totalized shell/interface defect. By the physical balance, exclusion affects it only through a nonnegative expenditure comparison or series. These are clauses (A1)–(A6); stochastic and successor-path semantics are exactly (A7)–(A8). Thus the grammar covers every input to the contact-versus-expenditure proof. $\square$

Theorem II.1.14 (Closed-input rule). *The value of $\mathfrak{J}_{A,m}^{\Sigma,r}$ is determined by the fixed PDE signature, target, physical balance, and the closures generated from them. An auxiliary lemma may prove that value but cannot be added as data defining it. Any proposed object not generated by this algebra either:*

- (E1) *is derived from the fixed signature, in which case the source compiler and property compiler assign it to existing rows; or*
- (E2) *changes the signature, in which case the conclusion concerns the enlarged equation/presentation and is not a conclusion about the original PDE.*

These alternatives exhaust the generated continuum algebra.

Proof. The evaluation compactification is generated by Definition I.3.1; the source syntax by Definition I.2.5; and the analytic statement syntax by Definition II.1.12. Membership in either syntax is therefore determined before an invariant is evaluated. Structural induction gives (E1). Failure of membership means that a new primitive map, topology, law, or balance has been appended, which is precisely a change of signature and gives (E2). $\square$

Theorem II.1.15 (Ledger propagation and no re-estimation). *At each fast-track row, exactly one of the closed and complementary branches is selected by its invariant value. A closed branch propagated by (76) remains zero in every later descended target quotient. If a downstream object*

does not descend, its descent defect is retained in the appropriate existing source/mode cell. Therefore later rows neither assume nor re-estimate any property closed by an earlier row.

Proof. For closed subspaces the third isomorphism theorem gives $(Y/\mathcal{K}_i)/(\mathcal{K}_{i+1}/\mathcal{K}_i) \cong Y/\mathcal{K}_{i+1}$. Hence every propagated class is zero in all later quotients. The same statement for closed convex ledgers follows by passing to the associated quotient seminorm. Failure of descent is precisely a nonzero commutator between the downstream map and the quotient map, which is a totalized graph defect. Sourcewise closure retains its ancestry. The exhaustive two-branch statement follows from Theorem II.1.11. $\square$

Theorem II.1.16 (Fast-track branch-to-successor completeness). *For every target-active source/mode cell, the tuple $\mathfrak{J}_{A,m}^\Sigma$ returns one of the following exhaustive structural branches.*

- (F1) *Each form, gradient, chain, boundary, tail, and covariance defect quotient vanishes or survives as a target-visible total-graph defect with the same source ancestry.*
- (F2) $\mathfrak{H}_{A,m}^\Sigma = 0$: *the cell is target-null and closes. Otherwise its nonzero quotient is retained.*
- (F3) $\alpha_{A,m}^* \leq 1$: *feeding is dominated by the generated carrier. If $\alpha_{A,m}^* > 1$, or $D = 0 < N_+$ for a member of the cell, the maximizing sequence becomes a target-feeding activity profile.*
- (F4) $\delta_{A,m}^* > 0$: *the cell has positive normalized price. If $\delta_{A,m}^* = 0$, a zero-price sequence exists in the closure and is retained with the same ancestry.*
- (F5) $\mathfrak{H}_{\text{eq},A,m}^\Sigma = 0$: *the sharp equality branch closes. Otherwise its target-visible equality remainder survives.*
- (F6) $\text{Cap}_{A,m}(\Sigma) = 0$: *the target is polar for that cell. Positive capacity supplies a normalized target-active activity profile.*
- (F7) $\mathcal{R}_{A,m}(\Sigma) < \infty$: *the source can be routed through the generated geometry. Infinite resistance supplies its harmonic component to the activity lift.*
- (F8) *For a scale cascade, the pulled-back series $\sum_j r_j^{\kappa_{A,m}} \delta_{A,m,j}$ either diverges, excluding that finite-budget cascade, or is finite, yielding vanishing-price windows for the ordered successor and recurrent-saturation compiler.*
- (F9) *The saturation quotient $\mathfrak{H}_{\text{sat},A,m}^\Sigma$ vanishes or retains the target-visible equality remainder. Every Pohozaev, Noether, virial, frequency, or entropy identity already generated by the equation refines its nonnegative descendants; the remaining equality record becomes a successor.*
- (F10) *The fluctuation invariant $\chi_{A,m}^\Sigma$ is target-null, physically routed at finite resistance, or survives as a target-visible covariance/bracket class in the existing lifted source cell.*
- (F11) *The total weak-solution graph retains ordinary values and every graph-boundary witness. Both enter the successor graph; persistent boundary values survive only through an infinite zero-price witness path.*
- (F12) *An inapplicable row returns $\{\perp_{n/a}\}$; failure of a type-present map to descend returns its graph defect and activates the same successor operator.*

Every complementary branch is a value of the same invariant used by the closing branch. The tuple is not a verdict: after each row, the canonical activity map sends every nonzero target quotient to $\mathcal{R}_\eta$, and the viability sets Z_n either remove it at finite depth or construct its recurrent zero-price mechanism.

Proof. The graph-defect row follows from totalization. The target, radius, gap, and equality rows follow from Theorem II.1.7 and the definitions of the closed quotient, supremum, infimum, and equality remainder. In the radius row, the convention $\alpha^* = +\infty$ records a positive D -null feeding direction. If $\delta^* = 0$, the definition of the infimum gives a sequence with prices tending to zero; sourcewise closure retains its limit records.

Zero capacity is the polar branch of the associated form. Its complement is positive capacity and supplies an activity profile. The spectral theorem gives the finite resistance/range branch and the orthogonal harmonic remainder. Finally a nonnegative series either diverges or has finite sum. The physical pullback identity makes these precisely the excluded branch and the successor-activation branch. The latter is closed by Theorem II.2.20; it is not a terminal formation class. The saturation and fluctuation alternatives are Theorem II.1.11. The singleton inapplicable value and totalized descent defect exhaust the remaining interface cases. Applying the activity lift and Theorem III.2.8 to each nonzero quotient proves the final constructive statement. $\square$

Corollary II.1.17 (Estimate compression after structural derivation). *Once a graph relation, positive-form identity, target-null quotient, or zero-set relation has been derived from the symbolic presentation, subsequent rows use only that structural output. Formal identities and universal functional calculus refine the generated positive cone directly. When they do not remove a row, its graph-boundary, positive-null, or zero-price witnesses are generated and passed to the successor calculus. Closed outputs are propagated by Theorem II.1.15.*

Definition II.1.18 (Canonical symbolic structural transcript). For the fixed PDE, complete continuation target, carrier, and derived ceiling, define

$$\Theta_{\Sigma}^{\text{syn}}(\mathfrak{P}, B_{\text{phys}}) := \left(\text{Syn}(\mathfrak{P}), \text{Graph}^{\text{tot}}(\mathfrak{P}), \text{Pos}(\mathcal{K}), \Pi_{\Sigma}, \text{Cov}(\text{Op}), \text{Anc} \right). \quad (78)$$

The entries are respectively the typed equation syntax, total graphs of its generated operations, the cone of nonnegative terms occurring in the proved carrier identity, the generated target morphism, the total covariance graphs, and the event/source ancestry maps. Exact extrema, capacities, resistances, realization images, contact orbits, continuation verdicts, and Liouville conclusions are not entries. They are semantic outputs of rows constructed from $\Theta_{\Sigma}^{\text{syn}}$.

Theorem II.1.19 (Public-signature structural sufficiency). *The symbolic transcript functorially determines the source cells, totalized interfaces, target-null quotient, carrier complex, activity measures, and successor relation used by the regularity argument:*

$$\text{Compiler}_{\Sigma} = \overline{\text{Compiler}}_{\Sigma} \circ \Theta_{\Sigma}^{\text{syn}}. \quad (79)$$

Every row is constructed together with its ordinary and complementary witness graphs. No exact row value or actual-contact image is consulted in this construction. Two presentations with isomorphic symbolic transcripts therefore have isomorphic prospective successor and zero-cost fixed-point algebras. An analytic derivation can only prove a relation already present in the transcript or refine its generated positive cone; it cannot insert a favorable value.

Proof. Proceed by structural induction over the typed syntax and then by the fixed row order (61). Each constructor supplies its ordinary graph, explicit failure value, source transport, and event ancestry. Graph closure supplies the corresponding boundary witnesses. The target morphism forms the null quotient, and the positive carrier cone supplies the expenditure coordinate. Quotient descent either factors or produces its commutator graph. These operations construct both branches of every row without deciding which branch contains a given future orbit record.

The activity and successor constructions use only the generated metric, target shell, graph witnesses, and event ancestry. The prospective fixed-point construction below uses their closed images, not the image of actual contact orbits. This proves (79) and the isomorphism statement. $\square$

Corollary II.1.20 (Closed structural input). *A PDE specialization supplies only the target-blind public signature of Definition I.1.3. The continuum algebra generates (78). Every identity obtainable by formal manipulation of that signature or by the fixed universal library is inserted automatically as a graph relation or positive-descendant refinement. The ordinary graph and every first-failure witness are inputs to the canonical activity lift and successor operator of Definitions II.2.8 and II.2.10. Their descending zero-cost closure is computed by Definition II.2.13. Thus every generated branch is reduced to a target-null class or to an explicit recurrent zero-cost mechanism; there is no unevaluated row and no additional invariant value to be supplied.*

Proposition II.1.21 (Positive-separator evaluation of an equality branch). *Totalize the separator relation on $\hat{\mathcal{E}}_{\alpha_*}$. On its zero graph-defect branch the identity is*

$$0 = Q_\Sigma(\Pi_\Sigma \mathcal{R}_{\text{eq}}(Z)) + \sum_{j=1}^m q_j(Z), \quad (80)$$

where $Q_\Sigma \geq 0$ is positive definite on Z_Σ and every $q_j \geq 0$. Then $\mathfrak{H}_{\text{eq}}^\Sigma = 0$.

Proof. Every term on the right of (80) is nonnegative, so each term vanishes. Positive definiteness gives $\Pi_\Sigma \mathcal{R}_{\text{eq}}(Z) = 0$ for every Z . Taking the closed linear span proves the claim. $\square$

II.2 Coherent carrier passivity and structural successor closure

The scalar price infimum of Definition II.1.6 is not the terminal evaluator for a finite-price cascade. A finite measure on infinitely many disjoint target windows produces a vanishing-price subsequence. The continuum analogue of the residual closure in the discrete algebra is to follow that subsequence through all compact, concentrating, escaping, diffuse, gauge, and equality outputs. This section constructs that closure directly.

Definition II.2.1 (Coherent carrier complex). Let u be one physical orbit. A *coherent carrier complex* consists of a locally finite rooted tree $\mathcal{T}$, cells Q_α , and nonnegative localization tests ψ_α with the following properties.

(C1) Every child is a subevent of its parent and, on the region where the local balance is evaluated,

$$\psi_\alpha = \sum_{\beta \in \text{ch}(\alpha)} \psi_\beta. \quad (81)$$

(C2) All cells are restrictions, chart images, or normalized descendants of the same orbit. Their transition maps satisfy the cocycle composition law. Galilean, gauge, and quotient representatives are fixed compatibly on overlaps.

(C3) The weak equation gives a local carrier relation, linear in the test,

$$K_\psi(t) + D_I(\psi) + M_I(\psi) = K_\psi(s) + J_I(\psi) + E_I(\psi). \quad (82)$$

Here $D_I, M_I \geq 0$ are respectively the intrinsic production and the weak energy-defect measure, J_I is the signed transport current, and E_I is true exterior input. An inequality is written as equality by adjoining its nonnegative defect to M_I .

(C4) The normalized target observable is constant under the allowed transition maps. Consequently target retention is inherited by a child selected from a target-retaining parent.

The complex is generated from the shell cascade by taking the maximal locally finite family of compatible target-retaining descendants. A failure of local finiteness, the partition relation, transition coherence, or gauge compatibility is a totalized Lift/Bdry successor and is not an additional admissibility condition.

Theorem II.2.2 (Signed carrier gluing). *Let $\mathcal{T}_0 \subset \mathcal{T}$ be a finite rooted subtree. Sum (82) over $\mathcal{T}_0$ before taking positive parts. Every current through an internal face cancels with the same current with opposite orientation in the adjacent cell. In particular, conservative transport, pressure or constraint transport, chart translation, gauge motion, cutoff motion, and exact scale motion contribute only through the root, the leaves, or the true exterior boundary. The result has the form*

$$K_{\text{leaf}}(\mathcal{T}_0) + \mu_{\text{Geom}}(\mathcal{T}_0) + \mu_{\text{def}}(\mathcal{T}_0) = K_{\text{root}} + I_{\text{ext}}(\mathcal{T}_0) + H_{\text{sc}}(\mathcal{T}_0), \quad (83)$$

where the first two measures on the left are nonnegative. The term H_{sc} is the cohomology class of scale motion that is not an edge coboundary in the chosen scale-critical carrier. It has inherited Caus/Lift ancestry. It vanishes when the scale current admits a coherent carrier potential. No sum of the absolute values of the internal currents occurs in (83).

Proof. Linearity in ψ and (81) imply that the carrier, production, defect, transport, and exterior terms of a parent are the sums of the corresponding child terms. Give every common face the orientation induced by its parent. The child has the opposite induced orientation, so its signed flux cancels the parent flux. Pressure-gauge constants cancel because their weak pairing with the divergence-free or constraint-compatible test is zero. Transition coherence makes translation, gauge, and exact scale terms edge coboundaries; the sum of a coboundary over a finite tree is its root value minus its leaf values. The only possible scale term left after this cancellation is its class modulo the range of the edge coboundary, which is H_{sc} . The intrinsic production and weak balance defect retain their signs and add over the cells. This proves (83). $\square$

Definition II.2.3 (Carrier cohomology and aggregate feeding). Let $\mathcal{W}_\eta$ be the saturated closure of normalized windows whose target observable is at least $\eta > 0$, after all generated symmetry and gauge quotients. Include the values of the carrier, production, exterior input, and every totalized graph coordinate in the evaluation topology. Let S be the successor or unit-window shift on its ordinary graph.

On a coherent ordinary branch, write F_{sc} for the aggregate signed inward scale-space feed left after Theorem II.2.2. Its target-active carrier cohomology is

$$\mathfrak{H}_{\text{car}}^\Sigma := \overline{\Pi_\Sigma F_{\text{sc}}} / \overline{\Pi_\Sigma \{\Psi - \Psi \circ S : \Psi \in C(\mathcal{W}_\eta)\}}. \quad (84)$$

The quotient is totalized when S , the carrier, or a transition is partial. Its graph-boundary values retain their source ancestry. The ordinary carrier-deficit cocycle is

$$g_{\text{car}} := F_{\text{sc}} - D - M. \quad (85)$$

For a finite block, its sum is the aggregate inward feeding minus production, up to root–leaf storage and true exterior input. Coboundaries disappear in block averages and under invariant measures. If $[F_{\text{sc}}] = 0$ in (84), then for every $\varepsilon > 0$ there is a continuous Ψ_ε whose coboundary approximates the feed class within ε ; no exact correction is asserted.

Theorem II.2.4 (Target-relative passivity). *Let $\mathcal{K}_\eta \subseteq \mathcal{W}_\eta$ be a compact S -invariant ordinary carrier class generated by the activity lift. On the ordinary signed-carrier branch, finite exterior*

input is already part of the derived physical budget. Then every S -invariant Borel probability measure ν on $\mathcal{K}_\eta$ satisfies

$$\int_{\mathcal{K}_\eta} g_{\text{car}} \, d\nu \leq 0. \quad (86)$$

Since every invariant measure annihilates continuous coboundaries, this is equivalent to

$$\int F_{\text{sc}} \, d\nu \leq \int (D + M) \, d\nu. \quad (87)$$

Equality in (86) is the zero carrier-deficit branch. It enters the zero-cost successor fixed point. It implies vanishing of a particular production term only when the generated consequence algebra contains a domination identity $p = q + r$ with that production as q .

Proof. Apply (83) to the first N consecutive cells of a coherent path. Internal currents cancel and the carrier terms telescope. The carrier is bounded on the compact evaluation class. The exterior input has finite total mass, so its contribution divided by N tends to zero. Integrating the block identity against an invariant measure cancels the endpoint terms exactly and gives (86), which is (87). Continuous coboundaries and their uniform closure have zero integral against every invariant measure, justifying the cohomological quotient without producing an exact potential. $\square$

Corollary II.2.5 (Totalized passivity branch). *For every target-retaining carrier route, exactly one of the following is generated:*

- (T1) *the coherent ordinary branch of Theorem II.2.4;*
- (T2) *a first failure of local finiteness, carrier partition, transition coherence, gauge descent, compact retention, or carrier correction, together with its closed-graph witness descendants;*
- (T3) *a target-null output removed by the generated quotient.*

The second output enters $\mathcal{R}_\eta$; the third closes. Positive exterior variation has Bdry ancestry and is charged to the sole physical measure. Consequently compactness, coherence and carrier cohomology are processed by the successor operator.

Proof. Totalize each partial relation in the order displayed in Definitions II.2.1 and II.2.3. If all ordinary values occur, Theorem II.2.4 applies. The first nonordinary value has a witness sequence in its closed total graph and therefore defines the successor in Definition II.2.13. Positive exterior variation is part of the physical measure. The first-failure alternatives are exhaustive because all graph coordinates belong to the compact activity lift. $\square$

Definition II.2.6 (Carrier saturation algebra). The *carrier saturation algebra* is the closed target projection

$$\mathfrak{H}_{\text{eq,car}}^\Sigma := \overline{\text{span}} \left\{ \Pi_\Sigma \text{supp } \nu : \begin{array}{l} \nu \text{ is invariant on a generated compact successor core,} \\ \int g_{\text{car}} \, d\nu = 0 \end{array} \right\}. \quad (88)$$

The support is evaluated through all five source tags and all totalized zero-deficit equations. It may contain equilibria, relative equilibria, or nonstationary recurrent conservative dynamics. The subsequent zero-cost fixed point, rather than an assumed rigidity statement, decides its target projection.

Theorem II.2.7 (Ergodic strict passivity or saturation). *Let $\mathcal{K}_\eta$ be as in Theorem II.2.4, and set*

$$\beta_\eta := \sup_{\nu \in \mathcal{P}_S(\mathcal{K}_\eta)} \int g_{\text{car}} \, d\nu, \quad (89)$$

where $\mathcal{P}_S(\mathcal{K}_\eta)$ is the compact set of S -invariant Borel probability measures. Then $\beta_\eta \leq 0$, and exactly one of the following holds.

(P1) $\beta_\eta < 0$. There exist an integer $N_\eta \geq 1$ and $\delta_\eta > 0$ such that every N_η -block retained in $\mathcal{K}_\eta$ satisfies

$$-\sum_{k=0}^{N_\eta-1} g_{\text{car}}(S^k w) \geq \delta_\eta. \quad (90)$$

(P2) $\beta_\eta = 0$. The class $\mathcal{K}_\eta$ carries an invariant maximizing measure and its target projection belongs to $\mathfrak{H}_{\text{eq,car}}^\Sigma$.

If $\mathfrak{H}_{\text{eq,car}}^\Sigma = 0$, only (P1) is possible. The positive block price is therefore derived from the absence of a target-visible equality state; no contact margin or compactness gap is inserted.

Proof. Let $h = g_{\text{car}}$. Compactness of $\mathcal{P}_S(\mathcal{K}_\eta)$ and continuity of the integral give a maximizing measure. By Theorem II.2.4, $\beta_\eta \leq 0$.

Suppose $\beta_\eta < 0$ but (P1) fails. Then there are $w_n \in \mathcal{K}_\eta$ with block length $N_n = n$ and total carrier deficit smaller than n^{-1} . In particular the mean of h has lower limit at least zero. Form the empirical measures

$$\nu_n := \frac{1}{N_n} \sum_{k=0}^{N_n-1} \delta_{S^k w_n}.$$

Compactness of probability measures on $\mathcal{K}_\eta$ gives a weakly convergent subsequence. The endpoint identity

$$\int f \circ S \, d\nu_n - \int f \, d\nu_n = \frac{f(S^{N_n} w_n) - f(w_n)}{N_n}$$

shows that its limit ν is invariant. Continuity of the evaluation coordinates and the lower bound give $\int h \, d\nu \geq 0$, while Theorem II.2.4 gives the reverse inequality. Hence $\int h \, d\nu = 0$, contradicting $\beta_\eta < 0$. Thus (P1) holds. If $\beta_\eta = 0$, a maximizing invariant measure is a saturation measure and gives (P2). Conversely, an invariant saturation measure contradicts any uniform positive block deficit after integration. Vanishing of $\mathfrak{H}_{\text{eq,car}}^\Sigma$ excludes a target-retaining instance of (P2). $\square$

Definition II.2.8 (Canonical activity lift and ordered successor rows). Let $\mathfrak{C}_\eta$ be the closure, in the countable product of the generated evaluation compactification, of all indexed normalized shell cascades

$$\mathbf{R} = ((R_n, Q_n, p_n))_{n \geq 1}, \quad \text{Prog}_\Sigma(R_n) \geq \eta,$$

that satisfy the total weak-equation, shell, and event-order graphs. On an ordinary contact value, the Q_n are the canonical pairwise disjoint physical events, $p_n = \mu_u(Q_n)$, and the progress is one. At a graph boundary, p_n is the lower-semicontinuous carrier-price coordinate. Boundary points of $\mathfrak{C}_\eta$ retain the sequences that approximate them. The compact cascade coordinate $\mathbf{R}$ is an *origin tag*; every descendant below retains it, so successor chains cannot splice different orbits or cascades.

The generated compact observer space Ω_{ev} is the product of the state, spacetime, scale, source, chart, constraint, origin, and total-graph coordinates occurring in the symbolic transcript.

On an ordinary localizable branch, let q_n^c and η_n be the finite signed Radon measures representing the c -source shell current and the shell defect. After pushing them to Ω_{ev} , set

$$a_n := \sum_{c \in \mathfrak{C}} (q_n^c)^+ + (\eta_n)^+, \quad \nu_n := \frac{a_n}{a_n(\Omega_{\text{ev}})}. \quad (91)$$

Here η_n is the shell defect in (29). The target floor implies $a_n(\Omega_{\text{ev}}) \geq \eta$, and an ordinary normalized contact cascade has the stronger lower bound one. If a current has no finite Radon value, its

total-graph value is placed at the corresponding rough boundary coordinate; rational truncations of its variation give the same probability-valued lift. The lifted record consists of

$$\hat{R}_n = \left(\mathbf{R}, R_n, \nu_n, \frac{a_n(\Omega_{\text{ev}})}{1 + a_n(\Omega_{\text{ev}})}, \frac{p_n}{1 + p_n}, n \right). \quad (92)$$

Run the following total graphs in order on this lift:

$$\begin{aligned} &\text{form/constraint realization} \longrightarrow \text{chart/gauge/scale coherence,} \\ &\text{activity-measure lift} \longrightarrow \text{atomic/non-atomic/boundary disintegration,} \\ &\text{state/scale disintegration} \longrightarrow \text{Young, profile-tree, and tail coordinates,} \\ &\text{carrier cohomology} \longrightarrow \text{strict passivity/saturation} \longrightarrow \text{zero-cost recurrent closure.} \end{aligned} \quad (93)$$

The grammar includes the cofinal tail shift $(R_n, Q_n, p_n) \mapsto (R_{n+1}, Q_{n+1}, p_{n+1})$. Hence a target-retaining zero-price witness sequence always has a successor; an analytic failure is carried by its total-graph coordinate rather than by an artificial failure self-loop. The familiar realization modes

$$\text{atom, separated, nested, diffuse, tail, rough, gauge, constraint, equality.} \quad (94)$$

are canonical readouts of ν_n and its coordinate disintegrations; they are not asserted to be mutually exclusive alternatives. Every ordinary or graph-boundary output retains its five-channel ancestry, original event index, and target floor. Hence a failed analytic relation is immediately another lifted successor input, not an unevaluated row.

Theorem II.2.9 (Compact activity-profile exhaustion). *Every sequence of lifted records $\hat{R}_n$ has a convergent subsequence in*

$$\hat{X}_\Sigma := \mathfrak{C}_\eta \times X_{\text{ev}} \times \mathcal{P}(\Omega_{\text{ev}}) \times [0, 1]^2 \times \overline{\mathbb{N}}, \quad (95)$$

where $\mathcal{P}(\Omega_{\text{ev}})$ carries the narrow topology and $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$. Its limit retains the target floor (unit progress for a contact origin) and source ancestry. The limiting activity law has the canonical decomposition

$$\nu = \sum_{j \geq 1} m_j \delta_{z_j} + \nu_{\text{na}}, \quad m_j > 0, \quad (96)$$

and its projections to the compactified scale, spatial-boundary, state-law, and total-graph coordinates record, respectively, nested or separated profiles, tails, Young oscillation, and rough/gauge/constraint failures. These coordinates exhaust the limit because they are coordinates of the single probability law ν ; no separate compactness input or remainder class is required.

Proof. The spaces $\mathfrak{C}_\eta$, X_{ev} , and Ω_{ev} are compact metric. Hence $\mathcal{P}(\Omega_{\text{ev}})$ is compact metric, and so is the product in (95). Sequential compactness gives a convergent subsequence. Unit target progress and the finitely many source supports are closed coordinates; pass first to a constant-support subsequence.

Every probability measure has a unique at most countable atomic part and a nonatomic remainder, which proves (96). Coordinate projection and disintegration record all spatial, scale, state-law, and graph-boundary behavior. Boundary escape is mass on the compactification boundary, rather than failure of tightness. Since every generated observation is one of these countable coordinates, equality of all coordinates is equality of lifted records. Thus no unrecorded limit output remains. $\square$

Definition II.2.10 (Witness-indexed retained successor relation). Let $\mathfrak{A}_\eta \subset \widehat{X}_\Sigma$ be the closed set of lifted records with target progress at least $\eta > 0$. An elementary successor witness is a tuple

$$(w, w', \mathbf{R}, \sigma, O), \quad (97)$$

where $\mathbf{R} = ((R_n, Q_n, p_n))_n \in \mathfrak{C}_\eta$, $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing, and O is one of the ordered total-graph operations in (93). Enumerate these operations and compactify their index by $\overline{\mathbb{N}}$; an operation-index boundary is represented by the corresponding total-graph limit. The two endpoints must be limits along the *same* selector:

$$w = \lim_{k \rightarrow \infty} \widehat{R}_{\sigma(k)}, \quad w' = \lim_{k \rightarrow \infty} O^{\text{tot}}(\widehat{R}_{\sigma(k)}), \quad (98)$$

and both retain the origin coordinate $\mathbf{R}$. Let $\mathfrak{W}_\eta^0$ be the set of these tuples. Compactify the selector by its graph in $\overline{\mathbb{N}}^\mathbb{N}$, retain the compactified operation index, take the closure $\mathfrak{W}_\eta = \overline{\mathfrak{W}_\eta^0}$, and define

$$\mathcal{R}_\eta := \pi_{w, w'}(\mathfrak{W}_\eta) \subseteq \mathfrak{A}_\eta \times \mathfrak{A}_\eta. \quad (99)$$

The witness space is compact, so $\mathcal{R}_\eta$ is closed and every edge has a nonempty compact witness fiber. Every witness carries the common origin, approximating sequence, source tags, unit target coordinate, and original event indices. A graph-boundary value is therefore another edge limit. It is never promoted to an actual orbit state without its witness sequence, and edges from different origin cascades cannot be concatenated.

Lemma II.2.11 (Diagonal event ancestry and no double counting). *Let a finite or infinite chain of successors have one ordinary contact origin $\mathbf{R} = ((R_n, Q_n, p_n))_n$. For every k one can choose an original index N_k such that $N_{k+1} > N_k$ and the N_k -th original record, after the first k ordered operations in the witness chain, approximates the k -th successor coordinate within 2^{-k} . Consequently the physical events Q_{N_k} are pairwise disjoint and*

$$\sum_k \mu_u(Q_{N_k}) \leq \mu_u\left(\bigsqcup_n Q_n\right) \leq B_{\text{phys}}. \quad (100)$$

Proof. Choose a witness from the compact fiber of every edge. At depth k , use the same-origin limits (98) and a diagonal approximation of the first k witness tuples by elementary witnesses. Their selectors have cofinal image in the one origin sequence. Choose $N_k > N_{k-1}$ sufficiently far out that the composed depth- k graph value is within 2^{-k} of w_k . No single event is used to approximate two different depths. The events therefore belong to distinct members of the original disjoint shell family, and countable additivity gives (100). $\square$

Theorem II.2.12 (Successor recurrence). *The relation $\mathcal{R}_\eta$ is closed in the compact metric lifted topology. If there is an infinite chain*

$$w_0 \mathcal{R}_\eta w_1 \mathcal{R}_\eta w_2 \mathcal{R}_\eta \cdots, \quad (101)$$

then the path space of all such chains is compact and invariant under the left shift. The orbit closure of the given path carries a shift-invariant Borel probability measure. Every point in the support of its coordinate projection is a limit generated from one retained cascade origin with a diagonal witness sequence, retains the target floor, and retains the union of source ancestries. No claim of pointwise realization by a new PDE orbit is used.

Proof. The witness bundle is compact and coordinate projection is continuous, so $\mathcal{R}_\eta$ is compact and hence closed. The set $\mathfrak{A}_\eta$ is compact in the lifted topology, so $\mathfrak{A}_\eta^\mathbb{N}$ is compact. The path space is the intersection, over all j , of the closed conditions $(w_j, w_{j+1}) \in \mathcal{R}_\eta$; it is therefore compact and is nonempty by (101). Empirical measures of the shifted given path have a weakly convergent subsequence, and the endpoint calculation proves shift invariance. Apply Lemma II.2.11 to obtain the diagonal original-event witnesses. Sourcewise closure preserves ancestry and the target floor. $\square$

Definition II.2.13 (Prospective zero-cost successor fixed point). Let $p : \mathfrak{A}_\eta \rightarrow [0, \infty]$ be the lower-semicontinuous relaxation of the original-event price coordinate. For a closed $K \subseteq \mathfrak{A}_\eta$, define the viability operator

$$\mathcal{V}_\eta(K) := \{w \in K : p(w) = 0, \mathcal{R}_\eta(w) \cap K \neq \emptyset\}. \quad (102)$$

It maps closed sets to closed sets, because it is the projection of the compact set $\mathcal{R}_\eta \cap (K \times K) \cap (\{p = 0\} \times K)$. Starting with the target-active zero-price set $Z_0 = \mathfrak{A}_\eta \cap \{p = 0\}$, set

$$Z_{n+1} = \mathcal{V}_\eta(Z_n), \quad Z_\infty^\Sigma = \bigcap_{n=0}^{\infty} Z_n. \quad (103)$$

Thus Z_n consists exactly of the records from which a target-retaining zero-price successor path of length n can be generated. Compactness and closedness give

$$\mathcal{V}_\eta(Z_\infty^\Sigma) = Z_\infty^\Sigma. \quad (104)$$

Indeed, for $w \in Z_\infty^\Sigma$, choose $w_n \in Z_n$ with $w\mathcal{R}_\eta w_n$; a convergent subsequence has a limit in every Z_n , and the closed successor graph retains the edge. Consequently the countable finite-depth iteration is the complete viability calculation. If $Z_\infty^\Sigma = \emptyset$, nested compactness implies $Z_N = \emptyset$ for some finite N . The canonical finite pruning transcript below then gives a finite structural derivation of closure. If every Z_n is nonempty, their intersection contains an infinite zero-price path.

Let $\text{Path}(Z_\infty^\Sigma)$ be the compact path space of $\mathcal{R}_\eta$ restricted to this fixed point. Let $\mathcal{P}_{\mathcal{R}_\eta}^{0,\Sigma}$ be all shift-invariant Borel probability laws on this path space. The prospective successor-saturation algebra is

$$\mathfrak{H}_{\text{succ}}^\Sigma := \overline{\text{span}} \left\{ \Pi_\Sigma(\pi_0)_\# \nu : \nu \in \mathcal{P}_{\mathcal{R}_\eta}^{0,\Sigma} \right\}. \quad (105)$$

Every object in this definition is constructed prospectively from $\Theta_\Sigma^{\text{syn}}$. Neither actual contact orbits nor exact invariant values occur. A total-graph failure contributes only when it belongs to an infinite target-retaining zero-cost path; naming the failure supplies no self-loop.

Definition II.2.14 (Finite rational-cylinder pruning transcript). Fix the countable basis $\mathcal{B}_\eta = (B_\ell)_{\ell \geq 1}$ of relatively open rational cylinders in $\mathfrak{A}_\eta$ generated by finitely many evaluation coordinates, and include cylinders whose closures refine any prescribed open neighborhood. Put

$$E_n := \mathfrak{A}_\eta \setminus Z_n. \quad (106)$$

A *pruning transcript of rank at most N* is a finite rooted forest whose nodes carry a rank $k \leq N$ and a cylinder B_ℓ , subject to the following rules.

- (T1) A rank-zero node carries a rational $q > 0$ such that $p \geq q$ on $\overline{B}_\ell$; equivalently, $\overline{B}_\ell \cap Z_0 = \emptyset$. This is a relation of the generated relaxed physical-price coordinate.

(T2) At a rank- $k + 1$ node, the compact successor image satisfies

$$\mathcal{R}_\eta(\overline{B}_\ell \cap Z_0) \subseteq \bigcup_{i=1}^m B_{\ell_i}, \quad (107)$$

where the finitely many children B_{ℓ_i} have rank at most k . The inclusion is read in the closed witness bundle $\mathfrak{W}_\eta$, so every operation index, source word, origin selector, and graph-boundary value used by the node is retained.

(T3) The root cylinders cover $\mathfrak{A}_\eta$.

All node labels and relations come from the fixed enumeration of the public evaluation coordinates, total graphs, price coordinate, and successor witnesses. Among all valid finite forests use the least code in the induced enumeration. The transcript is therefore an output of the continuum algebra, not additional PDE data.

Theorem II.2.15 (Canonical finite pruning or same-origin mechanism). *The elimination sets are open and satisfy*

$$E_{n+1} = E_n \cup \{w \in Z_n : \mathcal{R}_\eta(w) \cap Z_n = \emptyset\}. \quad (108)$$

Exactly one of the following holds.

(P1) *Some Z_N is empty. Then the least-coded finite pruning transcript of rank at most N exists and, by finite descent of node ranks, proves that no target-retaining zero-price successor path exists.*

(P2) *Every Z_n is nonempty. Then the compact inverse limit supplies an infinite same-origin zero-price path, and empirical shift averages supply its stationary law.*

Thus the regularity branch requires no externally supplied emptiness proof: finite closure has a generated finite transcript, while failure of every finite pruning attempt constructs the retained dynamical mechanism.

Proof. The set E_0 is open because Z_0 is closed. If E_n is open, then

$$\{w : \mathcal{R}_\eta(w) \cap Z_n \neq \emptyset\} = \pi_1(\mathcal{R}_\eta \cap (\mathfrak{A}_\eta \times Z_n))$$

is compact and hence closed. This proves (108) and openness of E_{n+1} .

Inductively, every point of E_n has a rational-cylinder neighborhood with a finite transcript of rank at most n . If $w \in E_0$, then $p(w) > 0$; lower semicontinuity gives a rational $q > 0$ and, by regularity of the compact metric space, a basis cylinder $B_\ell \ni w$ whose closure lies in $\{p > q\}$. This is a rank-zero node. For a new point of $E_{n+1} \setminus E_n$, its compact successor fiber lies in the open set E_n . Closedness of $\mathcal{R}_\eta$ in the compact product gives a smaller cylinder whose entire successor image lies in E_n ; compactness of that image and the induction hypothesis give the finite child cover in (107).

If $Z_N = \emptyset$, then $E_N = \mathfrak{A}_\eta$. Compactness gives a finite root cover by the preceding cylinder neighborhoods, and the fixed enumeration selects a least code. Along a zero-price successor path, every edge from a node of positive rank must enter a child of strictly smaller rank, while a rank-zero node contains no point of Z_0 ; such a path cannot exist. If no Z_N is empty, nested compactness gives a point of Z_∞^Σ and the fixed-point identity (104) recursively gives an infinite path. Theorem II.2.12 supplies the stationary law and preserves the common origin coordinate. $\square$

Definition II.2.16 (Generated zero-cost dynamics). Let ν be a stationary target-retaining successor law with zero mean physical expenditure. Let $\text{Conseq}(\mathfrak{P})$ be the least countable consequence algebra containing the public weak equation and native carrier identity and closed under rational

linear combination, localization by the generated test core, integration by parts already encoded in the weak formulation, polarization, adjoint formation, quotient descent, chart pullback, Jordan decomposition, completion of squares, composition of total graphs, and sourcewise graph closure. If a product, chain, trace, or covariance operation lacks an ordinary value, the algebra inserts its total-graph defect rather than the desired identity. This grammar is enumerated from $\text{Pre}(\text{Eq})$ and has no additional input slot.

Let $\mathcal{P}_{\text{gen}}$ be the countable family of all generated nonnegative functions q for which $\text{Conseq}(\mathfrak{P})$ contains a domination identity

$$p = q + r_q, \quad q \geq 0, \quad r_q \geq 0. \quad (109)$$

Include $q = p$ and $r_p = 0$. This definition takes the union of all valid positive descendants and does not require a choice of a maximal or unique decomposition. The family is generated by the fixed equation and carrier: mobility squares, weak energy defects, positive commutator squares, Fisher-information or Hessian squares, constraint defects, and target-ray defects occur only when their defining identity is already a relation in $\text{Conseq}(\mathfrak{P})$. No extra positive term is inserted.

The *saturation equations* are

$$q = 0 \quad (q \in \mathcal{P}_{\text{gen}}), \quad \dot{u} = J_{\text{Geom}} + J_{\text{Caus}} + J_{\text{Abs}} + J_{\text{Lift}} + J_{\text{Bdry}}, \quad (110)$$

together with the totalized constitutive, constraint, chart, and covariance relations. No tangency to a symmetry orbit is asserted. Equilibria, relative equilibria, periodic or recurrent conservative orbits, scale breathers, and persistent graph-boundary dynamics are retained whenever they solve these relations. If $\mathcal{F}_{\text{gen}}$ is the generated finite or countable family of partial relations used on the current branch, let $\partial_{\infty}F$ denote the recurrent graph-boundary values obtained from infinite witness-successor paths for F , rather than the entire graph boundary. Define the totalized saturation set

$$\mathcal{Z}_{\text{sat}}^{\text{tot}} := \left(\bigcap_{q \in \mathcal{P}_{\text{gen}}} \ker q \cap \{\dot{u} = J_{\text{tot}}\} \right) \sqcup \bigsqcup_{F \in \mathcal{F}_{\text{gen}}} \partial_{\infty}F. \quad (111)$$

Thus a failed relation enters the saturation set only through a recurrent zero-cost graph-boundary path. The definition imposes only identities already derived from the equation and does not convert stationarity of a law into stationarity of a state.

Theorem II.2.17 (Saturation-equation compiler). *Every generator of $\mathfrak{H}_{\text{succ}}^{\Sigma}$ is supported on the totalized solution set of (110) and on the greatest viable set Z_{∞}^{Σ} . Conversely, every stationary law on the prospective path space supported on this solution set generates a class in $\mathfrak{H}_{\text{succ}}^{\Sigma}$. Hence*

$$\mathfrak{H}_{\text{succ}}^{\Sigma} = \overline{\text{span}} \Pi_{\Sigma} \left(Z_{\infty}^{\Sigma} \cap \mathcal{Z}_{\text{sat}}^{\text{tot}} \right), \quad (112)$$

where the right side denotes the coordinate supports of stationary path laws on the displayed set. It is generated entirely by algebraic kernels, the closed successor graph, source quotients, and the descending viability operator.

Proof. For a stationary zero-expenditure law, $\int p \, d\nu = 0$, so $p = 0$ almost surely. For every $q \in \mathcal{P}_{\text{gen}}$, (109) gives $0 \leq q \leq p$, hence $q = 0$ almost surely on the ordinary branch. The full weak current identity remains one of the total graph coordinates, so its ordinary support solves $\dot{u} = J_{\text{tot}}$ without any symmetry-tangency inference. On a graph-boundary branch, membership in the path fixed point supplies the recurrent value in $\partial_{\infty}F$. This proves the first inclusion in (112). Conversely, a stationary path law in the displayed ordinary joint kernel has $p = \mathcal{P}_0 = 0$; a recurrent graph-boundary law has the same zero price by membership in Z_{∞}^{Σ} . Both are generators of the successor-saturation algebra. Taking target projections and closed spans proves equality. $\square$

Corollary II.2.18 (Complete zero-cost structural reduction). *The finite-depth recursion (103) has as its intersection the largest target-retaining zero-cost successor set. Its recurrent projection satisfies the exact alternative*

$$\mathfrak{H}_{\text{succ}}^\Sigma = 0 \iff \Pi_\Sigma \text{supp } \nu = 0 \text{ for every stationary law on } Z_\infty^\Sigma. \quad (113)$$

At each successor stage the compiler applies the five-source quotient to the full zero-cost weak equation and to every recurrent graph-boundary coordinate. A target-null output is removed at the next stage. Any nonzero fixed-point output is an explicit recurrent conservative, relative-equilibrium, profile, or graph-boundary mechanism compatible with singular contact. No compactness gap or separate rigidity datum appears in this reduction. Moreover, closure is operational: either some finite iterate Z_N is empty and carries the canonical finite pruning transcript, or the compact inverse-limit construction returns an infinite path and a stationary law on its shift orbit closure.

Theorem II.2.19 (Zero-mean successor reduction). *Let an infinite witness-indexed successor chain originate from the disjoint shell sequence of a finite-expenditure contact orbit. Then every coordinate of its diagonal path has relaxed original-event price zero, and its path closure carries a stationary successor law with zero mean physical expenditure. Consequently its target projection lies in $\mathfrak{H}_{\text{succ}}^\Sigma$.*

Proof. By countable additivity, the original prices $p_n = \mu_u(Q_n)$ are summable and hence tend to zero. Every successor uses a subsequence of these same indexed events. The diagonal selection in Lemma II.2.11 therefore approximates each path coordinate by records whose prices tend to zero. Lower semicontinuity and nonnegativity give price zero at every path coordinate.

Theorem II.2.12 gives a compact shift-invariant path closure. A weak limit of its empirical path measures is shift invariant and has zero mean price. Target retention and source ancestry are closed coordinates, so the law is one of the generators in (105). $\square$

Theorem II.2.20 (Estimate-free finite-budget contact routing). *Let a contact orbit with finite physical expenditure generate infinitely many pairwise disjoint target-retaining windows I_j . The canonical activity lift and cofinal tail shift generate an infinite path in Z_∞^Σ , and*

$$\Pi_\Sigma Z_\infty^\Sigma \neq 0, \quad \mathfrak{H}_{\text{succ}}^\Sigma \neq 0. \quad (114)$$

Thus finite price is completely reduced to an explicit recurrent zero-cost mechanism; it is not a terminal scalar branch or an outstanding analytic row. No lower price, compactness estimate in the original state topology, coercivity constant, or Liouville input is used: the only numerical fact is countable additivity of the equation-generated physical measure.

Proof. Countable additivity gives $\sum_j \mu_u(I_j) < \infty$, hence after a subsequence $\mu_u(I_j) \rightarrow 0$. Apply the compact activity lift. Its limit has relaxed price zero, unit target progress, and inherited source support. Apply each ordered total graph to cofinal tails. Ordinary and boundary values are both coordinates of the compact lift, and the tail shift supplies a successor at every stage. This gives an infinite target-retaining zero-cost path.

The descending viability recursion retains exactly the points admitting all finite continuations of such a path; compactness supplies an infinite path in their intersection. Hence the path lies in Z_∞^Σ . Theorem II.2.19 supplies a stationary zero-cost law. Its unit target coordinate makes its target projection nonzero, proving (114). $\square$

Theorem II.2.21 (Constructive prospective regularity alternative). *For a fixed canonical presentation, run the total local-solution and carrier graphs, source compiler, canonical shells, signed carrier compiler, activity lift, and zero-cost successor fixed point. Every target-visible local-solution or carrier defect is inserted into the same successor graph; it is not retained as a static obstruction cell. Define the generated regularity mechanism algebra*

$$\mathfrak{H}_{\text{out}}^{\Sigma} := \mathfrak{H}_{\text{succ}}^{\Sigma}, \quad (115)$$

where the right side is the prospective fixed-point algebra (105). The evaluator returns one of the following constructive outputs:

- (R1) *for some finite depth N , every target-active viability set is empty; the canonical rational-cylinder pruning transcript records the finite descent, and every ordinary maximal finite-expenditure branch is then global and avoids the complete continuation target;*
- (R2) *there is a stationary law ν on Z_{∞}^{Σ} with $\Pi_{\Sigma}(\pi_0)_{\#}\nu \neq 0$. Its support supplies the full zero-cost weak equation, source ancestry, target pairing, profile law, and a diagonal sequence of origin-event witnesses; for a contact-generated law these are original disjoint physical events.*
- (R3) *if the public local-solution or carrier graph has no ordinary value, its total-graph witness tree is pruned by the same recursion and returns either a finite violated public identity or the recurrent mechanism of (R2). It is never returned merely as an unevaluated defect.*

Every actual finite-expenditure contact orbit constructs an output of type (R2) with its own diagonal event witnesses. Consequently

$\mathfrak{H}_{\text{out}}^{\Sigma} = 0 \implies \text{every ordinary maximal branch with derived finite carrier is globally regular.}$

(116)

This is an output identity of the generated recursion. The regularity track forms the finite-depth fixed point; a closing branch carries the finite transcript of Theorem II.2.15, and every nonzero survivor has the constructive data listed in (R2).

Proof. Insert ordinary and boundary values of the total local-solution and carrier graphs into the activity lift. Compact finite-depth viability gives (R1), (R2), or the explicit finite identity failure in (R3). On an ordinary finite-expenditure branch, suppose a maximal orbit contacts the complete target. Canonical shells give the disjoint unit progress cascade. A divergent generated lower-price series contradicts countable additivity; the finite-price tail instead enters the successor graph. Theorem II.2.20 constructs a stationary law in Z_{∞}^{Σ} with nonzero target coordinate and original shell-event witnesses, which is (R2). Thus finite pruning excludes contact, and terminal coverage makes every ordinary maximal branch global. The contrapositive proves (116). $\square$

Proposition II.2.22 (Covariant entropy as a carrier correction). *Let a covariant entropy cocycle $\mathcal{M}$ satisfy the gluing law of Theorem II.10.2. If its local balance has the form*

$$\mathcal{M}(Sw) - \mathcal{M}(w) = F_{\text{sc}}(w) - D(w) - M(w) - R(w), \quad R \geq 0, \quad (117)$$

then $\Psi = -\mathcal{M}$ is the canonical carrier correction of Definition II.2.3, and

$$D + M - F_{\text{sc}} + \Psi - \Psi \circ S = R \geq 0. \quad (118)$$

Every invariant saturation measure is supported on $R = 0$. It is supported on $D = M = 0$ only when the generated carrier transcript supplies a nonnegative domination identities expressing the zero-price deficit with D and M as positive descendants. Thus a Perelman-type formula refines

the carrier cohomology and zero-price equations. Its equality models are exactly the ordinary or recurrent graph-boundary solutions of those generated equations; relative equilibria are one possible subclass, not an automatic conclusion.

Proof. Rearrange (117) to obtain (118). The entropy cocycle glues on one coherent orbit, so its increments telescope and define a carrier coboundary. For an invariant saturation measure the nonnegative right side has zero integral and therefore $R = 0$ almost surely. Any further vanishing follows only by applying the generated positive-descendant rule of Definition II.2.16; the full weak equation remains in the support coordinates. $\square$

Corollary II.2.23 (Gradient-kernel equality reduction). *Let Ω be connected and let the ordinary saturation equations on a normalized successor core contain*

$$D_m(V) := \int_{K_m} |\nabla V|^2 \, dy = 0, \quad m \geq 1, \quad (119)$$

for an increasing compact exhaustion $K_m \Subset \Omega$. Then every ordinary zero-expenditure stationary law is supported on spatially constant states. If the generated gauge quotient sends spatial constants to $\mathcal{N}_\Sigma$, its ordinary equality contribution to $\mathfrak{H}_{\text{succ}}^\Sigma$ vanishes. Every nonordinary value of the gradient, exhaustion, connected-core, or gauge relation remains a witness-successor and is reduced by $\widehat{\mathcal{R}}_\eta$.

Proof. For a zero-expenditure stationary law ν , nonnegativity and the saturation-equation compiler give $\int D_m(V) \, d\nu = 0$ for every m . Hence $D_m(V) = 0$ for ν -almost every state, simultaneously for all m . The exhaustion gives $\nabla V = 0$ almost everywhere on Ω . The weak fundamental theorem on connected domains makes V spatially constant. The target projection then vanishes by the stated gauge quotient. Totalization of each relation gives the last assertion. $\square$

II.3 Covariant action and the physical budget

The fixed-Caus formulation separates the current selected by the PDE from the metric used to measure irreversible expenditure. This separation is physical: the action below is the quantity appearing in the energy identity. The explicit inverse notation in this section is used only when the mobility has the Hilbert realization stated next. In a general Gelfand triple, the same identities are read at the level of the closed mobility form.

Definition II.3.1 (Mobility current space). Fix a state u , a real Hilbert tangent space $\mathcal{H}_u$, and a bounded self-adjoint nonnegative mobility $\mathbb{K}(u) \in \mathcal{L}(\mathcal{H}_u)$. Let $\mathbb{K}(u)^{-1/2}$ denote the Moore–Penrose inverse on $(\ker \mathbb{K}(u))^\perp$. The mobility current space is the Cameron–Martin space

$$\mathfrak{J}_u := \text{Ran } \mathbb{K}(u)^{1/2}$$

with inner product and norm

$$(J_1, J_2)_{u,*} := \langle \mathbb{K}(u)^{-1/2} J_1, \mathbb{K}(u)^{-1/2} J_2 \rangle_{\mathcal{H}_u}, \quad \|J\|_{u,*}^2 := \|\mathbb{K}(u)^{-1/2} J\|_{\mathcal{H}_u}^2.$$

Equivalently, this is the completion of $\text{Ran } \mathbb{K}(u)$ in the displayed norm. Covectors are represented in $\mathcal{H}_u$, with components in $\ker \mathbb{K}(u)$ identified when they act on currents. For a velocity v , put

$$W(u, v) := v + A_{\text{Caus}}(u) - f_{\text{Bdry}}(u).$$

The dissipation Lagrangian is

$$\mathcal{L}(u, v) := \begin{cases} \frac{1}{2} \|W(u, v)\|_{u,*}^2, & W(u, v) \in \mathfrak{J}_u, \\ +\infty, & \text{otherwise.} \end{cases} \quad (120)$$

Its physical action on an interval I is $\mathcal{A}_I(u) = \int_I 2\mathcal{L}(u, \dot{u}) \, dt$.

Definition II.3.2 (Covariant dissipation Hamiltonian). For $p \in \mathcal{H}_u$, define

$$\mathcal{H}_{\text{cov}}(u, p) := -\langle p, A_{\text{Caus}}(u) - f_{\text{Bdry}}(u) \rangle + \frac{1}{2} \|\mathbb{K}(u)^{1/2} p\|^2. \quad (121)$$

This is the Legendre–Fenchel dual of $v \mapsto \mathcal{L}(u, v)$.

Proposition II.3.3 (Fenchel equality and the selected current). *Whenever the displayed quantities are finite,*

$$\mathcal{L}(u, v) + \mathcal{H}_{\text{cov}}(u, p) - \langle p, v \rangle = \frac{1}{2} \|W(u, v) - \mathbb{K}(u)p\|_{u,*}^2 \geq 0. \quad (122)$$

Equality holds precisely when $W(u, v) = \mathbb{K}(u)p$ in the current quotient. For the fixed-Caus gradient flow (7), the selected velocity has

$$W(u, \dot{u}) = -\mathbb{K}(u)D\mathcal{E}(u),$$

so equality holds at $p = -D\mathcal{E}(u)$, modulo the mobility kernel.

Proof. Substitute (120) and (121); since $v = W - A_{\text{Caus}} + f_{\text{Bdry}}$, the linear terms combine to $-\langle p, W \rangle$. Completing the square in the $\mathbb{K}^{-1}$ -metric gives (122). The equality and selected-current statements follow directly. $\square$

Definition II.3.4 (Physical carrier balance). Let $u : [0, T_*) \rightarrow X$ be an ordinary maximal branch. Expand the native signed carrier graph of Definition I.1.7 and take the Jordan decomposition of its signed currents. Denote the resulting scalar by $\mathcal{K}_u$, the gross positive production by $\mathfrak{D}_u$, the returned internal feeding by Φ_u^+ , and true positive exterior input by R_u^+ . Its ordinary graph value is, for $0 \leq s < t < T_*$,

$$\mathcal{K}_u(t) + \mathfrak{D}_u((s, t]) \leq \mathcal{K}_u(s) + \Phi_u^+((s, t]) + R_u^+((s, t]). \quad (123)$$

Here all four quantities are generated from the one signed identity. The generated passivity radius is the least extended-real β for which the measure domination

$$\Phi_u^+ \leq \beta \mathfrak{D}_u. \quad (124)$$

holds. Whether $\beta \leq 1$, whether $\mathcal{K}_u$ has a lower bound, and whether the exterior input is finite are values of generated relations, not additional data. All terms are expressed in the physical clock. A normalized-clock identity is routed through the totalized clock-comparison relation in Proposition II.10.5.

Theorem II.3.5 (Carrier compiler for the single physical budget). *If the generated values satisfy $\beta \leq 1$, $\inf_t \mathcal{K}_u(t) = K_- > -\infty$, and $R_u^+([0, T_*)) < \infty$, then*

$$\mu_u := \mathfrak{D}_u - \Phi_u^+ \quad (125)$$

is a nonnegative countably additive physical expenditure measure and

$$\mu_u([0, T_*)) \leq \mathcal{K}_u(0) - K_- + R_u^+([0, T_*)) =: B_{\text{phys}} < \infty. \quad (126)$$

If $\beta < 1$, then

$$\mu_u \geq (1 - \beta)\mathfrak{D}_u. \quad (127)$$

For $\beta = 1$, μ_u is the exact flux-deficit measure. If any displayed relation has the opposite value, its total-graph witness is sent directly to the activity lift and successor recursion. No alternative expenditure or independent budget is appended later.

Proof. The domination $\Phi_u^+ \leq \mathfrak{D}_u$ makes the difference in (125) a nonnegative Radon measure. Rearrange (123) with $s = 0$, use $\mathcal{K}_u(t) \geq K_-$, and let $t \uparrow T_*$. Monotone convergence gives (126). Finally, $\Phi_u^+ \leq \beta\mathfrak{D}_u$ gives (127). $\square$

Definition II.3.6 (Reduced passivity radius). Let $\mathcal{P}_{\text{car}}^{\text{red}}$ be a normalized carrier profile space obtained after zero-descent-defect source reductions. Let $D(V) \geq 0$ and $\Phi_+(V) \geq 0$ be the descended gross-expenditure and feeding-flux densities in the same physical units. The reduced passivity radius is

$$\beta_*^{\text{red}} := \alpha_*(\Phi_+ \mid D) = \sup_{\substack{V \in \mathcal{P}_{\text{car}}^{\text{red}} \\ D(V) > 0}} \frac{\Phi_+(V)}{D(V)}, \quad (128)$$

with the null-mode convention of Definition II.1.6. A bound $\beta_*^{\text{red}} \leq 1$ supplies (124) whenever the carrier identity is local in the normalized profiles and its physical-clock pullback is proved.

Proposition II.3.7 (Quadratic passivity is a spectral invariant). *On the quadratic-form realization branch of the reduced Hilbert profile space,*

$$D(V) = \langle AV, V \rangle, \quad \Phi_+(V) = \max\{\langle BV, V \rangle, 0\},$$

where $A \geq 0$, $B = B^*$, all D -null target-visible feeding modes have been separated, and B is bounded in the A -form norm. Let $S_{\text{flux}} = A^{-1/2}BA^{-1/2}$ be the descended self-adjoint form operator on the completion modulo $\ker A$. Then

$$\beta_*^{\text{red}} = \max\{\sup \sigma(S_{\text{flux}}), 0\}. \quad (129)$$

Thus every multiplier or casewise proof of $\Phi_+ \leq D$ is, on this reduced quadratic interface, an upper bound for one spectral radius.

Proof. The A -form completion is a Hilbert space after quotienting by $\ker A$. The form induced by B is represented there by S_{flux} . Rayleigh–Ritz identifies the supremum of $\langle BV, V \rangle / D(V)$ with $\sup \sigma(S_{\text{flux}})$; taking the positive part gives (129). $\square$

II.4 Hamiltonian meanings and chart covariance

Definition II.4.1 (Geometric spectral Hamiltonian). Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric form on a real Hilbert space $\mathcal{H}$. Its geometric Hamiltonian is the unique self-adjoint operator $H \geq 0$ satisfying

$$\mathfrak{q}(u, v) = \langle H^{1/2}u, H^{1/2}v \rangle, \quad u, v \in D(\mathfrak{q}).$$

This Hamiltonian records only the **Geom** form; the fixed antisymmetric current remains in **Caus**.

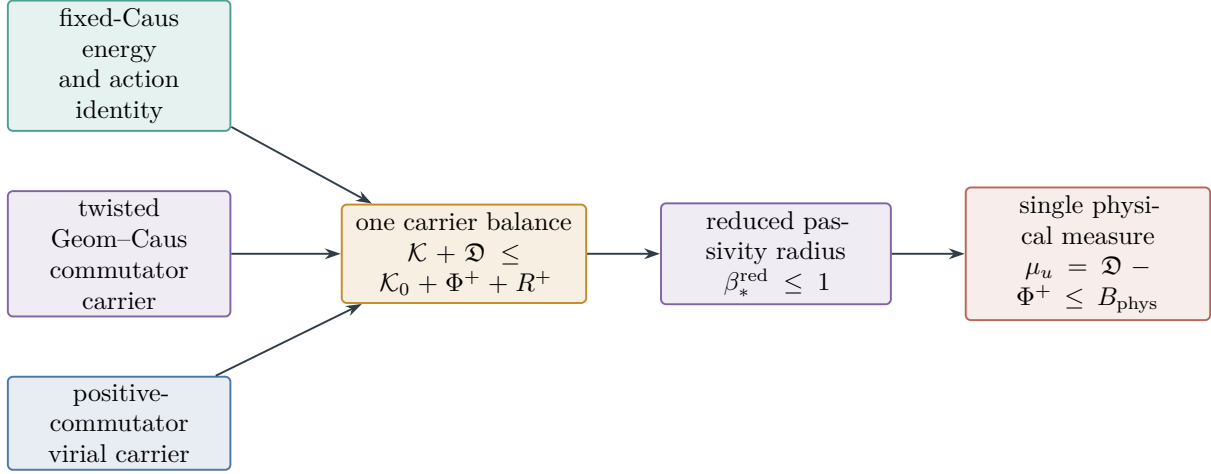


Figure 11: Carrier-to-budget compiler. Gradient-flow, hypocoercive, and positive-commutator identities are different ways to expose the same structural object: a lower-bounded carrier whose feeding flux is dominated by gross expenditure. The output is the sole physical budget measure.

Proposition II.4.2 (Separation of Hamiltonian meanings). *The following constructions require distinct data.*

- (i) A closed symmetric form determines the spectral Hamiltonian H of Definition II.4.1.
- (ii) A canonical conservative Hamiltonian vector field requires a Poisson tensor $\mathbb{J}$, a domain on which the Jacobi identity holds, and a functional $\mathcal{H}$; the field is $\mathbb{J}D\mathcal{H}$.
- (iii) A positive mobility and a covariant connection determine the Fenchel Hamiltonian $\mathcal{H}_{\text{cov}}$ of Definition II.3.2.

An antisymmetric Caus form alone supplies directed transport but does not provide the Poisson or Jacobi data in item (ii).

Proof. The representation theorem for closed nonnegative forms gives item (i) [5, Chapter VI]. Item (ii) is the definition of a Poisson Hamiltonian system; antisymmetry is only one of the Poisson requirements. The square completion in Proposition II.3.3 proves item (iii). $\square$

Theorem II.4.3 (Same-current covariance). *Let $v = \Phi_t(u)$ be a generated C^1 chart relation. Totalize its diffeomorphism and tangent-map relations. On the branch where $T_t = D\Phi_t(u) : \mathcal{H}_u \rightarrow \mathcal{H}_v$ is a bounded linear isomorphism with bounded inverse, with every expression on the right evaluated at $u = \Phi_t^{-1}(v)$, define*

$$\tilde{\mathbb{K}} = T_t \mathbb{K} T_t^*, \quad \tilde{A}_{\text{Caus}} = T_t A_{\text{Caus}}, \quad A_{\text{Lift}}^\Phi = -\partial_t \Phi_t, \quad \tilde{f}_{\text{Bdry}} = T_t f_{\text{Bdry}}. \quad (130)$$

Then

$$\dot{v} + \tilde{A}_{\text{Caus}}(v) + A_{\text{Lift}}^\Phi(v) - \tilde{f}_{\text{Bdry}}(v) = T_t(\dot{u} + A_{\text{Caus}}(u) - f_{\text{Bdry}}(u)). \quad (131)$$

The map T_t is an isometry between the corresponding mobility-current spaces equipped with their transported metrics, so the Lagrangian and physical action are invariant. The chart has Lift ancestry, and quotienting any declared symmetry additionally has Abs ancestry.

Proof. Differentiate $v = \Phi_t(u)$ and substitute (130); the $\partial_t \Phi_t$ terms cancel, giving (131). For $\tilde{J} = T_t J$, the variational identity

$$\|J\|_{\mathbb{K}^{-1}}^2 = \sup_{p \in \mathcal{H}_u} \{2\langle p, J \rangle - \langle p, \mathbb{K}p \rangle\}$$

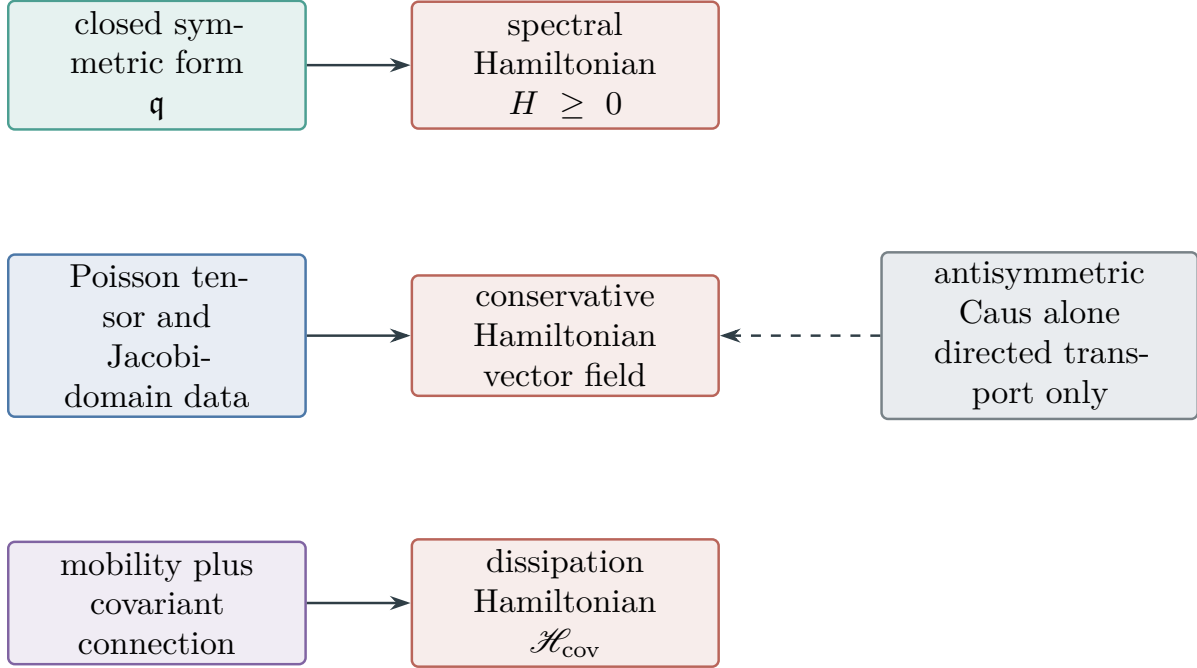


Figure 12: Three noninterchangeable Hamiltonian constructions. The dashed line marks a missing implication: skew transport alone does not establish a Poisson structure.

and the bijective change of covector $p = T_t^* \tilde{p}$ give $\|\tilde{J}\|_{\tilde{\mathbb{K}}^{-1}} = \|J\|_{\mathbb{K}^{-1}}$. Integrating proves action invariance. The source labels follow from Definition I.2.1. $\square$

Proposition II.4.4 (Chart covariance of the dissipation Hamiltonian). *On the ordinary covariance-graph branch of Theorem II.4.3, let $p = T_t^* \tilde{p}$, and define the transformed Hamiltonian using the total connection $\tilde{A}_{\text{Caus}} + A_{\text{Lift}}^\Phi$. Then*

$$\tilde{\mathcal{H}}_{\text{cov}}(t, v, \tilde{p}) = \mathcal{H}_{\text{cov}}(u, p) + \langle \tilde{p}, \partial_t \Phi_t(u) \rangle. \quad (132)$$

The additional term is the time-dependent Lift connection. It is not a new physical Caus term.

Proof. Insert (130) into (121). The quadratic term is invariant because $\tilde{\mathbb{K}} = T_t \mathbb{K} T_t^*$, while

$$-\langle \tilde{p}, T_t(A_{\text{Caus}} - f_{\text{Bdry}}) - \partial_t \Phi_t \rangle = -\langle p, A_{\text{Caus}} - f_{\text{Bdry}} \rangle + \langle \tilde{p}, \partial_t \Phi_t \rangle.$$

$\square$

Remark II.4.5 (A change of physical clock is additional data). Theorem II.4.3 keeps the physical time variable fixed. If a blow-up chart also introduces $\tau = \theta(t)$, the current, mobility, and action density must be transformed with the exact Jacobian $dt/d\tau$. Action invariance, or comparison with the original physical measure, is not automatic; the required one-sided comparison is the content of Proposition II.10.5.

II.5 Killed resolvents, resistance, and target realization

Definition II.5.1 (Killed geometric Hamiltonian). Let $\mathbf{q}$ be a densely defined closed nonnegative symmetric Markovian form and let Σ be quasi-closed for a regular or quasi-regular realization.

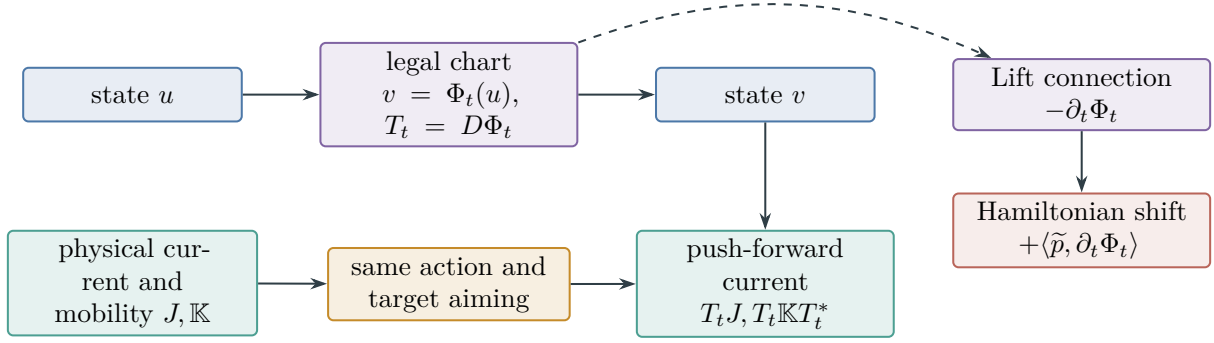


Figure 13: Same-current chart covariance. A chart transports the physical current and metric; its time dependence is a Lift connection with the exact Hamiltonian shift shown.

The killed form is the restriction

$$D(\mathbf{q}_\Sigma) = \{u \in D(\mathbf{q}) : \tilde{u} = 0 \text{ q.e. on } \Sigma\}, \quad \mathbf{q}_\Sigma = \mathbf{q}|_{D(\mathbf{q}_\Sigma)},$$

and its self-adjoint operator, when the restricted form is closed, is $H_\Sigma \geq 0$. Adding an admitted nonnegative killing potential instead defines a *soft-killed* form. It is a separate construction unless convergence to the hard restriction is proved.

Proposition II.5.2 (Existence of the killed Hamiltonian). *If $D(\mathbf{q}_\Sigma)$ is closed in the form norm, then $\mathbf{q}_\Sigma$ is closed and determines a unique self-adjoint $H_\Sigma \geq 0$.*

Proof. A form-norm closed subspace of the domain of a closed form is complete in that norm. The representation theorem for closed nonnegative forms gives H_Σ . $\square$

Proposition II.5.3 (Committor is the canonical binary-contact quotient). *Let A and B be disjoint quasi-closed sets for an associated right process, and let*

$$h(x) := \mathbb{P}_x\{\tau_A < \tau_B\}$$

be its quasi-continuous committor. Define $q_h(x) = h(x)$ outside the exceptional set. If a quotient $q : X \rightarrow Q$ preserves this binary contact law in the precise sense that $h = \bar{h} \circ q$ quasi-everywhere for some $\bar{h} : Q \rightarrow [0, 1]$, then

$$q_h = \bar{h} \circ q \quad \text{quasi-everywhere.} \quad (133)$$

Hence q_h is the coarsest quotient preserving the scalar event $\{\tau_A < \tau_B\}$: every other such quotient is finer than its level-set partition. This assertion concerns the binary hitting probability, not the full hitting-time distribution.

Proof. The preservation branch is exactly (133). Therefore $q(x) = q(y)$ implies $h(x) = h(y)$, so every fiber of q lies in a fiber of q_h . Conversely, q_h itself preserves the event by the defining identity for h . $\square$

Theorem II.5.4 (Killed-resolvent stopwatch). *Let L_Σ generate a killed semigroup P_t^Σ with growth bound ω . Let ℓ_0 be an initial functional and let k_Σ be a terminal-flux vector such that*

$$\mathbb{P}_{\ell_0}\{\tau_\Sigma \in dt, \tau_\Sigma < \infty\} = \langle \ell_0, P_t^\Sigma k_\Sigma \rangle dt \quad (134)$$

as finite measures. Then, for $\lambda > \omega$,

$$\mathbb{E}_{\ell_0}[e^{-\lambda\tau_\Sigma}; \tau_\Sigma < \infty] = \langle \ell_0, (\lambda - L_\Sigma)^{-1} k_\Sigma \rangle. \quad (135)$$

Without (134), the right side remains a cross-resolvent pairing and has no automatic first-contact interpretation.

Proof. Integrate (134) against $e^{-\lambda t}$. The Bochner Laplace transform of the semigroup is $\int_0^\infty e^{-\lambda t} P_t^\Sigma dt = (\lambda - L_\Sigma)^{-1}$, which proves the identity. $\square$

Theorem II.5.5 (Resolvent Fenchel dual and resistance). *Let $K \geq 0$ be self-adjoint with closed form $\mathfrak{q}$, and let $e \in \mathcal{H}$. For $\lambda > 0$,*

$$\langle e, (K + \lambda)^{-1} e \rangle = \sup_{v \in D(\mathfrak{q})} \left\{ 2 \operatorname{Re} \langle e, v \rangle - \mathfrak{q}(v, v) - \lambda \|v\|^2 \right\}. \quad (136)$$

If $P_{\ker K} e = 0$, then monotone convergence gives

$$\lim_{\lambda \downarrow 0} \langle e, (K + \lambda)^{-1} e \rangle = \int_{(0, \infty)} s^{-1} d\nu_e(s) =: \mathcal{R}_K(e) \in [0, \infty]. \quad (137)$$

Finiteness of $\mathcal{R}_K(e)$ is equivalent to boundedness of $v \mapsto \langle e, v \rangle$ on the energy space with norm $\mathfrak{q}(v, v)^{1/2}$; in that case it has a unique weak energy representer modulo $\ker K$.

Proof. For $R_\lambda = (K + \lambda)^{-1}$, completing the square in the form norm of $K + \lambda$ shows that the supremum is attained at $v = R_\lambda e$ and equals $\langle e, R_\lambda e \rangle$. The spectral theorem gives $\langle e, R_\lambda e \rangle = \int (\lambda + s)^{-1} d\nu_e(s)$. Monotone convergence proves (137). The last equivalence is the Riesz theorem on the completion of $D(\mathfrak{q})/\ker K$. $\square$

Theorem II.5.6 (Harmonic defect equals resistance error). *Let $K \geq 0$ be self-adjoint with form $\mathfrak{q}$, and let*

$$\mathcal{E}_K := \overline{D(\mathfrak{q})/\ker K}^\mathfrak{q}$$

be its energy space. For $\ell \in \mathcal{E}_K^$, totalize the weak solution relation $\mathfrak{q}(x, \phi) = \ell(\phi)$ modulo $\ker K$. On its ordinary branch let x denote the solution and let $v \in D(\mathfrak{q})$ be a trial state with residual*

$$e_v(\phi) := \mathfrak{q}(v, \phi) - \ell(\phi). \quad (138)$$

Then

$$\|e_v\|_{\mathcal{E}_K^*}^2 = \|[v - x]\|_{\mathcal{E}_K}^2 = \mathfrak{q}(v - x, v - x). \quad (139)$$

If e_v is represented by a vector $e \perp \ker K$ in $\mathcal{H}$, then

$$\|e_v\|_{\mathcal{E}_K^*}^2 = \langle e, K^+ e \rangle = \mathcal{R}_K(e). \quad (140)$$

Thus an approximate committor, elliptic corrector, or resolvent state is controlled by one resistance invariant of its harmonic defect; no $\ker K$ component of the trial state enters the error. When that kernel is a closed gauge or target-null space, the corresponding discarded motion requires no estimate.

Proof. The weak equation and (138) give

$$e_v(\phi) = \mathfrak{q}(v - x, \phi).$$

The Riesz map of $\mathcal{E}_K$ is an isometry onto its dual, proving (139). When the residual is represented by e , the spectral theorem identifies its squared energy-dual norm with $\int_{(0, \infty)} s^{-1} d\nu_e(s)$, which is (140); the middle pairing is understood as this spectral quadratic-form value. $\square$

Corollary II.5.7 (Defect-aligned drift bound). *If $e \perp \ker K$ and $\mathcal{R}_K(e) < \infty$, then for every $w \in D(\mathfrak{q})$,*

$$|\langle e, w \rangle| \leq \mathcal{R}_K(e)^{1/2} \mathfrak{q}(w, w)^{1/2}. \quad (141)$$

Only the component of w paired with the routable defect is charged.

Proof. Use the energy representer from Theorem II.5.5 and apply Cauchy–Schwarz in $\mathcal{E}_K$. $\square$

Proposition II.5.8 (Routable and zero-mode resistance split). *For arbitrary $e \in \mathcal{H}$, write*

$$e = e_0 + e_\perp, \quad e_0 = P_{\ker K} e, \quad e_\perp = P_{(\ker K)^\perp} e.$$

Then

$$\langle e, (K + \lambda)^{-1} e \rangle = \frac{\|e_0\|^2}{\lambda} + \int_{(0, \infty)} \frac{1}{s + \lambda} d\nu_{e_\perp}(s). \quad (142)$$

Consequently the zero-discount resistance is finite precisely when $e_0 = 0$ and $\int_{(0, \infty)} s^{-1} d\nu_{e_\perp}(s) < \infty$. A nonzero target-visible zero mode is an inherited Geom/Caus zero-frequency occurrence, not a residual coordinate. It is removed only when it belongs to a declared gauge or to $\mathcal{N}_\Sigma$; otherwise a separate carrier or price row evaluates it.

Proof. The spectral projection of K at 0 contributes $\lambda^{-1} \|e_0\|^2$; the remaining spectral measure is supported on $(0, \infty)$, which gives (142). Monotone convergence yields the finiteness criterion. The source statement follows from Theorems I.3.4 and I.4.5. $\square$

Proposition II.5.9 (Capacity and committor realization). *Let A and B be disjoint quasi-closed sets for a regular or quasi-regular Dirichlet form. The condenser capacity*

$$\text{Cap}(A, B) = \inf\{\mathfrak{q}(v, v) : v \in D(\mathfrak{q}), \tilde{v} = 1 \text{ on } A, \tilde{v} = 0 \text{ on } B\}$$

is an analytic Geom invariant. On the zero realization-defect branch, if the minimizer is unique and the associated right process realizes the weak Dirichlet problem, then its quasi-continuous version is the committor $x \mapsto \mathbb{P}_x\{\tau_A < \tau_B\}$ quasi-everywhere. The probabilistic conclusion is the ordinary value of the right-process realization row; positive capacity alone is only its prospective value [3, 7].

Theorem II.5.10 (Polar-target capacity criterion). *Let $\mathfrak{q}$ be a symmetric quasi-regular Dirichlet form with associated right process, and let Σ be quasi-closed. Write*

$$\text{Cap}_{\mathfrak{q}}(\Sigma) := \inf\{\mathfrak{q}(v, v) + \|v\|_2^2 : v \in D(\mathfrak{q}), \tilde{v} \geq 1 \text{ q.e. on a neighborhood of } \Sigma\}.$$

If $\text{Cap}_{\mathfrak{q}}(\Sigma) = 0$, then Σ is polar:

$$\mathbb{P}_x\{\tau_\Sigma < \infty\} = 0 \quad \text{for quasi-every } x. \quad (143)$$

This is a direct Geom nonreachability criterion for the Markov realization and requires no cascade argument. Positive capacity selects the complementary row but supplies no hitting lower bound without a starting-law or nonpolarity statement.

Proof. For a quasi-regular Dirichlet form, sets of zero 1-capacity are properly exceptional up to a capacity-zero enlargement. The associated right process avoids every properly exceptional set for quasi-every starting point, which gives (143) [3, 7]. $\square$

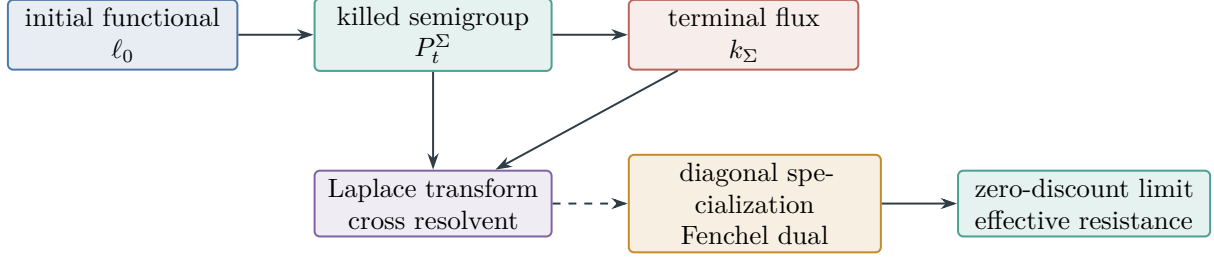


Figure 14: Killed arrival and resistance are related but distinct. Arrival is a cross pairing that requires a flux realization. The Fenchel and resistance identities concern a diagonal self-adjoint specialization.

Proposition II.5.11 (Quotient pullback of zero capacity). *Let $q : X \rightarrow Q$ identify a target $\Sigma = q^{-1}(\Sigma_Q)$ quasi-everywhere, and let $\mathfrak{q}_X, \mathfrak{q}_Q$ be symmetric quasi-regular Dirichlet forms. Totalize the lift relation which, for every admissible quotient potential f with $\tilde{f} \geq 1$ near Σ_Q , there is a lifted potential ψ_f with $\tilde{\psi}_f \geq 1$ near Σ and*

$$(\mathfrak{q}_X)_1(\psi_f, \psi_f) \leq C(\mathfrak{q}_Q)_1(f, f) + \varepsilon(f). \quad (144)$$

If a minimizing family for $\text{Cap}_{\mathfrak{q}_Q}(\Sigma_Q)$ can be chosen with $\varepsilon(f) \rightarrow 0$, then

$$\text{Cap}_{\mathfrak{q}_X}(\Sigma) \leq C \text{Cap}_{\mathfrak{q}_Q}(\Sigma_Q). \quad (145)$$

In particular, zero quotient capacity pulls back to zero capacity on X .

Proof. Every ψ_f is an admissible competitor for the capacity of Σ . Apply (144) along the stated minimizing family and take the liminf. $\square$

II.6 Full-generator response and physical aiming

Definition II.6.1 (Target-active full resolvent). Let $L = -H + A_{\text{Caus}} + B_{\text{Bdry}}$ be a closed generator with a declared resolvent point λ . For closed mathematical input and target-output subspaces $\mathcal{J}$ and $\mathcal{O}_\Sigma$, with bounded projections $P_{\mathcal{J}}$ and $P_{\mathcal{O}}$, define

$$\Psi_\Sigma(\lambda; L) := \|P_{\mathcal{O}}(\lambda - L)^{-1}P_{\mathcal{J}}\|. \quad (146)$$

This quantity measures nonnormal amplification visible to the fixed target interface. It evaluates the existing Geom–Caus–Bdry generator, not an additional source.

Proposition II.6.2 (Nonnormal source invariance). *Let L_0 be closed on a Banach space and let every K^c be bounded. Put $L_\theta = L_0 + \sum_c \theta_c K^c$ on $D(L_0)$. On the branch $\lambda \in \rho(L_\theta)$ locally in θ , this is the open ordinary value of the totalized resolvent relation; on it the resolvent is norm differentiable and*

$$\partial_{\theta_c}(\lambda - L_\theta)^{-1} = (\lambda - L_\theta)^{-1}K^c(\lambda - L_\theta)^{-1}. \quad (147)$$

Thus every first variation of target-active response is factored through the same source term K^c . Large pseudospectral response can expose a large Geom–Caus interaction, but it creates no new nonnormal source.

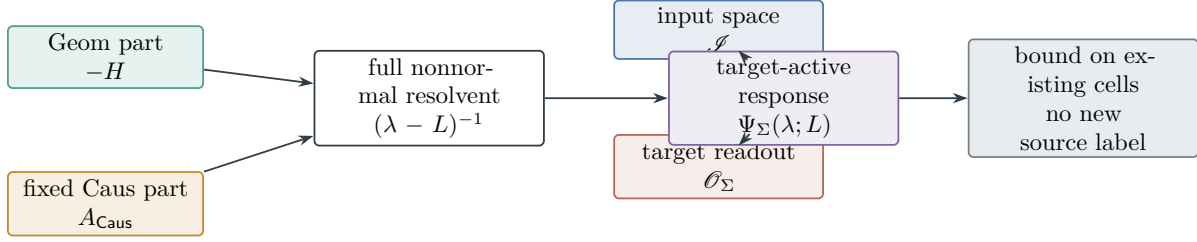


Figure 15: Full nonnormal response retains both Geom and fixed Caus. Input and target compression select a response coefficient; they do not change the operator’s source ancestry.

Proof. Differentiate $(\lambda - L_\theta)(\lambda - L_\theta)^{-1} = I$ on the common domain and solve for the derivative. Bounded input/output compression preserves the factorization. $\square$

Definition II.6.3 (Target-gradient ray). Let F be a differentiable target shell. Totalize membership of its mobility gradient

$$G_F := \mathbb{K}(u)DF(u)$$

in $\mathfrak{J}_u$. On the membership branch, for a physical current $J \in \mathfrak{J}_u$, define its positive target power and aiming contribution by

$$P_F(J) := (J, G_F)_{u,*}, \quad \text{Align}_F(J) := \frac{(P_F(J)_+)^2}{\|G_F\|_{u,*}^2}, \quad (148)$$

with value zero when $G_F = 0$. By the Cameron–Martin pairing,

$$P_F(J) = \langle DF(u), J \rangle$$

for every $J \in \mathfrak{J}_u$; components of DF in $\ker \mathbb{K}(u)$ annihilate that current space.

Theorem II.6.4 (Projection onto the target ray). *For every $J, G \in \mathfrak{J}_u$,*

$$\|J\|_{u,*}^2 = \frac{((J, G)_{u,*})_+^2}{\|G\|_{u,*}^2} + \text{dist}_{u,*}^2(J, \mathbb{R}_+G), \quad (149)$$

with the first term zero when $G = 0$. Hence

$$(P_F(J)_+)^2 \leq \|J\|_{u,*}^2 \|G_F\|_{u,*}^2. \quad (150)$$

Proof. The orthogonal projection of J onto the closed ray $\mathbb{R}_+G$ has coefficient $((J, G)_{u,*})_+ / \|G\|_{u,*}^2$. Pythagoras gives (149); dropping the nonnegative distance gives (150). $\square$

Proposition II.6.5 (Integrated action–aiming bound). *Let I be a pre-contact interval. On the finite ordinary branch of the two totalized squared-norm integrals below,*

$$\int_I P_F(J(t))_+ dt \leq \left(\int_I \|J(t)\|_{u(t),*}^2 dt \right)^{1/2} \left(\int_I \|G_F(u(t))\|_{u(t),*}^2 dt \right)^{1/2}. \quad (151)$$

For the selected Geom current of Definition I.1.9, the square of the first factor is the physical action and is bounded by the right-hand side of (10).

Proof. Apply (150) and then Cauchy–Schwarz in time. The final assertion is Proposition I.1.10. $\square$

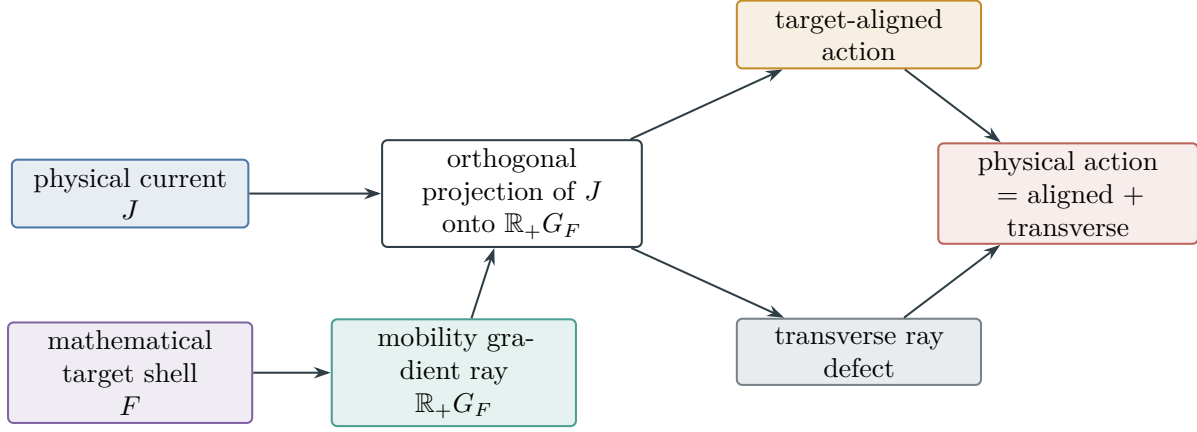


Figure 16: Physical aiming geometry. The target shell is a mathematical test of the already selected current. Projection splits the physical action into target-aligned and transverse expenditure.

II.7 Degenerate geometry and commutator-generated carriers

Direct resistance controls only the positive spectral subspace of the Geom operator. Fixed Caus transport may nevertheless move a Geom-null direction into a dissipative direction. The correct reduced invariant is then a positive commutator Gramian together with a carrier identity realizing that Gramian along the actual evolution.

Definition II.7.1 (Degenerate Geom–Caus core). Let $\mathcal{H}$ be a complex Hilbert space and

$$L = -S + A, \quad S = S^* \geq 0, \quad A^* = -A,$$

on a common invariant core $\mathcal{D}$. Let $\mathcal{G} \subseteq \ker S$ be a closed A -invariant gauge subspace, and write $P_{\mathcal{G}}^\perp$ for its orthogonal complement. Put $C_0 = S^{1/2}$. Totalize the iterated commutator relations on $\mathcal{D}$; their ordinary graph branch sets

$$C_{k+1} = [A, C_k] \quad \text{on } \mathcal{D}.$$

For fixed K and $c_k > 0$, totalize the quadratic form

$$q_K(v) := \sum_{k=0}^K c_k \|C_k v\|^2. \quad (152)$$

On its closable graph branch its positive self-adjoint operator is denoted by Q_K , and its reduced bracket gap is

$$\gamma_K := \inf_{\substack{v \in D(q_K) \cap \mathcal{G}^\perp \\ \|v\|=1}} q_K(v). \quad (153)$$

Failure of a domain, closure, or gauge relation is its totalized graph defect; algebraic commutator span alone does not select the closing branch.

Proposition II.7.2 (Bracket resistance invariant). *If $\gamma_K > 0$, then $Q_K \geq \gamma_K I$ on $\mathcal{G}^\perp$, and for every $e \in \mathcal{G}^\perp$,*

$$\mathcal{R}_K^{\text{br}}(e) := \langle e, Q_K^{-1} e \rangle \leq \gamma_K^{-1} \|e\|^2. \quad (154)$$

The quotient kernel

$$\mathfrak{h}_K := \mathcal{G}^\perp \cap \ker Q_K$$

is the structural rank deficit when $\gamma_K = 0$. Every target-visible element of $\mathfrak{h}_K$ retains its Geom–Caus ancestry and remains an existing source occurrence for the routing, carrier, and price rows.

Proof. The first assertion is the definition of the form lower bound in (153); functional calculus gives (154). The kernel is the intersection of the kernels of the closed commutator forms in (152). Source preservation and the target-null residual theorem give the final assertion. $\square$

Definition II.7.3 (Twisted-carrier realization evaluator). A bracket Gramian Q_K has a twisted-carrier realization when there is a bounded self-adjoint operator M on $\mathcal{G}^\perp$, constants $0 < m_- \leq m_+ < \infty$ and $c_M > 0$, and a common evolution core on which

$$m_- \|v\|^2 \leq \langle Mv, v \rangle \leq m_+ \|v\|^2, \quad (155)$$

$$2 \operatorname{Re} \langle Mv, Lv \rangle \leq -c_M q_K(v). \quad (156)$$

For a forced equation, the positive part of $2 \operatorname{Re} \langle Mu, f_{\text{Bdry}} \rangle dt$ is included in the input measure R_u^+ .

Theorem II.7.4 (Hypocoercive carrier produces physical expenditure). *On the zero graph-defect branch of Definition II.7.3, the evolution and carrier chain-rule relations for an ordinary maximal branch have ordinary values $\dot{u} = Lu + f_{\text{Bdry}}$ and $d\langle Mu, u \rangle / dt$. On this branch,*

$$\mathcal{K}_u(t) := \langle Mu(t), u(t) \rangle, \quad \mathfrak{D}_u(dt) := c_M q_K(u(t)) dt, \quad \Phi_u^+ = 0$$

satisfy the carrier balance of Definition II.3.4. Hence

$$\mu_u([0, T_*)) \leq \langle Mu_0, u_0 \rangle + R_u^+([0, T_*)) \leq m_+ \|u_0\|^2 + R_u^+([0, T_*)) \quad (157)$$

after subtracting any known lower carrier level. If $\gamma_K > 0$, this expenditure controls every nongauge direction in $\mathcal{G}^\perp$. The commutator calculation is therefore a structural compiler for a physical budget in a partially dissipative system [10].

Proof. Differentiate $\langle Mu, u \rangle$ and use (156); the positive forcing pairing gives the declared input measure. Apply Theorem II.3.5. When $\gamma_K > 0$, $q_K(u) \geq \gamma_K \|P_{\mathcal{G}}^\perp u\|^2$. $\square$

Definition II.7.5 (Positive-commutator carrier). Let the evolution be $\dot{u} = -iHu$, with $H = H^*$, and let $P = \mathbf{1}_\Delta(H)$ be a reducing spectral projection. A self-adjoint conjugate operator C supplies a positive-commutator carrier when the form

$$B := i[H, C]$$

is well defined and nonnegative on a common invariant core in $P\mathcal{H}$, and the expectation $\langle u(t), Cu(t) \rangle$ has a finite upper bound C_+ along the orbit. The strict Mourre branch has $PBP \geq \theta P$ modulo the declared gauge for some $\theta > 0$.

Theorem II.7.6 (Conservative carrier produces physical expenditure). *Under Definition II.7.5, let $u_0 \in P\mathcal{H}$. On the zero chain-defect branch, the totalized commutator relation gives*

$$\frac{d}{dt} \langle u(t), Cu(t) \rangle = \langle u(t), Bu(t) \rangle. \quad (158)$$

Consequently

$$\mu_u(Q) := \int_Q \langle u(t), Bu(t) \rangle dt, \quad \mu_u([0, T_*)) \leq C_+ - \langle u_0, Cu_0 \rangle. \quad (159)$$

When $PBP \geq \theta P$, this budget controls $\theta \int_0^{T_*} \|Pu(t)\|^2 dt$. Thus a virial, Morawetz, or Mourre identity contributes to the same physical-budget calculus whenever its carrier variation is finite [4].

Proof. Differentiate the expectation on the common core to obtain (158). Nonnegativity of B makes the right side a measure density. Integrate and use the upper carrier bound. The strict estimate follows from $PBP \geq \theta P$. On the complementary branch the distributional difference between the two sides of (158) is the retained $\mathfrak{H}_{\text{chain}}^\Sigma$ -class, so the theorem does not presume the chain rule in the fast track. $\square$

Corollary II.7.7 (Complete routing alternatives for a zero mode). *Let e_0 be the zero mode in Proposition II.5.8. Direct resistance controls $e_\perp$. A target-visible e_0 may be controlled by a twisted-carrier evaluator or a positive-commutator carrier, with its price charged to the same μ_u . The complementary values of both evaluators select the harmonic nonroutable branch of Theorem II.1.16; it remains a Geom-Caus source class and does not move to J_{res} .*

II.8 Compactness and rigidity as a price compiler

Compactness is not an additional continuum source and is not part of the definition of the residual. Its role is quantitative: it converts a qualitative zero-price rigidity statement into a uniform positive price on a closed class of target-approaching windows. The topology used in this conversion must be generated from seminorms or graph coordinates derived from the fixed PDE. The following lemma is one method for evaluating δ_* ; its tuple is not input to the fast track. Each failed clause is totalized into the boundary, chain, or price row.

Definition II.8.1 (Canonical compactness-rigidity row). Fix a physical ceiling B_{phys} , and let $\mathcal{W}_{B_{\text{phys}}}$ be the set of every normalized pre-contact window generated by the equation's Abs/Lift grammar. Inside the canonical evaluation compactification X_{eval} , define

$$K_{B_{\text{phys}}} := \overline{\mathcal{W}_{B_{\text{phys}}}}, \quad B_\Sigma := \overline{\mathcal{W}_{B_{\text{phys}}, \Sigma}}, \quad (160)$$

where $\mathcal{W}_{B_{\text{phys}}, \Sigma}$ consists of the generated windows that carry a normalized target-shell crossing. Thus $K_{B_{\text{phys}}}$ is compact and B_Σ is closed by construction; neither is an analytic premise.

Let d_0 be the nonnegative window expenditure read from the fixed physical balance. Its canonical lower-semicontinuous relaxation is

$$\bar{d}(w) := \sup_{U \ni w} \inf \{d_0(v) : v \in U \cap \mathcal{W}_{B_{\text{phys}}}\} \in [0, \infty], \quad (161)$$

where U ranges over neighborhoods of w , and the infimum of an empty set is $+\infty$. Equivalently, this is the largest lower-semicontinuous functional whose restriction to $\mathcal{W}_{B_{\text{phys}}}$ is bounded above by d_0 . The compactness-rigidity value is the exact extended number

$$\delta_{\text{rig}}(B_{\text{phys}}) := \inf_{w \in B_\Sigma} \bar{d}(w), \quad (162)$$

with value $+\infty$ when $B_\Sigma = \emptyset$. The branches $\delta_{\text{rig}} > 0$ and $\delta_{\text{rig}} = 0$ are exhaustive; the latter is the closed zero-price target-approach class. Every topology, normalization, target class, and price in this row is therefore generated from the fixed PDE, target, and physical balance.

Definition II.8.2 (Canonical physical-transfer radius). For each generated event Q , write $w_Q = \mathcal{N}_{Qu}$ and let μ_u be the orbit's physical expenditure measure. Define

$$c_{\text{pay}}^*(B_{\text{phys}}) := \inf_{\substack{Q: w_Q \in B_{\Sigma} \\ 0 < \bar{d}(w_Q) < \infty}} \frac{\mu_u(Q)}{\bar{d}(w_Q)} \in [0, \infty], \quad (163)$$

using $+\infty$ for an empty indexing family. A window with $\bar{d}(w_Q) = \infty$ and finite $\mu_u(Q)$, or a failure of the normalization/pullback relation, is retained by the price graph-defect row. For every ordinary finite-price window,

$$\mu_u(Q) \geq c_{\text{pay}}^*(B_{\text{phys}}) \bar{d}(w_Q). \quad (164)$$

Consequently the physical-transfer question has the exact complementary values $c_{\text{pay}}^* > 0$ and $c_{\text{pay}}^* = 0$, together with its typed graph-defect value. No comparison constant is supplied as data.

Lemma II.8.3 (Compactness–rigidity row evaluation). *The value in (162) is positive exactly when the canonical closed target-approach class contains no sequence with relaxed physical price tending to zero. If $B_{\Sigma} \neq \emptyset$, the infimum is attained.*

Proof. The set B_{Σ} is compact and $\bar{d}$ is lower semicontinuous by relaxation. Hence the infimum is attained. Its value is zero exactly when a minimizing sequence has price tending to zero; compactness then supplies a convergent subsequence in B_{Σ} , and lower semicontinuity identifies the zero-price branch. The converse follows from the constant sequence at a zero-price minimizer. $\square$

Theorem II.8.4 (Uniform physical price branch). *On the invariant branch $c_{\text{pay}}^*(B_{\text{phys}}) > 0$ and $\delta_{\text{rig}}(B_{\text{phys}}) > 0$, every ordinary generated event Q whose normalized window belongs to B_{Σ} satisfies*

$$\mu_u(Q) \geq \delta_{\text{phys}}(B_{\text{phys}}), \quad \delta_{\text{phys}}(B_{\text{phys}}) := c_{\text{pay}}^*(B_{\text{phys}}) \delta_{\text{rig}}(B_{\text{phys}}) > 0. \quad (165)$$

On either zero branch, the corresponding zero-transfer or zero-price sequence is retained in the same source/mode cell.

Proof. Apply (164) and then (162). The complementary statements follow from the two infimum definitions and sourcewise closure. $\square$

Theorem II.8.5 (Borderline passivity price compiler). *On the invariant branch $\beta_*^{\text{red}} = 1$ in Definition II.3.6, define the normalized carrier deficit*

$$\Delta_{\text{car}}(V) := D(V) - \Phi_+(V) \geq 0. \quad (166)$$

Let $\bar{\Delta}_{\text{car}}$ be its canonical relaxation (161) on the generated compact sets (160), and define

$$\delta_{\text{car}}^*(B_{\text{phys}}) := \inf_{V \in B_{\Sigma}} \bar{\Delta}_{\text{car}}(V). \quad (167)$$

Define the clock-transfer radius by the same infimum construction as (163), with $\bar{d} = \bar{\Delta}_{\text{car}}$; call it $c_{\text{clock}}^(B_{\text{phys}})$. On the positive values of both rows, physical pullback gives*

$$\mu_u(Q) \geq c_{\text{clock}}^* \bar{\Delta}_{\text{car}}(\mathcal{N}_{Qu}), \quad (168)$$

and every selected target-bad event has price

$$\mu_u(Q) \geq c_{\text{clock}}^* \delta_{\text{car}}^*(B_{\text{phys}}) > 0. \quad (169)$$

Thus the critical value $\beta_^{\text{red}} = 1$ is closed by an equality–rigidity invariant on the reduced saturation class. If either infimum is zero, its minimizing sequence is the complementary price class.*

Proof. The canonical relaxation is lower semicontinuous on the compact generated class, so Lemma II.8.3 evaluates δ_{car}^* . The transfer-radius definition gives (168); multiplication yields (169). $\square$

Corollary II.8.6 (Separator closure of the borderline branch). *On the zero separator-defect branch, the target projection of every zero-deficit profile vanishes by Proposition II.1.21. Hence the canonical target-bad closure contains no zero of $\underline{\Delta}_{\text{car}}$, so compactness gives $\delta_{\text{car}}^* > 0$. The positive clock-transfer branch of Theorem II.8.5 then gives the uniform event price; its zero branch retains the exact zero-transfer sequence.*

Proposition II.8.7 (A standard local compactness compiler). *Let X_0, X, X_1 be Banach spaces with $X_0 \subseteq X \hookrightarrow X_1$, the first embedding compact and the second continuous. If $1 \leq p < \infty$, $1 < q \leq \infty$, and a family of windows is bounded in*

$$L^p(0, 1; X_0) \cap W^{1,q}(0, 1; X_1),$$

then it is relatively compact in $L^p(0, 1; X)$. Consequently, its closure agrees with the corresponding compact branch of $K_{B_{\text{phys}}}$. The target and price rows are still evaluated by the canonical target closure and lower-semicontinuous relaxation; the lemma supplies no separate closedness or price premise.

Proof. The relative compactness is the Aubin–Lions lemma in the stated range [1, 6, 9]. The final assertion follows by mapping that relatively compact family into the generated evaluation closure. $\square$

Remark II.8.8 (Quotient compactness row). If a compact group acts continuously on a compact normalized set, its quotient is compact. For a noncompact symmetry group, fixing a gauge only defines a slice. The generated closure of that slice and the totalized gauge graph select the compact and graph-defect branches, respectively. In either case the quotient operation has Abs-ancestry and does not alter the source of the limiting physical record.

Remark II.8.9 (What failure of compactness means). A nonordinary value of a normalization, target-persistence, lower-semicontinuity, or physical-pullback graph is pushed into the compact activity lift with the relevant Geom/Caus/Abs/Lift/Bdry ancestry. A zero value of either sharp infimum supplies its minimizing sequence to the same lift. Finite-depth viability then removes the descendants or constructs their recurrent zero-price path. No target-visible element is transferred to J_{res} , and this optional price accelerator creates no new obligation in the core proof.

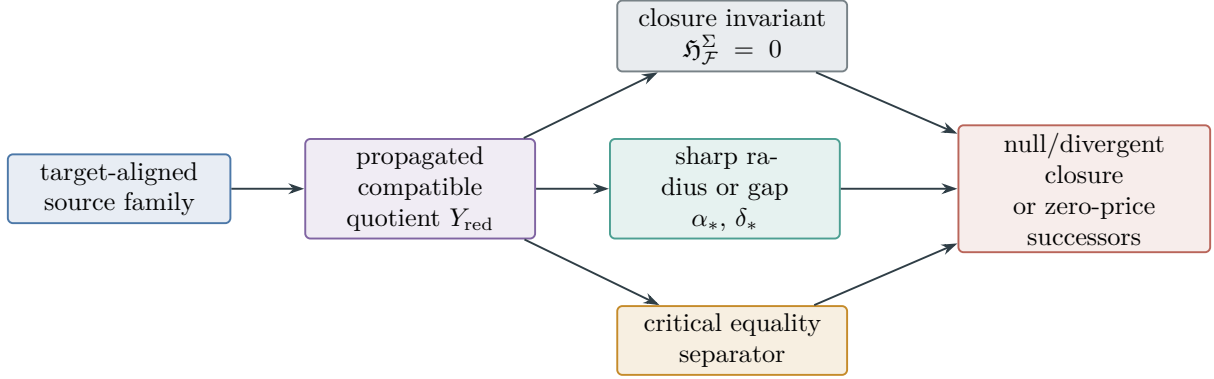


Figure 17: Identity-to-successor compiler. Once closed null directions are propagated out, generated identities expose a target closure, domination radius, physical-price gap, or critical equality relation. Every nonclosing value is normalized and passed to the successor operator. Each evaluator returns both its closing and complementary value.

II.9 Constraint tails as finite structural obstructions

Nonlocal constraints often enter a local evolution through an elliptic multiplier, potential, or pressure. A direct estimate of the whole nonlocal field obscures the smaller structural question: which part is fixed by the local source, which part is a gauge, which part decays from the exterior, and which finite-dimensional harmonic mode remains visible to the target? The following interface makes that separation explicit.

Definition II.9.1 (Elliptic constraint–tail interface). Let $U_0 \Subset U$ be open sets, let

$$\mathcal{C}p = S(V) \quad \text{in } U$$

be a linear elliptic constraint in a declared weak sense, and let $\mathcal{G}_{U_0}$ be its gauge space on U_0 . The compiler forms the product of the following five total graphs. Its ordinary value consists of:

- (T1) a local solution operator $\mathcal{L}_U$ satisfying $\mathcal{C}\mathcal{L}_U S(V) = S(V)$ on U ;
- (T2) a decomposition on U_0 ,

$$p|_{U_0} = p_{\text{loc}} + h, \quad p_{\text{loc}} := \mathcal{L}_U S(V)|_{U_0}, \quad \mathcal{C}h = 0 \text{ in } U_0; \quad (170)$$

- (T3) a Banach space $\mathcal{H}(U_0)$ of homogeneous solutions with a continuous splitting

$$\mathcal{H}(U_0) = \mathcal{H}_{\text{low}} \oplus \mathcal{H}_{\text{dec}}, \quad (171)$$

where $\mathcal{H}_{\text{low}}$ is finite dimensional;

- (T4) a dyadic exterior expansion $P_{\text{dec}}h = \sum_{k \geq 0} h_k$ convergent in $\mathcal{H}_{\text{dec}}$, with

$$\mathfrak{s}_{\text{tail}}(h) := \sum_{k \geq 0} \|h_k\|_{\mathcal{H}_{\text{dec}}} < \infty; \quad (172)$$

- (T5) a linear gauge realization $\Gamma : \mathcal{G}_{U_0} \rightarrow \mathcal{H}_{\text{low}}$ with closed range.

The residual low-mode obstruction is the cokernel class

$$\mathfrak{o}_{\text{tail}}(p) := [P_{\text{low}}h] \in \mathcal{H}_{\text{low}} / \text{Ran } \Gamma. \quad (173)$$

The local term retains the ancestry of $S(V)$; gauge removal appends **Abs**, and the exterior expansion and its limiting trace append **Bdry**. None of (T1)–(T5) is additional specialization data. Candidate operators, splittings, and dyadic pieces are enumerated from the typed weak constraint and the fixed universal elliptic library. The first relation without an ordinary value is retained as its total-graph successor with the displayed ancestry.

Theorem II.9.2 (Constraint–tail structural normal form). *On the ordinary value of the elliptic constraint–tail product graph, every field has, on U_0 , the decomposition*

$$p = p_{\text{loc}} + \Gamma g + h_{\text{dec}} + h_{\text{obs}}, \quad (174)$$

where $g \in \mathcal{G}_{U_0}$, $h_{\text{dec}} \in \mathcal{H}_{\text{dec}}$, and h_{obs} is any representative of the class $\mathfrak{o}_{\text{tail}}(p)$. The first three terms are closed by, respectively, the local source law, a propagated quotient, and the summability invariant (172). Consequently:

- (C1) if $\mathfrak{o}_{\text{tail}}(p) = 0$, the entire constraint field closes using only its inherited source, **Abs**, and **Bdry**;
- (C2) if $\mathfrak{o}_{\text{tail}}(p) \neq 0$ and its target projection vanishes, its remaining class lies in $\mathcal{N}_\Sigma$;
- (C3) if $\mathfrak{o}_{\text{tail}}(p) \neq 0$ and is target-visible, it is a finite-dimensional **Bdry/Abs**-ancestry occurrence evaluated by the carrier and price rows. It is not a new residual source.

Thus the nonlocal constraint is reduced internally to the local source graph, the summability graph, and a finite-dimensional target-projected cokernel. Every nonordinary value is already a witness-indexed successor rather than an additional analytic task.

Proof. Apply the continuous projections in (171) to h . Closed range and finite dimensionality give $\mathcal{H}_{\text{low}} = \text{Ran } \Gamma \oplus H_{\text{obs}}$ for a finite-dimensional complement H_{obs} . This gives (174); changing the complement changes h_{obs} only by a gauge term and hence preserves the cokernel class. Absolute convergence in (172) closes the decaying series. The three routing conclusions follow from quotient propagation, Theorem I.3.4, and the definition of $\mathcal{N}_\Sigma$. $\square$

Proposition II.9.3 (Constraint tails enter the physical budget only through a carrier law). *On the ordinary constraint-tail branch, let the signed tail contribution be Ψ_{tail} to a localized carrier balance. It is charged to the physical expenditure measure only if the equation supplies a measure inequality of one of the forms*

$$(\Psi_{\text{tail}})_+ \leq \Phi_u^+, \quad \text{or} \quad (\Psi_{\text{tail}})_+ \leq R_u^+, \quad (175)$$

with the right side appearing in (123). Tail summability alone proves analytic closure but supplies no physical price and no budget bound.

Proof. The carrier compiler uses only the measure terms displayed in (123). Either inequality in (175) places the positive tail contribution in that balance. Without such a comparison, (172) is a norm statement on a homogeneous field and has no implication for μ_u . $\square$

II.10 Covariant entropy cocycles and physical production

An entropy enters the reachability argument through the physical production that it controls along one orbit. Local entropy formulas in changing blow-up charts must therefore be joined by an explicit cocycle law. Independent windowwise monotonicity does not telescope. Perelman’s entropy formula is a classical model of covariant monotonicity with a rigid equality case [8]. The formulation below derives the required equation-independent cocycle contracts directly from a local balance and its reset relation.

Definition II.10.1 (Physical entropy cocycle). Let $I_j = [a_j, b_j]$, $j \geq 1$, be ordered normalized windows obtained from pairwise disjoint physical events Q_j . A physical entropy cocycle on this family consists of

- (E1) scalar primitives $\mathcal{M}_j : I_j \rightarrow \mathbb{R}$;
 - (E2) nonnegative Radon production measures $\mathfrak{D}_j$ and nonnegative error measures r_j on I_j ;
 - (E3) reset errors $e_j \geq 0$ and a constant $c_{\mathcal{M}} > 0$;
 - (E4) constants $M_+ < \infty$ and $M_- > -\infty$;
- such that, for $a_j \leq s < t \leq b_j$,

$$\mathcal{M}_j(t) + c_{\mathcal{M}} \mathfrak{D}_j((s, t]) \leq \mathcal{M}_j(s) + r_j((s, t]), \quad (176)$$

$$\mathcal{M}_{j+1}(a_{j+1}) \leq \mathcal{M}_j(b_j) + e_j, \quad (177)$$

$$\mathcal{M}_1(a_1) \leq M_+, \quad \mathcal{M}_j(t) \geq M_-. \quad (178)$$

The measures $\mathfrak{D}_j$ are coordinate expressions of one physical production law on the events Q_j ; chart Jacobians and time changes are included in that law. The half-open convention in (176) prevents endpoint atoms from being counted twice.

Theorem II.10.2 (Entropy-cocycle gluing). *For every $N \geq 1$, a physical entropy cocycle satisfies*

$$c_{\mathcal{M}} \sum_{j=1}^N \mathfrak{D}_j(I_j) \leq M_+ - M_- + \sum_{j=1}^N r_j(I_j) + \sum_{j=1}^{N-1} e_j. \quad (179)$$

If the two error series are finite, then $\sum_j \mathfrak{D}_j(I_j) < \infty$. If in addition every selected bad window satisfies $\mathfrak{D}_j(I_j) \geq \delta > 0$, their number is at most

$$\frac{M_+ - M_- + \sum_j r_j(I_j) + \sum_j e_j}{c_{\mathcal{M}} \delta}. \quad (180)$$

Proof. Apply (176) on every whole window. Summing gives

$$c_{\mathcal{M}} \sum_{j=1}^N \mathfrak{D}_j(I_j) \leq \sum_{j=1}^N (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) + \sum_{j=1}^N r_j(I_j).$$

The reset inequalities telescope the first sum from above:

$$\sum_{j=1}^N (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) \leq \mathcal{M}_1(a_1) - \mathcal{M}_N(b_N) + \sum_{j=1}^{N-1} e_j.$$

Use (178). This proves (179). Monotone convergence gives the countable statement, and $N\delta \leq \sum_{j=1}^N \mathfrak{D}_j(I_j)$ gives (180). $\square$

Definition II.10.3 (Perelman-type covariant invariant row). A Perelman-type construction is evaluated by the totalized tuple $\mathfrak{J}_{\text{Per}}$ whose six coordinates are the graph defects of the following relations for the fixed PDE and its generated window atlas:

- (P1) Each local primitive is a quotient-invariant scalar, typically a localized energy or variance plus an adjoint entropy and explicit pressure, drift, modulation, and boundary corrections.
- (P2) Differentiation along the *fixed* PDE gives (176); all unsigned remainders are placed in the displayed error measure, not suppressed formally.

- (P3) The target-visible quotient of production by its nonnegative physical floor $\mathfrak{d}$ is recorded by the price-gap row.
- (P4) The primitives have the lower bound in (178). Its sign must match the orientation of (176).
- (P5) Transport along the same physical orbit, chart pullback, and any change of a finite-horizon adjoint datum satisfy (177), with summable reset errors.
- (P6) Production either defines the expenditure measure μ_u , as in Corollary II.10.6, or is dominated by the declared μ_u through an explicit change-of-variables inequality of the form (164).

The entropy-cocycle branch is the common zero set of these six target-active graph defects. Any nonzero coordinate is retained with its inherited source ancestry and sent to the form, chain, price, tail, or boundary row indicated by the failed relation. Thus the construction neither presumes the entropy scalar nor accepts it as additional favorable data. These coordinates isolate the production-and-cocycle mechanism of Perelman's entropy formula [8].

Proposition II.10.4 (Weighted zero-production rigidity). *Let $D \subset \mathbb{R}^d$ be connected and open, let $v \in L^2(I; W_{\text{loc}}^{1,2}(D))$, and let $m \geq m_* > 0$ almost everywhere on $D \times I$. If a nonnegative production functional obeys*

$$\mathfrak{d}(w) \geq c_0 \int_I \int_D m(x, t) |\nabla v(x, t)|^2 dx dt \quad (181)$$

with $c_0 > 0$, then $\mathfrak{d}(w) = 0$ implies that $v(\cdot, t)$ is spatially constant on D for almost every t . Consequently, any closed bad class carrying a strictly positive mean-subtracted oscillation is disjoint from $\{\mathfrak{d} = 0\}$.

Proof. Nonnegativity and (181) imply $\nabla v = 0$ almost everywhere. A Sobolev function with zero weak gradient on a connected domain is almost everywhere constant. Its mean-subtracted oscillation is therefore zero, which excludes membership in the stated bad class. $\square$

Proposition II.10.5 (Physical-clock comparison). *Let μ be a physical measure on an event. On the absolutely-continuous clock branch, let a normalized production measure have density $w \geq 0$: $d\nu = w d\mu$. On a measurable set Q ,*

$$w \leq C, \quad 0 < C < \infty, \quad \text{a.e. on } Q \implies \mu(Q) \geq C^{-1} \nu(Q), \quad (182)$$

$$w \geq c > 0 \quad \text{a.e. on } Q \implies \mu(Q) \leq c^{-1} \nu(Q). \quad (183)$$

Thus a lower bound for normalized production yields a lower physical price only from an upper bound such as (182); the opposite density bound has the opposite implication.

Proof. Integrate the pointwise bounds in $\nu(Q) = \int_Q w d\mu$. $\square$

Corollary II.10.6 (Entropy as the physical expenditure measure). *On the finite-error branch of Definition II.10.1 and (179), on the atomic σ -algebra generated by the disjoint events Q_j , define*

$$\nu_u \left(\bigsqcup_{j \in S} Q_j \right) := \sum_{j \in S} \mathfrak{D}_j(I_j), \quad S \subseteq \mathbb{N}.$$

Then ν_u is a finite, countably additive physical measure on that event family. It may therefore be chosen as μ_u in Definition I.1.7. If other expenditures must also be charged, the measure may be enlarged only after a carrier or global balance proves a finite bound for the combined sum; that combined measure is fixed before the single ceiling B_{phys} is set.

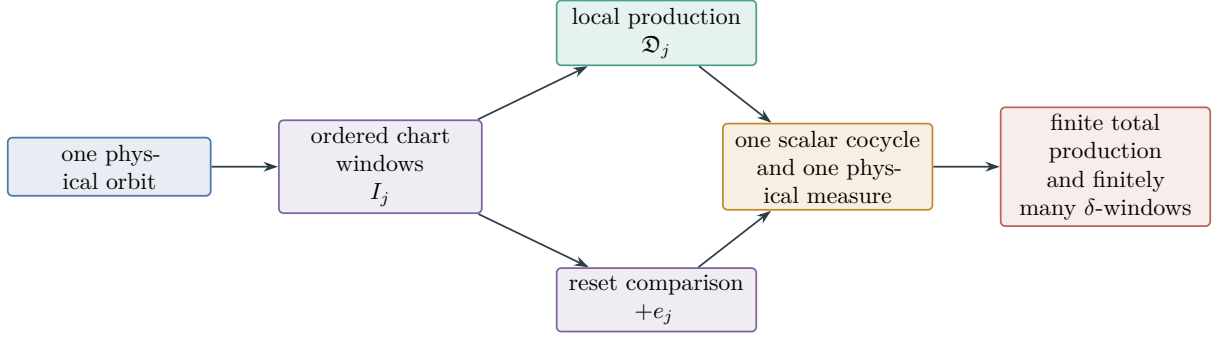


Figure 18: Covariant entropy-cocycle accounting. Local production estimates become global only after chart and adjoint-reset comparisons identify them along the same physical orbit. The output is a finite physical production measure that can serve as the orbit expenditure μ_u .

Proof. Countable additivity follows directly from the nonnegative series defining ν_u ; finiteness is Theorem II.10.2. $\square$

Remark II.10.7 (Equality models are row-local). A Gaussian may be the equality state of an adjoint Fisher-information term, while a constant field is the equality state of a weighted Dirichlet floor. These statements concern different nonnegative terms. A valid rigidity argument uses the equality set of the particular floor that is both compactly stable and linked to target badness; it does not merge the two equality models.

Part III

Structural Reachability and Regularity

III.1 Singular targets and cascade compilers

The algebra and normalized shell identity first evaluate whether target-aligned structure exists. Its absence excludes contact directly. On the surviving aligned branch, contact itself generates the canonical infinite disjoint shell family, and (196) generates its sharp lower-price series in the single physical measure. Divergence excludes the source cell; finite total price generates zero-price tails and activates the compact successor relation. Shell accountability, cascade extraction, physical pricing, and finite-depth viability are consecutive constructive operations.

Definition III.1.1 (Complete continuation target). Let Cont^{tot} be the total continuation relation of Definition I.1.1. For a finite maximal orbit, its terminal germ is the tuple of all limits of the generated evaluation coordinates along times increasing to T_* ; compact metrizable supplies such germ subsequences. Let $\mathcal{B}_{\text{cont}} \subset X_{\text{ev}}$ be the set of terminal germs for which Cont^{tot} has no ordinary regular extension. Define

$$\Sigma_{\text{sing}} := \overline{\mathcal{B}_{\text{cont}}}^{X_{\text{ev}}}. \quad (184)$$

This is the complete singular target. Contact means convergence of a preterminal subsequence to Σ_{sing} . Every finite maximal noncontinuation is therefore contact with this target. An empty local-solution fiber remains the separate form defect $\perp_{\text{loc}}$; it is not target avoidance.

Proposition III.1.2 (Terminal coverage). *Every ordinary maximal orbit has exactly one of the following outcomes: $T_* = \infty$, regular continuation through its terminal germ, or contact with Σ_{sing} . In particular there is no continuation remainder outside the singular target.*

Proof. Compact metrizability gives a terminal germ subsequence at finite T_* . The total continuation relation has either an ordinary extension value or its failure value. In the latter case the germ belongs to $\mathcal{B}_{\text{cont}} \subseteq \Sigma_{\text{sing}}$. An ordinary extension contradicts maximality unless it is recorded as continuation. These values are disjoint by totalization. $\square$

Definition III.1.3 (Singularity-formation support). For an ordinary maximal orbit u , let $\mathcal{D}_\Sigma(u)$ be a family of pre-contact shell records

$$R_r = (I_r, \Phi_r, (Q_{c,r}(u))_{c \in \mathfrak{C}}, \eta_{r,u}(I_r))$$

attached to that same orbit. The family is *shell-complete* when contact implies that it contains such a record for every sufficiently small r . Every record generated by contact then satisfies

$$\text{Prog}_\Sigma(R_r) = \sum_{c \in \mathfrak{C}} Q_{c,r}(u) + \eta_{r,u}(I_r) = 1. \quad (185)$$

The singularity-formation support is

$$\mathfrak{S}_\Sigma(u) := \bigcup_{R \in \mathcal{D}_\Sigma(u)} \text{Align}_\Sigma(R) \subseteq \mathfrak{C}. \quad (186)$$

It answers the target-relative question: which inherited source currents actually have positive pairing with the covectors descending toward the singular set? It does not ask which terms of the PDE are large in a target-independent norm.

Proposition III.1.4 (A singularity requires aligned structure). *If u is generated by the compiled continuum algebra and $\mathcal{D}_\Sigma(u)$ is shell-complete, then*

$$u \text{ contacts } \Sigma \implies \mathfrak{S}_\Sigma(u) \neq \emptyset. \quad (187)$$

Consequently $\mathfrak{S}_\Sigma(u) = \emptyset$ implies target avoidance. This implication is independent of the size or compactness of the full orbit.

Proof. Contact supplies normalized records satisfying (185). By Theorems I.5.2 and I.5.14, each has a nonempty aligned support, contained in the union (186). The contrapositive gives target avoidance. $\square$

Theorem III.1.5 (Canonical disjoint shell cascade). *Let a compiled ordinary maximal branch start outside Σ and contact Σ through a cofinal pre-contact sequence at its first contact or maximal time, and let (U_r, Φ_r) be a target-shell presentation. There are scales $r_j \downarrow 0$ and pairwise disjoint, chronologically ordered, strictly pre-contact intervals $I_j = [a_j, b_j]$ such that*

$$u(a_j) \notin U_{2r_j}(\Sigma), \quad u(b_j) \in U_{r_j}(\Sigma), \quad \text{Prog}_\Sigma(I_j, \Phi_{r_j}) = 1. \quad (188)$$

Every interval therefore has a nonempty aligned support and a source occurrence with positive target progress. On the ordinary current graph one of the five elementary contributions is at least $1/5$; on a boundary graph the positive shell defect retains its compiled nonempty support. No compactness, price gap, or contact margin is used in this extraction.

Proof. Set $b_0 = 0$ and $r_0 = 1$. Since every strictly pre-contact state is outside $\Sigma = \bigcap_{r>0} U_{2r}(\Sigma)$, after b_{j-1} choose a regular time $a_j > b_{j-1}$ and then a scale $r_j < \min\{r_{j-1}/4, 1/j\}$ for which $u(a_j) \notin U_{2r_j}(\Sigma)$. Contact supplies a later strictly pre-contact time at which the orbit belongs to $U_{r_j}(\Sigma)$; choose one as b_j . This inductively gives $b_{j-1} < a_j < b_j$, hence pairwise disjoint intervals. No continuity premise is needed: the two membership choices give $\Phi_{r_j}(u(a_j)) = 0$ and $\Phi_{r_j}(u(b_j)) = 1$ directly. Apply Theorem I.5.2 on each interval to get the final assertions. $\square$

Definition III.1.6 (Generated renormalization-scale row). Let z_* be a contact point or germ. The normalization grammar is the least family containing the identity and the space, time, amplitude, gauge, and quotient transformations for which the fixed PDE's totalized covariance graphs have ordinary values. Each failed covariance value is a **Lift**- or **Abs**-tagged graph defect. Applying every ordinary grammar value to the canonical shell cascade produces scales $r_j \downarrow 0$, physical pre-contact events $Q_j = Q(z_*, r_j)$, and records

$$V_j = \mathcal{S}_{z_*, r_j} u \in X_{\text{eval}},$$

where every $\mathcal{S}_{z,r}$ has its generated **Abs**/**Lift** ancestry. The row returns:

- (R1) the geometric-separation value obtained, after the canonical dyadic subsequence, from $r_{j+1} \leq \vartheta r_j$ for some $0 < \vartheta < 1$ and the events Q_j are pairwise disjoint;
- (R2) the totalized target-persistence inclusion $\{\overline{V_j}\} \subseteq B_\Sigma$, whose complementary value is a target-persistence graph defect;
- (R3) the total-graph value of the physical pullback relation

$$\mu_u(Q_j) \geq c_{\text{sc}}^* r_j^\kappa \overline{d}_{\text{sc}}(V_j), \quad (189)$$

Here c_{sc}^* is the sharp transfer infimum, κ is the exponent returned by the covariance graph, and $\overline{d}_{\text{sc}}$ is the canonical lower-semicontinuous relaxation. If no scalar exponent represents the pullback, the row returns the full scale-dependent weight sequence or its graph defect. Thus scale covariance is evaluated rather than postulated.

Theorem III.1.7 (Scale-cascade price compiler). *Evaluate the totalized scale interface. Its normalized price value is*

$$\delta_{\text{sc}} := \inf_{V \in B_\Sigma} \overline{d}_{\text{sc}}(V) \in [0, \infty]. \quad (190)$$

For the (j) -th generated scale stratum, define its sharp physical lower price exactly as in (196) and denote it by $\underline{p}_{\text{sc},j}$. The exact scale-price value is

$$\mathcal{S}_\Sigma := \sum_{j=1}^{\infty} \underline{p}_{\text{sc},j} \in [0, \infty]. \quad (191)$$

On the ordinary scalar-pullback branch, whenever the displayed factors are finite, (189) gives

$$\underline{p}_{\text{sc},j} \geq c_{\text{sc}}^* r_j^\kappa \delta_{\text{sc}}. \quad (192)$$

The branch $\mathcal{S}_\Sigma = +\infty$ gives

$$B_{\text{phys}} \geq \mu_u \left(\bigsqcup_{j=1}^{\infty} Q_j \right) = \sum_{j=1}^{\infty} \mu_u(Q_j) \geq \sum_{j=1}^{\infty} \underline{p}_{\text{sc},j} = \mathcal{S}_\Sigma = \infty, \quad (193)$$

contradicting a finite physical budget. The stronger condition $a_* := \inf_j r_j^\kappa > 0$ gives the uniform event price $\mu_u(Q_j) \geq c_{\text{sc}}^* a_* \delta_{\text{sc}} > 0$ on the positive transfer and price branches. On the positive transfer and normalized-price branches, after restricting to $r_j \leq 1$, divergence is automatic when $\kappa \leq 0$, and the scale-critical case $\kappa = 0$ transfers the normalized gap without loss. When $\kappa > 0$, the exact criterion is divergence of the generated lower-price series; a normalized gap alone supplies no such conclusion. The complementary value $\mathcal{S}_\Sigma < \infty$ produces vanishing-price windows in its inherited cell and is routed by Theorem II.2.20. In particular, a positive dissipative pullback exponent cannot be converted into a scale-independent price: the scale motion is retained in the signed carrier cohomology and then tested by the successor-saturation algebra.

Proof. The definition of each sharp lower price gives $\mu_u(Q_j) \geq \underline{p}_{\text{sc},j}$. Sum over the pairwise disjoint events to obtain (193); countable additivity and the finite physical budget give the contradiction. On the scalar pullback branch, insert (190) into (189) to obtain (192). If $0 < r_j \leq 1$, then $r_j^\kappa \geq 1$ for $\kappa \leq 0$. For $\kappa > 0$, the individual weights tend to zero, so their summability must be decided from the actual scale sequence. $\square$

Proposition III.1.8 (Canonical blow-up invariant image). *Every contact cascade and every ordinary value of the generated normalization grammar has the canonical invariant image*

$$\left(\vartheta, \overline{\{V_j\}}, \mathfrak{H}_{\{V_j\}}^\Sigma, \kappa, \delta_{\text{sc}} \right). \quad (194)$$

Here $\vartheta \leq 1/4$ is produced by the canonical shell subsequence, the closure is taken in the generated evaluation compactification, $\mathfrak{H}_{\{V_j\}}^\Sigma$ records whether the cascade remains target-visible, κ or its full weight-sequence replacement is the physical pullback value, and δ_{sc} is the canonical price infimum. A failed normalization or pullback occupies the corresponding Lift/Abs graph-defect cell. Thus every blow-up outcome has an image; none of the five coordinates is additional problem data.

Definition III.1.9 (Normalized cascade refinement). For a contact orbit, combine the canonical intervals from Theorem III.1.5 with every ordinary value of the generated normalization grammar. The resulting normalized cascade refinement consists of

- (C1) pairwise disjoint measurable pre-contact events $Q_1, Q_2, \dots$ in the domain of the same physical expenditure measure μ_u ;
- (C2) generated normalizations $\mathcal{N}_{Q_j}$, carrying their Abs/Lift ancestry, such that $w_j := \mathcal{N}_{Q_j} u \in B_\Sigma$;
- (C3) for every j , a normalized shell-crossing record R_j carried by w_j for which

$$\text{Prog}_\Sigma(R_j) = 1, \quad \max_{c \in \mathfrak{C}} (\text{Prog}_\Sigma^c(R_j))_+ \geq \frac{1}{5}. \quad (195)$$

By Theorem I.4.5 and the applicable prospective shell-accountability theorem, every occurrence in clause (C3) has a nonempty aligned source support $A_j := \text{Align}_\Sigma(R_j) \subseteq \mathfrak{C}$. Every nonordinary normalization value is already a typed graph-defect record. The refinement asserts no positive price; it generates the family on which the price row is evaluated.

Proposition III.1.10 (Persistent channel extraction). *For every cascade generated by Definition III.1.9, some nonempty support $A \subseteq \mathfrak{C}$ occurs for infinitely many indices. In particular, at least one of the five elementary source labels belongs to the ancestry of infinitely many target-approaching windows.*

Proof. There are only $2^5 - 1$ nonempty supports. The infinite pigeonhole principle gives an infinite constant-support subsequence. Any member of that support has the second asserted property. $\square$

Definition III.1.11 (Target-aligned source ledger). Fix an ordinary maximal branch u and a normalized cascade family $(Q_j, R_j)_{j \geq 1}$. For every nonempty $A \subseteq \mathfrak{C}$, let $\mathcal{F}_A(u)$ be the family of source-current occurrences extracted from those R_j for which $\text{Align}_\Sigma(R_j) = A$. The source ledger is generated as follows. For $j \in I_A := \{j : \text{Align}_\Sigma(R_j) = A\}$, let $\mathcal{Q}_{A,j}$ be the family of all generated ordinary events under the same physical ceiling and in the same propagated source/mode and shell/scale stratum as Q_j , normalized to the same unit target progress. Equality of stratum means equality of every transcript entry preceding the price row, so the equivalence relation is canonical. Define the sharp physical lower price

$$\underline{p}_{A,j} := \inf\{\mu_v(Q) : (v, Q) \in \mathcal{Q}_{A,j}\} \in [0, \infty], \quad (196)$$

with $+\infty$ for an empty stratum. The stratum contains (u, Q_j) , so $\mu_u(Q_j) \geq \underline{p}_{A,j}$. Each source row has exactly one of the following exhaustive entries:

(L1) a *null entry*,

$$\mathfrak{H}_{\mathcal{F}_A(u)}^\Sigma = 0; \quad (197)$$

(L2) a *divergent-price entry*, when the target quotient is nonzero and

$$|I_A| = \infty \implies \sum_{j \in I_A} \underline{p}_{A,j} = \infty; \quad (198)$$

(L3) a *finite-price successor entry*, when the target quotient is nonzero and

$$\sum_{j \in I_A} \underline{p}_{A,j} < \infty. \quad (199)$$

The quotient value and the dichotomy for a nonnegative series make the list exhaustive. Quotient, symmetry, conservation, carrier, resistance, compactness–rigidity, equality, scale, and entropy calculations are evaluators of these exact entries. A uniform positive lower price is a special divergent-price value. Since $|\mathfrak{C}| = 5$, the ledger has at most $2^5 - 1 = 31$ nonempty rows even when the cascade contains infinitely many scales.

Theorem III.1.12 (Physical expenditure exclusion). *Let μ be a nonnegative countably additive measure with $\mu(\Omega) \leq B_{\text{phys}} < \infty$. Let $Q_j \subseteq \Omega$ be pairwise disjoint measurable sets and let $\delta_j \geq 0$. If*

$$\mu(Q_j) \geq \delta_j \quad \text{for every } j, \quad \sum_{j=1}^{\infty} \delta_j = \infty, \quad (200)$$

then such a family cannot exist. The uniform-price criterion $\delta_j \geq \delta > 0$ is a special case.

Proof. For every N , finite additivity and monotonicity give

$$B_{\text{phys}} \geq \mu\left(\bigsqcup_{j=1}^N Q_j\right) = \sum_{j=1}^N \mu(Q_j) \geq \sum_{j=1}^N \delta_j.$$

The partial sums on the right are unbounded by (200), contradicting the fixed upper bound B_{phys} . $\square$

Corollary III.1.13 (Shell progress implies expenditure exclusion). *On the ordinary contact-cascade branch, let R_j be the normalized shell records on pairwise disjoint pre-contact events Q_j . On the physical-price row*

$$\text{Prog}_\Sigma(R_j) = 1 \implies \mu_u(Q_j) \geq \delta_j \geq 0. \quad (201)$$

If $\sum_j \delta_j = \infty$, then u cannot contact Σ .

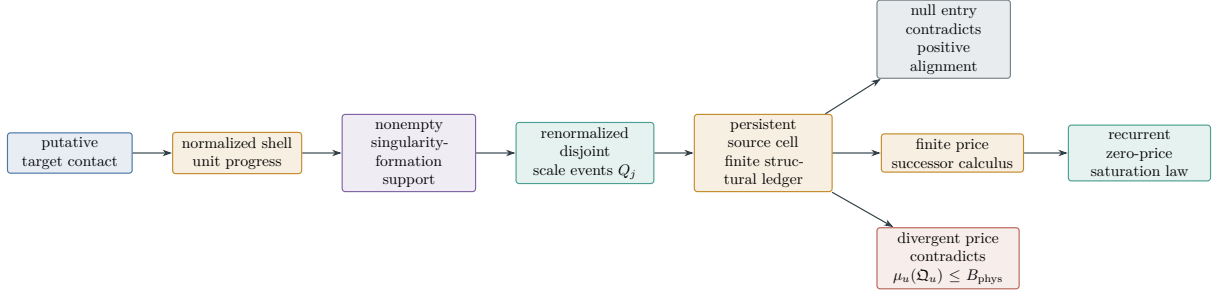


Figure 19: Singularity-formation accounting. Contact forces a unit normalized shell crossing and therefore exposes a target-aligned source. Repeated scales select a persistent source cell. A null ledger entry contradicts alignment, a divergent-price entry contradicts the sole finite physical budget, and a finite-price entry activates ordered successors and recurrence. Only the target projection of the generated saturation algebra enters finite-depth viability; survival at every depth constructs the singularity mechanism.

Proof. Contact supplies the unit identities by Theorem I.5.2. The price implication then gives (200), and Theorem III.1.12 gives the contradiction. $\square$

Theorem III.1.14 (Finite structural-ledger exclusion). *Let u have the canonical contact cascade of Theorem III.1.5. On the fast-track value where every persistent target-aligned support is either zero in the target quotient or has a divergent physical-price series, u does not contact Σ .*

Proof. On the contact branch, Proposition III.1.10 gives a nonempty $A \subseteq \mathfrak{C}$ for which $I_A := \{j : \text{Align}_\Sigma(R_j) = A\}$ is infinite. If A has a null entry, then every target projection of the corresponding source-current occurrences vanishes. This contradicts $A = \text{Align}_\Sigma(R_j) \neq \emptyset$. If A has a divergent-price entry, the pairwise disjoint events $(Q_j)_{j \in I_A}$ carry the divergent lower-price schedule $(\underline{p}_{A,j})_{j \in I_A}$, contradicting Theorem III.1.12. The ledger alternatives are exhaustive, so contact is impossible. $\square$

Corollary III.1.15 (Estimate-to-invariant reduction). *In the setting of Theorem III.1.14, the canonical transcript evaluates each persistent source support as null, divergent-price, or finite-price successor. By Theorem II.1.19, no global estimate of the full orbit remains in the verdict after these row values are known. In particular, arbitrarily large target-null motion is absent from the reduced calculation. The last value is not a verdict: it activates Theorem II.2.20.*

Corollary III.1.16 (Entropy exclusion is physical expenditure exclusion). *On the entropy-cocycle, finite-error, and positive-production branches for a canonical contact cascade, $\mathfrak{D}_j(I_j) \geq \delta > 0$ on every selected window and contact is impossible.*

Proof. By Corollary II.10.6, the glued production is a finite physical measure. Apply Theorem III.1.12 to its restrictions to the disjoint events Q_j . $\square$

III.2 The continuous structural regularity theorem

The five-channel conclusion is derived before any regularity estimate. The only equation-dependent step in target exclusion is the physical evaluation of the target-aligned class exposed by contact.

Definition III.2.1 (Totalization of a continuum interface). Let $F : D(F) \subset X_F \rightarrow Y_F$ be any partial interface used by the continuum argument: a constitutive realization, a gradient potential map, a target-shell derivative, a physical-price comparison, or a continuation map. Adjoin a failure value $\perp_F$ and define

$$F^{\text{tot}}(x) = \begin{cases} F(x), & x \in D(F), \\ \perp_F, & x \notin D(F). \end{cases}$$

Embed the graph of this total map in the structural test compactification and set

$$\partial F := \overline{\text{graph } F^{\text{tot}}} \setminus \text{graph } F. \quad (202)$$

Thus ∂F contains both limiting graph defects and the explicit failure records $(x, \perp_F)$. The *totalized interface record* of an input is its graph value when the input lies in $D(F)$, and otherwise its class in the closed defect cell generated by ∂F . The defect cell retains the ancestry of the approximating graph records; failure at a chart, weak-limit, or terminal interface appends **Lift** or **Bdry** according to Definition I.2.1. Its target-null part is quotiented by $\mathcal{N}_\Sigma$; its nonzero target quotient is a target-visible defect class in the same five-channel algebra.

Definition III.2.2 (Total continuum interface algebra). The target-relative continuum interface algebra is the finite coproduct

$$\mathfrak{H}_{\text{cont}}^\Sigma := \mathfrak{H}_{\text{form}}^\Sigma \sqcup \mathfrak{H}_{\text{grad}}^\Sigma \sqcup \mathfrak{H}_{\text{chain}}^\Sigma \sqcup \mathfrak{H}_{\text{price}}^\Sigma \sqcup \mathfrak{H}_{\text{cov}}^\Sigma. \quad (203)$$

Here the five cells are respectively the target-active quotients of:

- (D1) the graph boundary of the local-solution map and of the native constitutive or sector-form realization;
- (D2) $\overline{\text{Ran } G} \setminus \text{Ran } G$, together with the target-active part of $\ker G^*$, after the exact target-null quotient;
- (D3) failure of the target closure/shell presentation, or the chain-rule defect measure

$$\eta_{\Phi, u} := d(\Phi(u_t)) - \langle D\Phi(u_t), J_t \rangle dt; \quad (204)$$

- (D4) the total carrier graph, including failure to produce the finite physical measure dictated by the equation, and the zero-price limit of every finite-expenditure shell cascade;
- (D5) the generated chart, gauge, scale, and quotient-covariance graphs, including every commutator defect created when an operation fails to descend.

Every cell is source-tagged before its target-null quotient is taken. A missing property is therefore an input to the activity lift and successor operator, not a terminal verdict. Regular continuation is not a sixth interface cell: by Definition III.1.1 and Proposition III.1.2, a finite maximal germ without an ordinary continuation is itself a point of the complete target and is processed by the same shell and successor machinery.

Theorem III.2.3 (Defect totalization and five-channel closure). *For every fixed continuum evolution, target, and physical carrier, each of the five interfaces in Definition III.2.2 has an ordinary total-graph value or a graph-boundary value. Every target-visible value has ancestry in Geom , Caus , Abs , Lift , Bdry ; every target-invisible value lies in J_{res} . Every target-visible value, ordinary or defective, is converted into an activity measure and a witness-indexed successor. Hence lack of closed range, a classical chain rule, positive price, or covariance is not a stopping point: it is a concrete successor mechanism in the same closed algebra. Finite noncontinuation is target contact, not an unprocessed remainder.*

Proof. For each partial map, the total value is either an ordinary graph value or the failure value $\perp_F$. Closing the total graph adds its limiting defects, so (202) exhausts both kinds of failure. Sourcewise closure and Theorem I.3.4 retain the ancestry of every approximating record. The gradient cell is exhausted by Theorem I.3.8. For the shell interface, the distributional identity

$$d(\Phi(u_t)) = \langle D\Phi(u_t), J_t \rangle dt + \eta_{\Phi,u}$$

is tautological once $\eta_{\Phi,u}$ is defined by (204); singular and endpoint parts have Lift/Bdry ancestry. A finite-price cascade enters Definition II.2.8; a carrier-graph failure is lifted by the same total-graph coordinate. Covariance and descent failures retain the tags of the operation and commutator that generated them. Finally quotient each cell by its intersection with $\mathcal{N}_\Sigma$. The kernel is residual. Every nonzero quotient value enters $\mathcal{R}_\eta$, and Definition II.2.13 iterates it until it is eliminated or lies on an infinite zero-price path. Terminal coverage handles finite noncontinuation. This proves all assertions. $\square$

Theorem III.2.4 (No favorable analytic premise in constructive closure). *Fix only the canonical weak-PDE signature, its complete continuation target, and its equation-generated physical carrier. The source compiler, sourcewise closure, target quotient, total interface graphs, activity lift, successor relation, and finite-depth viability sets are defined without requiring any of the following properties:*

- a gradient/sector realization, local or global regularity, compactness, closed gradient range,*
- a classical chain rule, smooth shells, coercivity, a positive price gap, rigidity, polarity,*
- finite resistance, summable tails, orbit realization, or continuation.*

(205)

Each listed property is tested by a total graph generated from the equation. Both its ordinary and boundary values are supplied to the same activity lift, successor relation, and finite-depth viability recursion. A branch therefore cannot terminate at the sentence that a property fails: it is removed by the target-null quotient, charged by the physical carrier, or retained as an explicit zero-price recurrent mechanism. Consequently none of the listed properties is a premise of the totalization, contact-accountability, or constructive regularity theorems.

Proof. The compiler is structural induction on the fixed equation syntax. Sourcewise closure is defined separately in each tagged cell. Quotienting by the common target kernel is exact by definition. Totalization replaces every partial interface by the disjoint union of its graph values, explicit failure values, and graph-boundary values. The realization modes partition the resulting ordinary and defect records. The positive target-current part of each record defines its activity measure, whose normalized lift defines $\mathcal{R}_\eta$. Iterating $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ removes every record without arbitrarily long zero-price descendants. Compactness turns survival at every finite depth into an infinite path and a stationary path law. Thus both sides of every analytic relation undergo an explicit structural operation, and no favorable side of a dichotomy is selected. $\square$

Definition III.2.5 (Physical reachability profile). For a finite number B_{phys} arising from the carrier identity, let $\mathcal{O}_{\leq B_{\text{phys}}}^{\text{ord}}$ be the ordinary maximal solution branches whose derived expenditure satisfies $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}$. When this set is nonempty, define

$$\text{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}}) := \sup_{u \in \mathcal{O}_{\leq B_{\text{phys}}}^{\text{ord}}} \mathbf{1}_{\{u \text{ contacts } \Sigma\}}. \quad (206)$$

There is no empty-supremum convention. If the ordinary solution stratum is empty, the total local-solution or carrier graph returns its explicit defect record. Thus $\text{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}}) = 0$ means

that ordinary maximal branches exist and none contacts the target. The ceiling is not a second datum: it is only a stratum of the one physical expenditure generated by the equation.

Definition III.2.6 (Constructive zero-price mechanism cells). Fix a derived finite physical stratum B_{phys} . For nonempty $A \subseteq \mathfrak{C}$ and $m \in \mathfrak{M}$, let $Z_{\infty,A,m}^{\Sigma}(B_{\text{phys}})$ be the part of Z_{∞}^{Σ} whose inherited support is exactly A and whose primary mode is exactly m . Full ancestry and all nonprimary flags remain coordinates of the record. Define

$$\mathfrak{F}_{A,m}^{\Sigma}(B_{\text{phys}}) := \overline{\text{span}} \left\{ \Pi_{\Sigma}(\pi_0)_{\#} \nu : \nu \text{ stationary on } \text{Path}(Z_{\infty,A,m}^{\Sigma}(B_{\text{phys}})) \right\}, \quad (207)$$

$$\mathfrak{F}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) := \bigoplus_{\substack{\emptyset \neq A \subseteq \mathfrak{C} \\ m \in \mathfrak{M}}} \mathfrak{F}_{A,m}^{\Sigma}(B_{\text{phys}}). \quad (208)$$

These cells are not defined from contact orbits. They are the finite source/mode decomposition of the prospectively generated zero-price fixed point. The direct sum is provenance-tagged: analytically equal representatives with different ancestry remain distinct. Every generator carries its stationary path law, full weak equation or recurrent graph boundary, target pairing, and diagonal cascade-origin witnesses.

Theorem III.2.7 (Constructive source decomposition of zero-price dynamics). *For every derived finite physical stratum B_{phys} ,*

$$\boxed{\mathfrak{H}_{\text{succ}}^{\Sigma}(B_{\text{phys}}) = \mathfrak{F}_{\Sigma}^{\text{phys}}(B_{\text{phys}})}. \quad (209)$$

The right side has at most 31×5 provenance-tagged cells. Every finite-expenditure contact orbit produces a nonzero generator in one of them. Consequently, whenever the ordinary stratum is nonempty,

$$\mathfrak{F}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) = 0 \implies \text{Reach}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) = 0. \quad (210)$$

A nonzero cell is not a label awaiting a further estimate: its generator is a stationary zero-price successor law carrying the weak equation, recurrent graph defects, target pairing, source ancestry, and cascade-origin witness sequence; an ordinary contact origin makes these actual physical events.

Proof. Along a successor path the inherited support can only increase. Because the source alphabet is finite, it is eventually constant; stationarity makes it constant almost surely. The primary mode is likewise monotone in a finite ordered set and is stationary only when constant almost surely. Decomposing every stationary path law along these invariant finite parts proves (209).

If an ordinary branch contacts Σ , the total shell identity gives the disjoint unit-progress cascade without a chain-rule premise. Theorem II.2.20 converts that same cascade into a stationary zero-price law with nonzero target projection and diagonal origin-event witnesses. The finite tag argument places a nonzero component in one (A, m) -cell. The contrapositive proves (210); the stated data are supplied by Theorem II.2.17 and Lemma II.2.11. $\square$

Theorem III.2.8 (Finite-depth structural evaluator). *For every source/mode cell, start with its generated target-active zero-price records and form*

$$Z_{0,A,m} \supseteq Z_{1,A,m} \supseteq \cdots, \quad Z_{n+1,A,m} = \mathcal{V}_{\eta}(Z_{n,A,m}). \quad (211)$$

Exactly one of the following constructive outcomes occurs:

- (E1) *for some finite N , $Z_{N,A,m} = \emptyset$; the least-coded finite rational-cylinder pruning transcript of Theorem II.2.15 records the descendant failure and closes the cell;*

(E2) every $Z_{n,A,m}$ is nonempty; compactness constructs a point of their intersection, an infinite successor path, a stationary path law, and a nonzero generator of $\mathfrak{F}_{A,m}^\Sigma(B_{\text{phys}})$.

Applying the generated zero-cost equations to outcome (E2) removes each target-null component and recursively expands each ordinary or graph-boundary component. Thus every nonzero row value undergoes a further structural operation.

Proof. Each $Z_{n,A,m}$ is compact. If their intersection is empty, the finite intersection property gives an N for which $Z_{N,A,m}$ is empty, and Theorem II.2.15 generates the finite transcript. If all are nonempty, choose compatible successors at every finite depth. The inverse limit of these nonempty compact finite-path spaces is nonempty and provides an infinite path. Empirical measures of its shift orbit have a stationary weak limit. Target activity and zero price are closed coordinates, so the resulting law generates the stated cell. The final reduction is Theorem II.2.17 applied to every support component. $\square$

Theorem III.2.9 (End-to-end constructive closure). *For a derived finite physical stratum B_{phys} , apply source compilation, the target quotient, total interface graphs, shell activity lifting, witness-indexed successors, finite-depth viability pruning, and the generated zero-cost equations, in that order. The output is*

$$\boxed{\mathfrak{H}_{\text{out}}^\Sigma(B_{\text{phys}}) = \mathfrak{H}_{\text{succ}}^\Sigma(B_{\text{phys}}) = \bigoplus_{A,m} \mathfrak{F}_{A,m}^\Sigma(B_{\text{phys}}).} \quad (212)$$

On every ordinary local finite-carrier branch,

$$\bigoplus_{A,m} \mathfrak{F}_{A,m}^\Sigma(B_{\text{phys}}) = 0 \implies \text{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}}) = 0. \quad (213)$$

The compiler never consults an actual-contact image. A contact orbit itself constructs a nonzero cell. Conversely, a cell that survives the evaluator comes with an infinite zero-price path, stationary law, full generated saturation equations, source support, target pairing, and diagonal origin events, which are physical on an ordinary contact origin. These are the mechanism to be tested by the next algebraic refinement; every further relation is generated from the public equation.

Proof. Totalization inserts every target-visible local-solution or carrier boundary witness into the same successor graph. On an ordinary finite-carrier branch, source and mode tags partition the zero-price path laws and Theorem III.2.7 identifies their direct sum with $\mathfrak{H}_{\text{succ}}^\Sigma$, proving (212). If contact occurs, shell accountability, disjoint cascade extraction, and Theorem II.2.20 construct a nonzero stationary path law in one cell. Its contrapositive proves (213). The finite-depth evaluator and saturation-equation compiler supply the listed data for every surviving generator, without an actual-contact filter. $\square$

Theorem III.2.10 (Unconditional continuum contact mechanism). *Fix the canonical physical continuum presentation and close its orbit algebra sourcewise. Every target-active current, total-graph defect, and target-visible limit has ancestry in Geom, Caus, Abs, Lift, Bdry, while*

$$G_{B_{\text{phys}},\Sigma}^{J_{\text{res}},\text{src},\text{sat}} = 0. \quad (214)$$

Every contact orbit has a canonical sequence of pairwise disjoint pre-contact shell intervals I_j with unit target progress. After passage to a subsequence, one nonempty support $A \subseteq \mathfrak{C}$ occurs on every I_j and contributes positive target progress there. This statement does not require a constitutive realization, classical chain rule, compactness, or price gap: a failed relation contributes through its total-graph defect with the ancestry of the failed operation.

Proof. The continuum normal-form compiler and source-preserving completion give the five-channel statement for ordinary values. Defect totalization gives the same statement for every boundary value. The target quotient gives (214): by construction residual means target-null. The canonical disjoint shell-cascade theorem supplies the intervals. On each interval the total shell identity has increment one. Hence either an ordinary current has positive target pairing or the defect current does; in both cases its compiled five-channel support is nonempty. The infinite pigeonhole principle selects a persistent support. $\square$

Theorem III.2.11 (Constructive physical-price evaluation). *In the setting of Theorem III.2.10, restrict to the derived finite physical stratum $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}} < \infty$ of ordinary maximal branches beginning outside Σ . For each persistent support use the canonical sharp prices of (196). Then: (P1) if*

$$\sum_{j=1}^{\infty} p_{A,j} = \infty. \quad (215)$$

then that support cannot sustain contact;

(P2) *if the series is finite, the corresponding shell tails enter Z_0 , and finite-depth viability either removes them at some Z_N or constructs a stationary zero-price law in $\mathfrak{F}_{A,m}^{\Sigma}(B_{\text{phys}})$.*

If every persistent support is removed by one of the first alternatives, then

$$\boxed{\text{Reach}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) = 0.} \quad (216)$$

Thus a finite lower-price series is never the end of the global-regularity track.

Proof. If an ordinary maximal branch contacted Σ , Theorem III.2.10 would supply pairwise disjoint intervals and a persistent support A . On their selected subsequence, the definition of the sharp price and countable additivity give

$$B_{\text{phys}} \geq \sum_j \mu_u(I_j) \geq \sum_j p_{A,j} = \infty,$$

a contradiction in case (P1). In the finite-series case, use the actual prices as the price coordinate in the activity lift. Cofinal tails have price tending to zero, so Theorems II.2.20 and III.2.8 give precisely the two outcomes in (P2). If every persistent support is eliminated, contact has no possible support, contradicting shell accountability. This proves (216). $\square$

Remark III.2.12 (Orbitwise source ledgers). The divergent-price branch is evaluated only for the source support that persists along a putative singular cascade. Its canonical lower-price schedule may depend on the orbit stratum and may decay with scale. A uniform positive price is stronger and yields uniform window counts. Theorem III.2.11 uses only divergence of this generated series; five-channel exhaustion and cascade existence were proved earlier.

Theorem III.2.13 (Total constructive continuum regularity evaluator). *Let u_0 be a datum in the public state class of the fixed equation. Totalize the local-solution and carrier graphs and run the constructive closure of Theorem III.2.9 on every ordinary maximal branch and every target-visible graph-boundary witness. The evaluator returns:*

- (i) *a finite successor depth N at which every target-active cell is empty; then every ordinary maximal branch from u_0 is global and avoids Σ_{sing} ;*
- (ii) *an infinite zero-price successor path and a stationary law supported on the generated equations (110), with nonzero target pairing, source ancestry, and diagonal cascade-origin witnesses;*

- (iii) when the local-solution or finite-carrier graph itself has no ordinary value, its canonical total-graph witness tree, which is subjected to the same finite-depth test and therefore ends in a finite failed identity or in the recurrent mechanism of (ii).

Finite noncontinuation is already contact with the complete continuation target. It never produces a separate continuation class. Neither a failed chain rule nor finite physical price is terminal: each is a successor input.

Proof. The local-solution and carrier graphs give ordinary values or their explicit boundary witnesses. Apply the activity lift and viability recursion to the latter, proving (iii). On an ordinary branch, finite maximal time without continuation is contact by Proposition III.1.2. Contact then gives disjoint unit-progress shells and a persistent five-channel support by Theorem III.2.10. A divergent sharp price contradicts the finite carrier. A finite-price cascade enters Z_0 and, by Theorem II.2.20, yields an infinite zero-price path. The saturation-equation compiler supplies (ii), including every recurrent interface defect.

Conversely, if finite-depth pruning empties every target-active cell, a contact branch would construct a nonempty fixed-point cell by the preceding paragraph, a contradiction. Thus no finite maximal branch contacts the complete target; terminal coverage makes every ordinary maximal branch global. This is (i). $\square$

Corollary III.2.14 (Generated PDE reachability output). *Let $\mathfrak{U}_{\text{pub}}$ be the datum class in the public PDE presentation. Derive the physical expenditure from the carrier identity and run the source, signed-carrier, witness-successor, viability, and saturation compilers on every resulting finite stratum. For each stratum the generated output is exactly*

$$\boxed{\begin{array}{l} Z_N^{\Sigma_{\text{sing}}} = \emptyset \text{ for some finite } N \quad \text{with its canonical finite pruning transcript,} \\ \text{or} \\ Z_n^{\Sigma_{\text{sing}}} \neq \emptyset \text{ for every } n \quad \text{with its same-origin stationary mechanism.} \end{array}} \quad (217)$$

If the first output occurs for every datum and finite stratum, every ordinary maximal branch in $\mathfrak{U}_{\text{pub}}$ is global. Every second output is the explicit mechanism in (ii) of Theorem III.2.13. Neither output is a premise supplied with the PDE presentation.

Proof. Apply Theorem III.2.13 at the derived finite ceiling of each branch. Theorem II.2.15 gives the exhaustive alternative (217). Its first branch excludes contact; its second branch constructs the stated infinite path and stationary law. $\square$

Corollary III.2.15 (Quantitative finite-cascade bound). *If the physical expenditure satisfies $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}$ and every disjoint target-bad event costs at least $\delta_{\text{phys}}(B_{\text{phys}}) > 0$, then any such family has cardinality at most*

$$\left\lceil \frac{B_{\text{phys}}}{\delta_{\text{phys}}(B_{\text{phys}})} \right\rceil. \quad (218)$$

Proof. The proof of Theorem III.1.12 gives $N\delta_{\text{phys}}(B_{\text{phys}}) \leq B_{\text{phys}}$ for every finite subfamily. $\square$

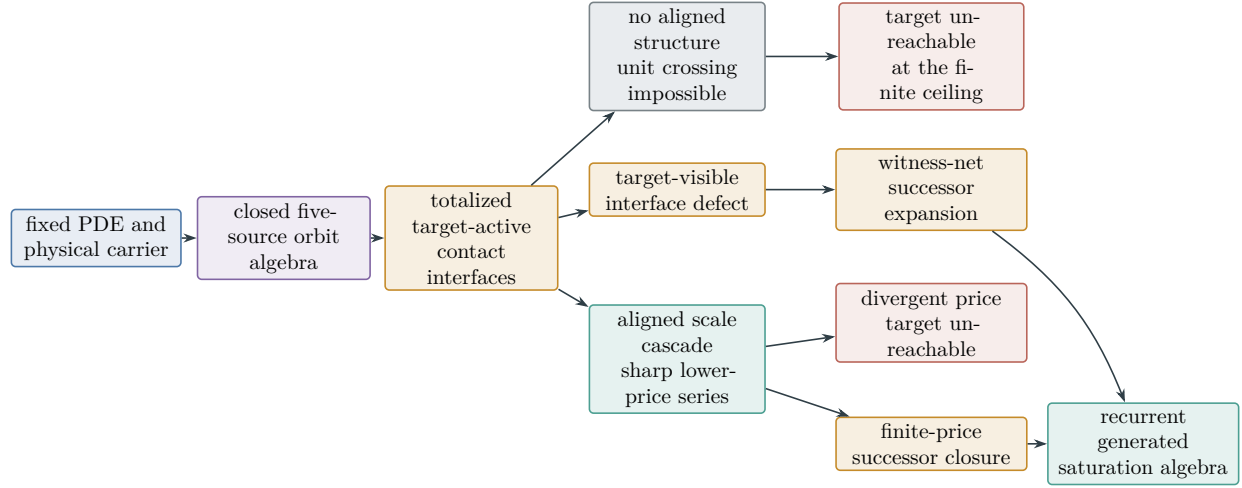


Figure 20: Constructive singularity evaluator. Only the target-active quotient is analyzed. If it contains no singularity-aligned structure, normalized shell contact is impossible. Otherwise the surviving source class is evaluated structurally: divergent physical price excludes it, while finite price forces ordered first-failure routing and a recurrent zero-production law. The viability recursion then generates either a finite rational-cylinder pruning transcript or an infinite zero-price path supporting the displayed singularity mechanism.

III.3 Structural evaluator coverage and failure semantics

The following table is the coverage proof for the fast track. Every classical analytic question is assigned a generated invariant and every opposite value activates a structural operation. No row accepts a desired analytic conclusion as input or returns a bare label.

Analytic question	Generated structural invariant	Successor action
Does a native, chain, trace, or covariance relation hold?	target-active boundary of its total graph	activity lift of the typed graph witness
Does a reduction descend through later operations?	commutator of the operation with the quotient map	typed descent defect
Is the gradient range closed where the target sees it?	$\mathfrak{H}_{\text{grad}}^\Sigma$	inherited closure or harmonic class
Is a tail structurally closed?	$\mathfrak{H}_{\text{tail}}^\Sigma$	target-visible tail cokernel
Is target feeding dominated?	α_*	supercritical or positive null-cost feeding profile
Is normalized target progress physically priced?	δ_*	zero-price sequence in the same source cell
Can the source be routed?	$\mathcal{R}(\Sigma)$	target-visible harmonic non-routable class
Is the target polar?	$\text{Cap}(\Sigma)$	positive-capacity target-compatible class
Does a scale cascade exceed finite expenditure?	$\mathcal{S}^\Sigma$	finite-price successor chain and recurrent saturation row
Do internal signed currents create net target feed?	carrier cohomology $[F_{\text{sc}} - D - M]$	nonzero scale-feed cohomology successor
Does a failed interface persist dynamically?	recurrent witness boundary $\partial_\infty F$	target-null loss or recurrent failure mechanism
Does a sharp branch close?	$\mathfrak{H}_{\text{sat}}^\Sigma$	nonzero saturation/equality remainder
Are stochastic fluctuations invisible or routed?	χ^Σ and fluctuation resistance	target-visible covariance/bracket class
Does a weak-solution graph value persist?	recurrent witness boundary $\partial_\infty F$	finite-depth removal or recurrent weak-equation path
Can zero-price target progress persist?	viability sets Z_n^Σ	finite pruning transcript or an infinite path and stationary law

Remark III.3.1 (Exact input closure). The fast-track data are constructed from the public equation, generated target, source grammar, and native physical carrier. A specialization cannot append compactness, coercivity, rigidity, nonpolarity, continuation, or a desired invariant value. Each relation and its complement are graph values. The complementary value is converted into a typed successor on the surviving target-active quotient, and its descendants are processed by finite-depth viability within the same generated recursion.

Remark III.3.2 (Separation of invariant roles). The logical interfaces enforce ten distinctions.

(N1) The fixed **Caus** term is transported by a chart; it is not optimized to point toward the target.

(N2) A killed spectral resolvent is mapped to arrival only through the totalized terminal-flux realization row; a nonzero realization defect is retained.

- (N3) Finiteness of a basic energy does not imply a positive cost at every singular scale; the validity or failure of (164) is the physical-price row.
- (N4) Local entropy monotonicity does not yield a global production bound without the reset relation; its failure is the corresponding cocycle graph defect.
- (N5) Preservation of a nonlinear badness predicate under weak limits is the boundary-defect value in the canonical evaluation closure.
- (N6) Failure of an interface creates its totalized graph-defect record. Target visibility makes that defect part of the singularity-formation algebra; target invisibility places it in J_{res} .
- (N7) Large or noncompact motion in $\mathcal{M}_\Sigma$ requires no downstream estimate because it cannot increase a declared prospective target functional.
- (N8) A quotient is propagated through the zero descent-defect branch; failure to factor is retained as a target-active descent defect.
- (N9) Summability of a constraint tail closes the analytic tail but supplies a physical price only through a carrier comparison such as (175).
- (N10) A positive normalized scale gap excludes an infinite cascade when the pulled-back lower prices have the divergent total mass required by (191); a nonvanishing infimum is a stronger sufficient condition.

III.4 A reusable PDE instantiation protocol

For a particular deterministic PDE, the specialization datum is only its public textbook presentation from Proposition I.1.2 and Corollary I.1.5. Recording that datum requires no new mathematical result. The compiler then performs the following generated operations.

- (I1) Read the displayed equation, datum class, domain, boundary conditions, weak test core, declared regularity class, native transformations, and formal carrier multiplier. Generate the total solution graph, continuation graph, and complete singular target; none is selected by the user after the dynamics are inspected.
- (I2) Expand the formal carrier multiplier against the weak equation, glue common faces with their signs, and take the Jordan decomposition. This generates the one measure μ_u and its derived ceiling. Every nonordinary value is inserted into the activity lift; no replacement measure is requested.
- (I3) Run the normal-form source compiler. It labels native mobility or reversible form **Geom**, fixed directed transport **Caus**, quotient operations **Abs**, generated charts and augmented variables **Lift**, and terminal or exterior fluxes **Bdry**.
- (I4) Form the sourcewise mathematical closure of every dynamically generated weak limit, concentration, oscillation, tail, defect, and blow-up germ.
- (I5) Generate and totalize the target-shell and chain-rule interfaces. Contact supplies unit shell progress by Theorem I.5.2. Killed-generator and predictable-compensator identities may supply additional source factorizations. Compute the resulting singularity-formation support $\mathfrak{S}_\Sigma$.
- (I6) Propagate every vanished target-null, gauge, conservation, and equality direction through the zero descent-defect branch of each later quotient; retain every nonzero descent defect. An empty formation support is removed by Proposition III.1.4; every nonempty support proceeds to the activity lift.
- (I7) Apply the totalized constraint-tail row to any nonlocal constraint. Its output is its local source, gauge image, exterior tail, and target-projected cokernel, without presuming summability. Apply the gradient, commutator, resistance, and capacity rows to degenerate **Geom** directions

selected by the fixed equation type.

- (I8) On the surviving aligned branch, apply Theorem III.1.5; disjoint shell events are derived from contact. Every normalization generated by the equation's symmetry grammar records its exact physical scaling exponent; failure of a normalization relation is a Lift defect.
- (I9) Build the coherent carrier complex and add all local currents with their signs before taking positive parts. Internal transport, pressure, gauge, chart, and compatible interface currents cancel by Theorem II.2.2; true exterior flux and scale cohomology remain explicit.
- (I10) Generate each persistent aligned source cell and its sharp physical price from the preceding stratum. Divergent price excludes the class. Finite price selects vanishing-price windows and activates the ordered atomic/separated/nested/diffuse/tail/rough/gauge/constraint/equality successor rows. Iterate every first-failure output rather than declaring it terminal.
- (I11) Pass an infinite retained route to the compact path-space recurrence theorem. Compile its stationary zero-mean law into the nonnegative production kernels, relative-equilibrium equation, and recurrent graph-boundary equations generated by the PDE. Reduce their target projection with the same five-source and successor algebra; every further reduction is generated from the public weak equation.
- (I12) Form $Z_{n+1} = \mathcal{V}_\eta(Z_n)$. If some finite Z_N is empty, generate the least-coded rational-cylinder pruning transcript of Theorem II.2.15; together with Theorem II.2.21 it gives global regularity on the ordinary finite-carrier branches. Otherwise compact inverse limits construct the infinite path, stationary law, saturation equations, source support, target pairing, and cascade-origin witnesses of the singular mechanism.

This protocol is equation-agnostic. Navier–Stokes, geometric flows, reaction–diffusion systems, kinetic equations, and dispersive–dissipative systems generate different row values in steps (I2), (I5), (I7)–(I11). They do not supply desired values as additional input. The closed source algebra and failure semantics do not change.

III.5 Conclusion

The continuum completion of structural complexity is a theory of physical reachability. Its closure contains every state, profile, measure, trace, and germ generated by ordinary finite-expenditure branches. Limit formation is source-preserving, and the residual is defined only after saturation as a target-null kernel. Hence a target-visible continuum residual cannot exist.

Normalized shell accountability writes unit target progress as an exact integral of the pre-contact current and proves that a singularity requires nonempty target-aligned source support. When that support is empty, global target avoidance follows without estimating the size of the target-null motion. When it is nonempty, propagated reductions remove every already closed direction, and Hamiltonian, resolvent, carrier, capacity, constraint-tail, compactness, scale, and entropy arguments reduce to a finite set of closure, radius, gap, and equality invariants.

The total continuum theorem is a constructive regularity compiler. The source stage exhausts native terms and every generated record by the inherited five sources. A contact cascade selects a persistent source/mode cell. A target-null class cannot form the singularity, and divergent physical price excludes an aligned class. Finite-price infinite progress supplies vanishing-price windows to the ordered successor calculus. Ordinary and graph-boundary values carry witness descendants; an infinite retained route yields a recurrent zero-price law. Signed carrier gluing and the generated saturation equations reduce that law again through the same five-source quotient. A finite rational-cylinder pruning transcript proves global target avoidance; survival at every depth constructs an infinite path, stationary law, full weak equations, source ancestry, target pairing,

and cascade-origin witnesses. Thus the paper never ends at the name of a failed analytic property. It either removes the target-active branch or returns the structural dynamics that sustain singular formation.

A Total branch table for the main conclusions

Generated row	Elimination value	Constructive successor action
Native and local evolution	zero form graph defect	target-visible form or local-existence witness enters the activity lift
Source completion	sourcewise evaluation closure	closure, concentration, harmonic, or boundary coordinate enters the successor graph
Gradient structure	$\mathfrak{H}_{\text{grad}}^{\Sigma} = 0$	target-visible nonclosed-range or harmonic activity profile
Shell accountability	$\mathfrak{H}_{\text{chain}}^{\Sigma} = 0$	chain or shell graph witness enters the activity lift
Target quotient	$\mathfrak{H}_A^{\Sigma} = 0$	target-active source cell
Quotient propagation	zero descent defect	target-active descent defect
Feeding	$\alpha_* \leq 1$	supercritical or null-cost feeding successor
Physical price	divergent sharp series	zero-price target-progress tails
Routing	$\mathcal{R}(\Sigma) < \infty$	harmonic nonroutable successor
Polarity	$\text{Cap}(\Sigma) = 0$	positive-capacity prospective activity profile
Scale accounting	$\mathcal{J}^{\Sigma} = +\infty$	finite-price infinite progress activates ordered successors
Sharp equality	$\mathfrak{H}_{\text{sat}}^{\Sigma} = 0$	equality remainder enters generated zero-price equations
Successor recurrence	target lost or strict block deficit	stationary zero-mean retained law
Generated saturation	totalized equality set target-null	nonzero relative equilibrium or recurrent graph-boundary mechanism
Stochastic lift	zero/routed χ^{Σ}	target-visible covariance/bracket successor
Total weak-solution graph	target-null boundary witness	recurrent graph-boundary path with full weak-equation coordinates
PDE regularity	finite pruning transcript for $Z_N^{\Sigma_{\text{sing}}} = \emptyset$	infinite zero-price path and stationary generated mechanism

B Notation index

Symbol	Meaning
$\mathfrak{C}$	the five-source alphabet $\{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}$
$\mathfrak{M}$	the five analytic realization flags: exact, closure, harmonic, concentration, and boundary
$\hat{X}_\Sigma, U_r(\Sigma), \Phi_r$	target state-or-germ space, normalized shell neighborhoods, and shell observables
B_{phys}	sole equation-generated physical expenditure ceiling for an ordinary maximal branch
$\mathcal{R}_A^{\text{phys}, \text{sat}}(B_{\text{phys}})$	sourcewise saturated physical record cell with support A
$\text{Anc}(R), \text{Src}(R)$	full ancestry word and its elementary source support
$\mathcal{K}_\Sigma$	common kernel of all prospective target functionals
J_{res}	target-null post-saturation residual coordinate; never a source
$\text{Align}_\Sigma(R)$	source labels with positive target progress in a prospective record
$Q_{c,r}(u), \text{Prog}_\Sigma^c(R)$	signed contribution of source c to a normalized shell crossing or another prospective factorization
$\mathfrak{S}_\Sigma(u)$	union of aligned labels in the orbit's shell-complete family; its singularity-formation support
μ_u	finite nonnegative localizable physical expenditure measure for one orbit
α_*, δ_*	sharp domination radius and normalized physical price gap on a reduced source space
$\mathfrak{H}_F^\Sigma$	closed target projection of a candidate source family
$\mathcal{I}_{A,m}^\Sigma$	fast-track invariant tuple for source/mode cell (A, m)
$\mathcal{R}_{\text{FT}}, \mathcal{K}_i^\Sigma$	ordered fast-track rows and the closed physical ledger propagated through row i
$\mathfrak{F}_{A,m}^\Sigma(B_{\text{phys}}),$	source/mode cells of stationary zero-price successor laws and their
$\mathfrak{F}_\Sigma^{\text{phys}}(B_{\text{phys}})$	provenance-tagged total algebra
β_*^{red}	reduced passivity radius of feeding flux relative to gross expenditure
$\mathfrak{o}_{\text{tail}}$	finite-dimensional gauge-cokernel class of an elliptic constraint tail
κ	physical pullback exponent for a normalized scale event
B_Σ	closed normalized target-approaching class in the generated evaluation topology
$\mathfrak{d}$	lower-semicontinuous physical price functional on normalized windows
δ_{phys}	positive lower physical price for a selected event
$\underline{p}_{A,j}$	canonical scale-dependent lower-price infimum for a persistent aligned source row; its series is finite or divergent
$g_{\text{car}}, \beta_\eta$	corrected signed carrier cocycle and its maximal invariant mean on the retained target-active class
$\hat{\mathcal{R}}_\eta$	totalized witness-successor relation for atomic, separated, nested, diffuse, tail, rough, gauge, constraint, and equality descendants
$\mathfrak{H}_{\text{succ}}^\Sigma$	target projection of stationary zero-mean successor laws after generated saturation reduction
$Z_n^\Sigma, Z_\infty^\Sigma$	finite-depth zero-price viability sets and their greatest compact successor fixed point

$E_n, \mathcal{B}_\eta$	open elimination-rank sets and the fixed rational cylinder basis used by the canonical finite pruning transcript
$\text{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}})$	physical target-contact profile under the single ceiling B_{phys}

References

- [1] Jean-Pierre Aubin. Un théorème de compacité. *Comptes Rendus de l'Académie des Sciences de Paris*, 256:5042–5044, 1963.
- [2] Guillem Duran Ballester. Presentation-relative structural complexity: Quantitative limits for general intelligent computation, 2026. Part I of this series.
- [3] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*. De Gruyter, Berlin, second revised and extended edition, 2011.
- [4] Arne Jensen, Eric Mourre, and Peter Perry. Multiple commutator estimates and resolvent smoothness in quantum scattering theory. *Annales de l'Institut Henri Poincaré, Physique Théorique*, 41(2):207–225, 1984.
- [5] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, Berlin, second edition, 1995.
- [6] Jacques-Louis Lions. *Quelques méthodes de résolution des problèmes aux limites non linéaires*. Dunod, Paris, 1969.
- [7] Zhi-Ming Ma and Michael Röckner. *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*. Springer, Berlin, 1992.
- [8] Grisha Perelman. The entropy formula for the ricci flow and its geometric applications, 2002.
- [9] Jacques Simon. Compact sets in the space $l^p(0, t; b)$. *Annali di Matematica Pura ed Applicata*, 146:65–96, 1987.
- [10] Cédric Villani. Hypocoercivity. *Memoirs of the American Mathematical Society*, 202(950), 2009.